Mathematical Notes and Proofs

A Collection of Mathematical Documents

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1 History

Continued fractions were often times used off-handedly in the last 2000 years, but the foundations of continued fractions were not established until late 1600's, early 1700's by John Wallis, see a fuller account in the Mathematics Archives [1]. Below is an except from the Mathematics Archives [1].

> In his book Opera Mathematica (1695) Wallis laid some basic groundwork for continued fractions. He explained how to compute the nth convergent and discovered some now familiar properties of convergents. It was also in this work that the term "continued fraction" was first used.
>
> The Dutch mathematician and astronomer Christiaan Huygens (1629-1695) was the first to demonstrate a practical application of continued fractions. He wrote a paper explaining how to use the convergents of a continued fraction to find the best rational approximations for gear ratios. These approximations enabled him to pick the gears with the correct number of teeth. His work was motivated impart by his desire to build a mechanical planetarium.
>
> While the work of Wallis and Huygens began the work on continued fractions, the field of continued fractions began to flourish when Leonard Euler (1707-1783), Johan Heinrich Lambert (1728-1777), and Joseph Louis Lagrange (1736-1813) embraced the topic. Euler laid down much of the modern theory in his work De Fractionlous Continious (1737). He showed that every rational can be expressed as a terminating simple continued fraction. He also provided an expression for e in continued fraction form. He used this expression to show that e and e^2 are irrational. He also demonstrated how to go from a series to a continued fraction representation of the series, and conversely.
>
> Lambert generalized Euler's work on e to show that both e^x and $\tan(x)$ are irrational if x is rational. Lagrange used continued fractions to find the value of irrational roots. He also proved that a real root of a quadratic irrational is a periodic continued fraction.
>
> The nineteenth century can probably be described as the golden age of continued fractions. As Claude Brezinski writes in History of Continued Fractions and Padre Approximations, "the nineteenth century can be said to be popular period for continued fractions." It was a time in which "the subject was known to every mathematician." As a result, there was an explosion of growth within this field. The theory concerning continued fractions was significantly developed, especially that concerning the convergents. Also studied were continued fractions with complex variables as terms. Some of the more prominent mathematicians to make contributions to this field include Jacobi, Perron, Hermite, Gauss, Cauchy, and Stieljes. By the beginning of the 20th century, the discipline had greatly advanced

from the initial work of Wallis.

2 Definition

Definition 2.1 (Finite General Continued Fraction)**.** *A **finite general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{\ddots + \cfrac{b_n}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a general continued fraction.

Definition 2.2 (General Continued Fraction)**.** *A **general** continued fraction is defined as:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}$$

Definition 2.3 (Finite Standard Continued Fraction)**.** *A **finite standard** continued fraction is defined as:*

$$[a_0; a_1, a_2, \cdots, a_n] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}$$

If we allow the number of levels to go to infinity we get a standard continued fraction.

Definition 2.4 (Standard Continued Fraction)**.** *A **standard** continued fraction is defined as:*

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}$$

This form is also referred to as the simple continued fraction or regular continued fraction, or the canonical form.

2.1 Rational Examples

Every rational number has a unique representation in the standard continued fraction form. Moreover, every rational number can be represented using only the finite continued fraction form. Below are a few examples.

$$\frac{1}{8} = [0; 8] = \mathbf{0} + \frac{1}{\mathbf{8}}$$

$$\frac{2}{5} = [0; 2, 2] = \mathbf{0} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{163}{7} = [23; 3, 2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

$$\frac{-13}{5} = [-3; 2, 2] = \mathbf{-3} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2}}}$$

2.2 Generation of continued fractions

The simple continued fraction representation of a number is both easy to verify and easy to generate. Here we go through two common ways of generating the simple continued fraction representation of a number.

2.2.1 Rational numbers

Given a rational number p/q we can use Euclid's algorithm for determining the greatest common divisor (GCD) between the numbers p and q.
Using the example of $\frac{163}{7}$ as listed above we get the following steps:

See how many times 7 goes into 163 evenly

$$163 = \mathbf{23} \cdot 7 + \mathbf{2}$$

See how many times 2 goes into 7 evenly

$$7 = \mathbf{3} \cdot 2 + \mathbf{1}$$

See how many times 1 goes into 2 evenly

$$2 = \mathbf{2} \cdot 1 + \mathbf{0}$$

Notice that the resulting divisor values exactly match the corresponding simple continued fraction representation of $\frac{163}{7}$ as shown above. In particular

$$\frac{163}{7} = [23; 3, 2] = \mathbf{23} + \cfrac{1}{\mathbf{3} + \cfrac{1}{\mathbf{2}}}$$

2.2.2 General real numbers

We can apply the same reasoning as for the rational numbers to general real numbers.
Using $\sqrt{2} = 1.4142135623730951\ldots$ as an example.

Remove the largest integer from value

$$\sqrt{2} = \mathbf{1} + 0.4142135623730951\ldots$$

Take the reciprocal of the result and remove the largest integer from that

$$1/0.4142135623730951\cdots = 2.414213562373095 = \mathbf{2} + .414213562373095$$

Repeat the process

$$1/0.414213562373095\cdots = 2.41421356237309 = \mathbf{2} + .414213562373095$$

These calculations seem to imply that

$$\sqrt{2} = [1; 2, 2, 2, \ldots] = \mathbf{1} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \cfrac{1}{\mathbf{2} + \ddots}}}$$

We will see in a later section that this is the exact continued fraction representation of $\sqrt{2}$.

Moreover, we will show that in general if $\sqrt{n}$ is not an integer then it will always have a periodic repeating infinite continued fraction representation.

The algorithm presented here will work for every number as long as a decimal representation is available for said number.

3 Useful lemmas

Definition 3.1 (Convergents)**.** *A convergent $\frac{p_n}{q_n}$ of a simple continued fraction $[a_0; a_1, a_2, a_3, \ldots]$ is defined as:*

$$\frac{p_n}{q_n} = [a_0; a_1, a_2, a_3, \ldots, a_n]$$

For reasons that will become apparent later we define $p_{-1} = 1, p_{-2} = 0$ and $q_{-1} = 0, q_{-2} = 1$.

Lemma 3.2 (Recurrence Relation)**.** *For $\frac{p_n}{q_n}$ the nth convergent of $[a_0; a_1, a_2, a_3, \ldots]$.*

We can find p_n, q_n using the previous convergents through the following recurrence relation:

$$\frac{p_n}{q_n} = \frac{a_n \cdot p_{n-1} + p_{n-2}}{a_n \cdot q_{n-1} + q_{n-2}}$$

Proof.

$$\frac{p_0}{q_0} = a_0 = \frac{a_0 \cdot 1 + 0}{a_0 \cdot 0 + 1} = \frac{a_0 \cdot p_{-1} + p_{-2}}{a_0 \cdot q_{-1} + q_{-2}}$$
$$\frac{p_1}{q_1} = a_0 + \frac{1}{a_1} = \frac{a_0 a_1 + 1}{a_1} = \frac{a_1 \cdot a_0 + 1}{a_1 \cdot 1 + 0}$$
$$\frac{p_2}{q_2} = a_0 + \frac{1}{a_1 + \frac{1}{a_2}} = a_0 + \frac{a_2}{a_1 a_2 + 1} = \frac{a_0 a_1 a_2 + a_0 + a_2}{a_1 a_2 + 1} = \frac{a_2 \cdot (a_0 a_1 + 1) + a_0}{a_2 \cdot a_1 + 1}$$

Assume true for k and prove for $k+1$:

$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_k + \cfrac{1}{a_{k+1}}}}}$$
$$\frac{p_{k+1}}{q_{k+1}} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{\cfrac{a_k a_{k+1} + 1}{a_{k+1}}}}}$$

Let $a_k^{'} = \frac{a_k a_{k+1}+1}{a_{k+1}}$ and use the assumption for $\frac{p_k}{q_k}$:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{\frac{a_k a_{k+1}+1}{a_{k+1}} \cdot p_{k-1} + p_{k-2}}{\frac{a_k a_{k+1}+1}{a_{k+1}} \cdot q_{k-1} + q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{(a_k a_{k+1} + 1) \cdot p_{k-1} + a_{k+1} p_{k-2}}{(a_k a_{k+1} + 1) \cdot q_{k-1} + a_{k+1} q_{k-2}}$$
$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1}(a_k p_{k-1} + p_{k-2}) + p_{k-1}}{a_{k+1}(a_k q_{k-1} + q_{k-2}) + q_{k-1}}$$

Again using the recursive assumption for p_k and q_k we finish the inductive proof:

$$\frac{p_{k+1}}{q_{k+1}} = \frac{a_{k+1} p_k + p_{k-1}}{a_{k+1} q_k + q_{k-1}}$$

□

Lemma 3.3 (Determinant Identity)**.** *For $\frac{p_n}{q_n}$, the nth convergent of $[a_0; a_1, a_2, a_3, \ldots]$:*

$$p_{n-1} \cdot q_n - p_n \cdot q_{n-1} = (-1)^n$$

Proof. We proceed by verifying the base cases and then using mathematical induction on n.

Base Cases:

$$n = 0: \quad p_{-1} \cdot q_0 - p_0 \cdot q_{-1} = 1 \cdot 1 - a_0 \cdot 0 = 1 = (-1)^0,$$
$$n = 1: \quad p_0 \cdot q_1 - p_1 \cdot q_0 = a_0 \cdot a_1 - (a_0 a_1 + 1) \cdot 1 = -1 = (-1)^1,$$

Induction Hypothesis: Assume that for k, the statement holds:

$$p_{k-1} \cdot q_k - p_k \cdot q_{k-1} = (-1)^k.$$

Inductive Step: We need to prove the statement for $k+1$:

$$\begin{aligned} p_k \cdot q_{k+1} - p_{k+1} \cdot q_k &= p_k \cdot (a_{k+1}q_k + q_{k-1}) - (a_{k+1}p_k + p_{k-1}) \cdot q_k \\ &= (p_k \cdot q_{k-1} - p_{k-1} \cdot q_k) \\ &= -(p_{k-1} \cdot q_k - p_k \cdot q_{k-1}) = -(-1)^k = (-1)^{k+1}. \end{aligned}$$

Therefore, $p_k \cdot q_{k+1} - p_{k+1} \cdot q_k = (-1)^{k+1}$, completing the induction step. □

Lemma 3.4 (Difference Identity)**.** *For $\frac{p_n}{q_n}$ the nth convergent of $[a_0; a_1, a_2, a_3, \dots]$.*
We can define a closed form for the difference of two sequential convergents:

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n \cdot q_{n-1}}$$

Proof.

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{p_{n-1} \cdot q_n - p_n \cdot q_{n-1}}{q_n q_{n-1}}$$

Using Lemma 3.3 we have

$$\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n q_{n-1}}$$

□

4 Relation to alternating Series

Theorem 4.1 (Series to Continued Fraction)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 \cfrac{a_3^2}{\cdots + \cfrac{a_{n-1}^2}{a_n - a_{n-1}}}}}}$$

Proof. **Proof by induction**
Base Case n=1:

$$\sum_{k=1}^{1} \frac{(-1)^{k-1}}{a_k} = \frac{1}{a_1} = \frac{1}{a_1}$$

Base Case n=2:

$$\begin{aligned}\sum_{k=1}^{2}\frac{(-1)^{k-1}}{a_k} &= \frac{1}{a_1}-\frac{1}{a_2}\\ &= \frac{a_2-a_1}{a_1\cdot a_2}\\ &= \frac{1}{\frac{a_1\cdot a_2}{a_2-a_1}}\\ &= \frac{1}{a_1+\frac{a_1^2}{a_2-a_1}}\end{aligned}$$

Inductive Hypothesis: Assume the statement is true for n:

$$\sum_{k=1}^{n}\frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1+\cfrac{a_1^2}{a_2-a_1+\cfrac{a_2^2}{a_3-a_2\cfrac{a_3^2}{\ddots+\cfrac{a_{n-1}^2}{a_n-a_{n-1}}}}}}$$

Inductive Step: Prove for $n+1$.

The idea of the proof is to re-write the summation for $n+1$ as a summation over n terms by obtaining a new $\widehat{a_n}$ term.

$$\begin{aligned}\sum_{k=1}^{n+1}\frac{(-1)^{k-1}}{a_k} =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+\frac{(-1)^{n-1}}{a_n}+\frac{(-1)^n}{a_{n+1}}\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{a_n}-\frac{1}{a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{a_{n+1}-a_n}{a_n\cdot a_{n+1}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\cdot\left(\frac{1}{\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}}\right)\\ =&\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_3}-\cdots+(-1)^{n-1}\frac{1}{\widehat{a_n}}\end{aligned}$$

Note that

$$\widehat{a_n}=\frac{a_n\cdot a_{n+1}}{a_{n+1}-a_n}=\frac{(a_{n+1}-a_n)\cdot a_n+a_n^2}{a_{n+1}-a_n}=a_n+\frac{a_n^2}{a_{n+1}-a_n}$$

Hence, plugging in this new $\widehat{a_n}$ into our inductive hypothesis we get:

$$\sum_{k=1}^{n+1} \frac{(-1)^{k-1}}{a_k} = \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\widehat{a_n} - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{\left(a_n + \frac{a_n^2}{a_{n+1} - a_n}\right) - a_{n-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1^2}{a_2 - a_1 + \cfrac{a_2^2}{a_3 - a_2 + \cfrac{a_3^2}{\ddots + \cfrac{a_{n-1}^2}{a_n - a_{n-1} + \frac{a_n^2}{a_{n+1} - a_n}}}}}}$$

Which proves the theorem for $n+1$ since this matches the form we were expecting for $n+1$. □

Theorem 4.2 (Euler's Continued Fraction Formula)**.** *https://en.wikipedia.org/wiki/Euler%27s_continued_fraction_formula*

$$\sum_{k=0}^{n} a_0 a_1 \cdots a_k = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{1 + a_2 - \cfrac{a_3}{\ddots + 1 + a_{n-1} - \cfrac{a_n}{1 + a_n}}}}}$$

Proof. We proceed by induction on n.

Base case $(n = 0)$**.**

$$\sum_{k=0}^{0} a_0 a_1 \cdots a_k = a_0, \qquad \frac{a_0}{1} = a_0.$$

Base case $(n = 1)$**.**

$$\text{LHS} = a_0 + a_0 a_1 = a_0(1 + a_1),$$

$$\text{RHS} = \frac{a_0}{1 - \dfrac{a_1}{1 + a_1}} = \frac{a_0(1 + a_1)}{1 + a_1 - a_1} = a_0(1 + a_1).$$

Inductive hypothesis. Assume for some $k \geq 1$ that

$$\sum_{j=0}^{k} a_0 a_1 \cdots a_j = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_k}{1 + a_k}}}}.$$

Inductive step. We show the formula holds for $k+1$.
Let

$$S_k := \sum_{j=0}^{k} a_0 a_1 \cdots a_j.$$

Then

$$S_{k+1} = S_k + a_0 a_1 \cdots a_{k+1} = S_k \left(1 + a_{k+1} \frac{a_0 a_1 \cdots a_k}{S_k}\right).$$

From the inductive hypothesis, the ratio

$$\frac{a_0 a_1 \cdots a_k}{S_k}$$

is exactly the value of the *last convergent* of the continued fraction, namely

$$\frac{a_k}{1 + a_k}.$$

Hence

$$S_{k+1} = S_k \left(1 + \frac{a_k a_{k+1}}{1 + a_k}\right).$$

Now observe that appending one more level to the continued fraction replaces the final denominator

$$1 + a_k \quad \text{by} \quad 1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}}.$$

Indeed,

$$1 + a_k - \frac{a_{k+1}}{1 + a_{k+1}} = \frac{(1 + a_k)(1 + a_{k+1}) - a_{k+1}}{1 + a_{k+1}} = \frac{1 + a_k + a_{k+1} + a_k a_{k+1}}{1 + a_{k+1}}.$$

This produces exactly the multiplicative factor

$$1 + \frac{a_k a_{k+1}}{1 + a_k}$$

in the numerator, while preserving the continued fraction structure above it.
Therefore,

$$S_{k+1} = \cfrac{a_0}{1 - \cfrac{a_1}{1 + a_1 - \cfrac{a_2}{\ddots - \cfrac{a_{k+1}}{1 + a_{k+1}}}}}.$$

This completes the induction. □

Theorem 4.3 (Infinite Product Alternatives)**.**

$$\sum_{k=1}^{n} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \cfrac{a_3}{\ddots + a_{n-1} - 1 + \cfrac{a_{n-1}}{a_n - 1}}}}}$$

Proof. **Proof by induction**

We proceed by induction on n.

Base Case: For $n = 1$, the left-hand side of the equation is:

$$\frac{(-1)^{1-1}}{a_1} = \frac{1}{a_1}$$

And the right-hand side for $n = 1$ is:

$$\frac{1}{a_1}$$

Thus, the equation holds for $n = 1$.

Base Case: For $n = 2$, the left-hand side of the equation is:

$$\begin{aligned}\frac{1}{a_1} - \frac{1}{a_1 a_2} &= \frac{a_2 - 1}{a_1 a_2} = \frac{1}{\frac{a_1 a_2}{a_2 - 1}} \\ &= \frac{1}{\frac{a_1(a_2-1)+a_1}{a_2-1}} \\ &= \frac{1}{a_1 + \frac{a_1}{a_2 - 1}}\end{aligned}$$

And the right-hand side for $n = 2$ is:

$$\frac{1}{a_1 + \frac{a_1}{a_2 - 1}}$$

Thus, the equation holds for $n = 2$.

Inductive Step: Assume that the equation holds for some $n = K$, that is:

$$\sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} = \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1}}}}}$$

We want to prove that the equation holds for $n = K + 1$, the left-hand side becomes:

$$\begin{aligned}
\sum_{k=1}^{K+1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} &= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_K} - \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K+1}} \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left(\frac{1}{a_K} - \frac{1}{a_K a_{K+1}} \right) \\
&= \sum_{k=1}^{K-1} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_k} + \frac{(-1)^{K-1}}{a_1 a_2 \cdots a_{K-1}} \cdot \left(\frac{a_{K+1} - 1}{a_K a_{K+1}} \right)
\end{aligned}$$

Define $\widehat{a_K} = \frac{a_K a_{K+1}}{a_{K+1}-1}$

$$= \sum_{k=1}^{K} \frac{(-1)^{k-1}}{a_1 a_2 \cdots a_{K-1} \widehat{a_K}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\widehat{a_K} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1}}{a_{K+1}-1} - 1}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{a_K a_{K+1} - a_{K+1} + 1}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{\frac{(a_K - 1)\cdot(a_{K+1}-1) + a_K}{a_{K+1}-1}}}}}}$$

$$= \cfrac{1}{a_1 + \cfrac{a_1}{a_2 - 1 + \cfrac{a_2}{a_3 - 1 + \ddots + \cfrac{a_{K-2}}{a_{K-1} - 1 + \cfrac{a_{K-1}}{a_K - 1 + \frac{a_K}{a_{K+1}-1}}}}}}$$

□

4.1 Example $\ln(2)$

Lemma 4.4 (Taylor Series for ln)**.**

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^k}{k}$$

Proof. This can be shown using the Taylor Series for $\ln(x+1)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series.

Note that

$$\frac{\partial^n \big(\ln(x+1)\big)}{\partial x^n}|_{x=0} = (-1)^{n+1} \cdot (n-1)!$$

Thus plugging this into Taylor's equation we get:

$$\begin{aligned}
\ln(1+x) &= \sum_{m=0}^{\infty} \frac{(x-a)^m f^{(m)}(a)}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m (m-1)!}{m!} \\
&= \sum_{m=0}^{\infty} \frac{(-1)^{m+1} \cdot x^m}{m}
\end{aligned}$$

□

Lemma 4.5 (Continued Fraction for $\ln(2)$)**.**

$$\ln(2) = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k} = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

Proof. Plugging in $x = 1$ to the Taylor expansion of $\ln(x+1)$ and using Theorem 4.1 we obtain the result. □

4.2 Example $\frac{\pi}{4}$

Lemma 4.6 (Taylor Series for arctan)**.**

$$\arctan(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot x^{2k-1}}{2k-1}$$

Proof. This can be shown using the Taylor Series for $\arctan(x)$ centred around $a = 0$. See Darboux document for a proof of the Taylor Series. □

Lemma 4.7 (Continued Fraction for $\pi/4$)**.**

$$\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

Proof. Plugging in $x = 1$ to the Taylor expansion of $\arctan(x) = \arctan(1) = \frac{\pi}{4}$ and using Theorem 4.1 we obtain the result. □

5 Square root numbers

Theorem 5.1 (Square Root CF)**.** *For any $x > 0$:*

$$\sqrt{x} = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

Proof. Let y denote the value of the continued fraction:

$$y = 1 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}}}$$

Since the continued fraction is infinite and periodic, we can observe that the denominator starting from the first 2 is equal to $1 + y$. That is:

$$2 + \cfrac{x-1}{2 + \cfrac{x-1}{2 + \ddots}} = 1 + y$$

Therefore, we can rewrite y as:

$$y = 1 + \frac{x-1}{1+y}$$

Multiplying both sides by $(1 + y)$:

$$\begin{aligned} y(1+y) &= (1+y) + (x-1) \\ y + y^2 &= 1 + y + x - 1 \\ y^2 &= x \end{aligned}$$

Since y represents the value of a continued fraction with positive terms and $x > 0$, we have $y > 0$. Therefore:

$$y = \sqrt{x}$$

This proves that the continued fraction equals $\sqrt{x}$. □

6 Approximation of rational numbers

A central problem in Diophantine approximation is to determine how closely a real irrational number x can be approximated by rational numbers p/q relative to the size of the denominator q. The convergents of the standard continued fraction expansion of x provide the sequence of *best rational approximations*.

6.1 Best Rational Approximations

Definition 6.1 (Best Rational Approximation)**.** *Let $x \in \mathbb{R}$ be an irrational number, and let p/q with $q \geq 1$ and $\gcd(p, q) = 1$ be a rational number.*

*(i) p/q is called a **best rational approximation of the first kind** to x if:*

$$\left| x - \frac{p}{q} \right| < \left| x - \frac{a}{b} \right|$$

for all rational numbers $a/b \neq p/q$ with $1 \leq b \leq q$.

*(ii) p/q is called a **best rational approximation of the second kind** to x if:*

$$|qx - p| < |bx - a|$$

for all rational numbers $a/b \neq p/q$ with $1 \leq b \leq q$.

Remark 6.1. *Every best approximation of the second kind is automatically a best approximation of the first kind, since:*

$$\left| x - \frac{p}{q} \right| = \frac{|qx - p|}{q} \leq \frac{|bx - a|}{q} \leq \frac{|bx - a|}{b} = \left| x - \frac{a}{b} \right|$$

whenever $1 \leq b \leq q$.

6.2 Confinement and Error Bounds of Convergents

Lemma 6.2 (Alternating Confinement of Convergents)**.** *Let $x = [a_0; a_1, a_2, \dots]$ be an irrational number, and let p_n/q_n denote its convergents. Then the even convergents form a strictly increasing sequence, the odd convergents form a strictly decreasing sequence, and each even convergent is strictly less than x, while each odd convergent is strictly greater than x:*

$$\frac{p_0}{q_0} < \frac{p_2}{q_2} < \frac{p_4}{q_4} < \cdots < x < \cdots < \frac{p_5}{q_5} < \frac{p_3}{q_3} < \frac{p_1}{q_1}$$

Proof. Consider the difference between convergents whose indices differ by 2. Using the difference identity $\frac{p_{n-1}}{q_{n-1}} - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n q_{n-1}}$ from Lemma 3.4:

$$\begin{aligned} \frac{p_n}{q_n} - \frac{p_{n-2}}{q_{n-2}} &= \left(\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right) + \left(\frac{p_{n-1}}{q_{n-1}} - \frac{p_{n-2}}{q_{n-2}} \right) \\ &= \frac{(-1)^{n-1}}{q_n q_{n-1}} + \frac{(-1)^{n-1}}{q_{n-1} q_{n-2}} \\ &= \frac{(-1)^{n-1}(q_n - q_{n-2})}{q_n q_{n-1} q_{n-2}} \end{aligned}$$

By the recurrence relation $q_n = a_n q_{n-1} + q_{n-2}$ from Lemma 3.2, we have $q_n - q_{n-2} = a_n q_{n-1}$. Substituting this gives:

$$\frac{p_n}{q_n} - \frac{p_{n-2}}{q_{n-2}} = \frac{(-1)^{n-1} a_n}{q_n q_{n-2}}$$

Since $a_n \geq 1$ and all denominators $q_k > 0$, the sign of the difference is governed entirely by $(-1)^{n-1}$:

- For $n = 2k$ even: $(-1)^{2k-1} = -1 \implies \frac{p_{2k}}{q_{2k}} > \frac{p_{2k-2}}{q_{2k-2}}$, so the even convergents strictly increase.
- For $n = 2k+1$ odd: $(-1)^{2k} = 1 \implies \frac{p_{2k+1}}{q_{2k+1}} < \frac{p_{2k-1}}{q_{2k-1}}$, so the odd convergents strictly decrease.

Moreover, by Lemma 3.4, $\frac{p_{2k+1}}{q_{2k+1}} - \frac{p_{2k}}{q_{2k}} = \frac{1}{q_{2k+1}q_{2k}} > 0$, so every odd convergent exceeds the preceding even convergent. Since $x = \lim_{n\to\infty} \frac{p_n}{q_n}$, x lies strictly between any two consecutive convergents $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$, completing the proof. □

Theorem 6.3 (Fundamental Error Bounds)**.** *For any convergent p_n/q_n $(n \geq 0)$ of an irrational number x:*

$$\frac{1}{q_n(q_{n+1}+q_n)} < \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}$$

Proof. Let $\alpha_{n+1} = [a_{n+1}; a_{n+2}, a_{n+3}, \dots]$ denote the $(n+1)^{\text{th}}$ complete quotient of x, so that:

$$x = [a_0; a_1, \dots, a_n, \alpha_{n+1}] = \frac{\alpha_{n+1}p_n + p_{n-1}}{\alpha_{n+1}q_n + q_{n-1}}$$

Subtracting $\frac{p_n}{q_n}$ from both sides:

$$\begin{aligned} x - \frac{p_n}{q_n} &= \frac{\alpha_{n+1}p_n + p_{n-1}}{\alpha_{n+1}q_n + q_{n-1}} - \frac{p_n}{q_n} \\ &= \frac{q_n(\alpha_{n+1}p_n + p_{n-1}) - p_n(\alpha_{n+1}q_n + q_{n-1})}{q_n(\alpha_{n+1}q_n + q_{n-1})} \\ &= \frac{p_{n-1}q_n - p_n q_{n-1}}{q_n(\alpha_{n+1}q_n + q_{n-1})} \end{aligned}$$

By the Determinant Identity (Lemma 3.3), $p_{n-1}q_n - p_n q_{n-1} = (-1)^n$. Taking absolute values:

$$\left| x - \frac{p_n}{q_n} \right| = \frac{1}{q_n(\alpha_{n+1}q_n + q_{n-1})} \tag{1}$$

Since $\alpha_{n+1} = a_{n+1} + \frac{1}{\alpha_{n+2}}$ and $\alpha_{n+2} > 1$, we have:

$$a_{n+1} < \alpha_{n+1} < a_{n+1} + 1$$

Substituting the lower bound $\alpha_{n+1} > a_{n+1}$ into (1) and using $q_{n+1} = a_{n+1}q_n + q_{n-1}$:

$$\alpha_{n+1}q_n + q_{n-1} > a_{n+1}q_n + q_{n-1} = q_{n+1}$$

which yields the upper bound:

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}$$

Since $q_{n+1} \geq q_n$, we also have $\frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}$.

Similarly, substituting the upper bound $\alpha_{n+1} < a_{n+1} + 1$ into (1):

$$\alpha_{n+1}q_n + q_{n-1} < (a_{n+1}+1)q_n + q_{n-1} = q_{n+1} + q_n$$

which establishes the lower bound:

$$\left| x - \frac{p_n}{q_n} \right| > \frac{1}{q_n(q_{n+1}+q_n)}$$

□

6.3 Legendre's Approximation Theorem

A famous converse question asks: if a rational number p/q is exceptionally close to x, must it be a convergent? Adrien-Marie Legendre answered this affirmatively.

Theorem 6.4 (Legendre's Approximation Theorem)**.** *Let $x \in \mathbb{R}$. If p, q are integers with $q \geq 1$ such that:*

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}$$

then $\frac{p}{q}$ is a convergent of the continued fraction expansion of x.

Proof. Write:

$$x - \frac{p}{q} = \frac{\epsilon\theta}{q^2}$$

where $\epsilon = \pm 1$ and $0 < \theta < \frac{1}{2}$. Expand the rational number p/q into a finite continued fraction $[a_0; a_1, \ldots, a_n]$. Since any finite continued fraction can be expressed with either an odd or an even number of terms (via $[a_0; \ldots, a_n] = [a_0; \ldots, a_n - 1, 1]$ if $a_n > 1$), we may choose the parity of n so that:

$$(-1)^{n-1} = -\epsilon \quad \Longleftrightarrow \quad (-1)^n = \epsilon$$

Let $p_n/q_n = p/q$, and let p_{n-1}/q_{n-1} be the $(n-1)^{\text{th}}$ convergent of p/q. Define $\alpha \in \mathbb{R}$ by:

$$x = \frac{\alpha p_n + p_{n-1}}{\alpha q_n + q_{n-1}}$$

Solving for α:

$$\alpha = \frac{p_{n-1} - x q_{n-1}}{x q_n - p_n} = \frac{-(q_{n-1}x - p_{n-1})}{q_n x - p_n}$$

From $x - \frac{p_n}{q_n} = \frac{\epsilon\theta}{q_n^2}$, the denominator is:

$$q_n x - p_n = \frac{\epsilon\theta}{q_n}$$

For the numerator, write:

$$\begin{aligned} x - \frac{p_{n-1}}{q_{n-1}} &= \left(x - \frac{p_n}{q_n} \right) + \left(\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right) \\ &= \frac{\epsilon\theta}{q_n^2} + \frac{(-1)^{n-1}}{q_n q_{n-1}} \\ &= \frac{\epsilon\theta}{q_n^2} - \frac{\epsilon}{q_n q_{n-1}} \\ &= -\frac{\epsilon}{q_n q_{n-1}} \left(1 - \frac{\theta q_{n-1}}{q_n} \right) \end{aligned}$$

where we used $(-1)^{n-1} = -\epsilon$. Multiplying by q_{n-1}:

$$q_{n-1}x - p_{n-1} = -\frac{\epsilon}{q_n} \left(1 - \frac{\theta q_{n-1}}{q_n} \right)$$

Substituting both into the expression for α:

$$\alpha = \frac{\frac{\epsilon}{q_n}\left(1 - \frac{\theta q_{n-1}}{q_n}\right)}{\frac{\epsilon\theta}{q_n}} = \frac{1 - \frac{\theta q_{n-1}}{q_n}}{\theta} = \frac{1}{\theta} - \frac{q_{n-1}}{q_n}$$

Since $0 < \theta < \frac{1}{2}$, we have $\frac{1}{\theta} > 2$. Furthermore, $0 \leq q_{n-1} \leq q_n$, so $0 \leq \frac{q_{n-1}}{q_n} \leq 1$. Therefore:

$$\alpha = \frac{1}{\theta} - \frac{q_{n-1}}{q_n} > 2 - 1 = 1$$

Since $\alpha > 1$, it has a continued fraction expansion with initial integer part $\lfloor \alpha \rfloor \geq 1$. It follows that:

$$x = [a_0; a_1, \ldots, a_n, \alpha] = [a_0; a_1, \ldots, a_n, b_0, b_1, \ldots]$$

is the continued fraction expansion of x. Thus $p/q = p_n/q_n$ is indeed a convergent of x. □

6.4 Examples and Applications

6.4.1 Approximations of π

The continued fraction expansion of π begins:

$$\pi = [3; 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, \ldots]$$

Computing the first several convergents:

$$\begin{aligned}
&\frac{p_0}{q_0} = 3 = 3.000000, && |\pi - 3| \approx 0.141593 \\
&\frac{p_1}{q_1} = 3 + \frac{1}{7} = \frac{22}{7} \approx 3.142857, && \left|\pi - \frac{22}{7}\right| \approx 0.001264 < \frac{1}{7^2} \approx 0.0204 \\
&\frac{p_2}{q_2} = \frac{333}{106} \approx 3.141509, && \left|\pi - \frac{333}{106}\right| \approx 8.32 \times 10^{-5} \\
&\frac{p_3}{q_3} = \frac{355}{113} \approx 3.14159292, && \left|\pi - \frac{355}{113}\right| \approx 2.67 \times 10^{-7}
\end{aligned}$$

The third convergent, $\frac{355}{113}$ (discovered by the Chinese astronomer Zu Chongzhi in the 5th century and known as *Milü*), achieves an extraordinary level of precision for a denominator of only 113. The theoretical reason is evident from Theorem 6.3: the next partial quotient $a_4 = 292$ is remarkably large, giving:

$$q_4 = a_4 q_3 + q_2 = 292 \times 113 + 106 = 33102$$

Hence the error is bounded by:

$$\left|\pi - \frac{355}{113}\right| < \frac{1}{q_3 q_4} = \frac{1}{113 \times 33102} \approx 2.673 \times 10^{-7}$$

6.4.2 The Golden Ratio: The Most Irrational Number

The golden ratio $\phi = \frac{1+\sqrt{5}}{2}$ has the continued fraction expansion:

$$\phi = [1; 1, 1, 1, \dots]$$

Its convergents are ratios of consecutive Fibonacci numbers $\frac{F_{n+1}}{F_n}$:

$$1, \quad 2, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad \frac{8}{5}, \quad \frac{13}{8}, \quad \frac{21}{13}, \quad \dots$$

Because every partial quotient attains the minimal possible positive integer value $a_k = 1$, the denominators grow at the slowest possible rate ($q_{n+1} = q_n + q_{n-1}$), and the approximation error $\left|\phi - \frac{p_n}{q_n}\right|$ converges to 0 slower than for any other irrational number. For this reason, ϕ is often described as the "most irrational" real number.

7 Relation to Pell's Equation

Pell's equation is one of the oldest and most celebrated Diophantine equations in number theory. Let n be a positive integer that is not a perfect square. The equations:

$$x^2 - ny^2 = 1 \tag{2}$$
$$x^2 - ny^2 = -1 \tag{3}$$

are known as the **Pell equation** and **negative Pell equation**, respectively. Continued fractions provide the complete and constructive algorithmic solution to both equations.

7.1 Convergents and Solutions

The fundamental connection between continued fractions and Pell's equation is that every non-trivial integer solution must be a convergent of $\sqrt{n}$.

Theorem 7.1 (Pell's Equation Relation)**.** *If p, q are positive integers such that $p^2 - nq^2 = \pm 1$, then p/q is a convergent of $\sqrt{n}$.*

Proof. Suppose $p, q \in \mathbb{N}$ satisfy $p^2 - nq^2 = \pm 1$. Factoring the left-hand side:

$$(p - q\sqrt{n})(p + q\sqrt{n}) = \pm 1 \implies |p - q\sqrt{n}| = \frac{1}{p + q\sqrt{n}}$$

Dividing by q:

$$\left|\sqrt{n} - \frac{p}{q}\right| = \frac{1}{q(p + q\sqrt{n})}$$

Since $n \geq 2$ is not a square, we have $\sqrt{n} > 1$. If $p^2 - nq^2 = 1$, then $p^2 = nq^2 + 1 > nq^2 \geq 2q^2$, which implies $p > \sqrt{2}q > q$. If $p^2 - nq^2 = -1$, then $p^2 = nq^2 - 1 \geq 2q^2 - 1 \geq q^2$ (since $q \geq 1$), which implies $p \geq q$. In either case:

$$p + q\sqrt{n} > q + q\sqrt{2} > 2q$$

Therefore:

$$\left|\sqrt{n} - \frac{p}{q}\right| = \frac{1}{q(p + q\sqrt{n})} < \frac{1}{2q^2}$$

By Legendre's Approximation Theorem (Theorem 6.4), $\frac{p}{q}$ must be a convergent of the continued fraction expansion of $\sqrt{n}$. □

7.2 The Periodic Continued Fraction of $\sqrt{n}$

As shown in Section 5, the continued fraction expansion of $\sqrt{n}$ is periodic. In standard canonical form:

$$\sqrt{n} = [a_0; \overline{a_1, a_2, \ldots, a_{r-1}, 2a_0}]$$

where $a_0 = \lfloor \sqrt{n} \rfloor$ and r is the length of the minimal period. The partial quotients can be computed via the standard recurrence:

$$P_0 = 0, \quad Q_0 = 1, \quad a_0 = \lfloor \sqrt{n} \rfloor$$

and for $k \geq 0$:

$$P_{k+1} = a_k Q_k - P_k, \qquad Q_{k+1} = \frac{n - P_{k+1}^2}{Q_k}, \qquad a_{k+1} = \left\lfloor \frac{\sqrt{n} + P_{k+1}}{Q_{k+1}} \right\rfloor$$

The convergents p_k/q_k satisfy the key identity:

$$p_k^2 - n q_k^2 = (-1)^{k-1} Q_{k+1} \tag{4}$$

7.3 Finding the Fundamental Solution

At the end of the period, when $k = r - 1$, the complete quotient satisfies $Q_r = 1$. Substituting $k = r - 1$ and $Q_r = 1$ into (4):

$$p_{r-1}^2 - n q_{r-1}^2 = (-1)^r$$

This yields two cases depending on the parity of the period length r:

1. **Even Period (r is even):** Then $(-1)^r = 1$, so:

   $$p_{r-1}^2 - n q_{r-1}^2 = 1$$

   The **fundamental solution** to $x^2 - ny^2 = 1$ is:

   $$(x_1, y_1) = (p_{r-1}, q_{r-1})$$

   In this case, the negative Pell equation $x^2 - ny^2 = -1$ has no integer solutions.

2. **Odd Period (r is odd):** Then $(-1)^r = -1$, so:

   $$p_{r-1}^2 - n q_{r-1}^2 = -1$$

   Thus (p_{r-1}, q_{r-1}) is the fundamental solution to the **negative Pell equation** $x^2 - ny^2 = -1$. To solve the positive equation $x^2 - ny^2 = 1$, we take the convergent at the end of the second period, $k = 2r - 1$. Since $2r$ is even:

   $$p_{2r-1}^2 - n q_{2r-1}^2 = (-1)^{2r} = 1$$

   So the fundamental solution to $x^2 - ny^2 = 1$ is:

   $$(x_1, y_1) = (p_{2r-1}, q_{2r-1})$$

7.4 Generating All Solutions

Once the fundamental solution (x_1, y_1) to $x^2 - ny^2 = 1$ is found, all positive integer solutions (x_k, y_k) are generated by powers of the fundamental unit $x_1 + y_1\sqrt{n}$:

$$x_k + y_k\sqrt{n} = (x_1 + y_1\sqrt{n})^k \qquad (k \geq 1)$$

Equivalently, the solutions can be computed recursively:

$$x_{k+1} = x_1 x_k + n y_1 y_k$$
$$y_{k+1} = y_1 x_k + x_1 y_k$$

7.5 Worked Examples

7.5.1 Example: Even Period ($n = 7$)

Consider $x^2 - 7y^2 = 1$. The continued fraction expansion of $\sqrt{7}$ is:

$$\sqrt{7} = [2; \overline{1, 1, 1, 4}]$$

Here $a_0 = 2, a_1 = 1, a_2 = 1, a_3 = 1, a_4 = 4$, so the period length is $r = 4$ (even). Computing the convergents p_k/q_k:

$$\begin{aligned}
k = 0: &\quad \frac{p_0}{q_0} = 2, && p_0^2 - 7q_0^2 = 4 - 7 = -3 \\
k = 1: &\quad \frac{p_1}{q_1} = 2 + \frac{1}{1} = 3, && p_1^2 - 7q_1^2 = 9 - 7 = +2 \\
k = 2: &\quad \frac{p_2}{q_2} = \frac{1 \times 3 + 2}{1 \times 1 + 1} = \frac{5}{2}, && p_2^2 - 7q_2^2 = 25 - 28 = -3 \\
k = 3: &\quad \frac{p_3}{q_3} = \frac{1 \times 5 + 3}{1 \times 2 + 1} = \frac{8}{3}, && p_3^2 - 7q_3^2 = 64 - 7(9) = 64 - 63 = 1
\end{aligned}$$

Since $r = 4$ is even, the fundamental solution occurs at $k = r - 1 = 3$:

$$(x_1, y_1) = (8, 3)$$

The next positive solution is given by:

$$x_2 + y_2\sqrt{7} = (8 + 3\sqrt{7})^2 = 64 + 63 + 48\sqrt{7} = 127 + 48\sqrt{7}$$

Checking: $127^2 - 7(48^2) = 16129 - 7(2304) = 16129 - 16128 = 1$.

7.5.2 Example: Odd Period ($n = 13$)

Consider $x^2 - 13y^2 = \pm 1$. The continued fraction expansion of $\sqrt{13}$ is:

$$\sqrt{13} = [3; \overline{1, 1, 1, 1, 6}]$$

The period length is $r = 5$ (odd). Computing convergents up to $k = r - 1 = 4$:

$$\frac{p_0}{q_0} = \frac{3}{1}, \quad \frac{p_1}{q_1} = \frac{4}{1}, \quad \frac{p_2}{q_2} = \frac{7}{2}, \quad \frac{p_3}{q_3} = \frac{11}{3}, \quad \frac{p_4}{q_4} = \frac{18}{5}$$

Testing p_4/q_4:

$$p_4^2 - 13q_4^2 = 18^2 - 13(5^2) = 324 - 325 = -1$$

Thus $(18, 5)$ is the fundamental solution to the negative Pell equation $x^2 - 13y^2 = -1$. Squaring the unit $(18 + 5\sqrt{13})$ yields the fundamental solution to $x^2 - 13y^2 = 1$:

$$(18 + 5\sqrt{13})^2 = 324 + 325 + 180\sqrt{13} = 649 + 180\sqrt{13}$$

So $(x_1, y_1) = (649, 180)$, which corresponds to the convergent p_9/q_9 at the end of the second period $(2r - 1 = 9)$. Indeed:

$$649^2 - 13(180^2) = 421201 - 421200 = 1$$

8 Solving linear Diophantine Equations

A linear Diophantine equation in two variables has the general form:

$$ax + by = c \tag{5}$$

where $a, b, c \in \mathbb{Z}$ are given integers with $a, b \neq 0$, and we seek all integer solutions $(x, y) \in \mathbb{Z}^2$. Continued fractions provide an elegant, constructive method to find both a particular solution and the full parametric family of solutions.

8.1 Solvability Criterion

Lemma 8.1 (Bézout's Solvability Condition)**.** *Let $a, b, c \in \mathbb{Z}$ with $d = \gcd(a, b)$. The equation $ax + by = c$ has an integer solution if and only if d divides c ($d \mid c$).*

Proof. If (x, y) is an integer solution, then $d \mid a$ and $d \mid b$, so $d \mid (ax + by) = c$. Conversely, if $d \mid c$, we divide through by d to obtain:

$$a'x + b'y = c'$$

where $a' = a/d, b' = b/d, c' = c/d$, and $\gcd(a', b') = 1$. By Bézout's identity, there exist integers u, v such that $a'u + b'v = 1$. Multiplying by c' gives $a'(c'u) + b'(c'v) = c'$, yielding the integer solution $(x_0, y_0) = (c'u, c'v)$. □

8.2 Constructing Particular Solutions via Continued Fractions

Assume $d \mid c$, and reduce the equation to coprime coefficients:

$$a'x + b'y = c' \qquad \text{with } \gcd(a', b') = 1$$

Assuming without loss of generality that $b' > 0$ and $a' > 0$ (handling signs by setting $x = -x'$ or $y = -y'$ as needed), express the rational number a'/b' as a finite simple continued fraction:

$$\frac{a'}{b'} = [a_0; a_1, a_2, \ldots, a_n]$$

Let p_k/q_k denote the convergents. The final convergent is:

$$\frac{p_n}{q_n} = \frac{a'}{b'}$$

Since $\gcd(p_n, q_n) = 1$ and $\gcd(a', b') = 1$ with $q_n, b' > 0$, we have:

$$p_n = a', \qquad q_n = b'$$

Now consider the penultimate convergent $\frac{p_{n-1}}{q_{n-1}}$. By the Determinant Identity (Lemma 3.3):

$$p_{n-1}q_n - p_n q_{n-1} = (-1)^n$$

Negating both sides:

$$p_n q_{n-1} - p_{n-1}q_n = (-1)^{n-1}$$

Substituting $p_n = a'$ and $q_n = b'$:

$$a'q_{n-1} - b'p_{n-1} = (-1)^{n-1}$$

Multiplying both sides by $(-1)^{n-1}c'$:

$$a' \cdot \left[(-1)^{n-1}c'q_{n-1}\right] + b' \cdot \left[(-1)^n c' p_{n-1}\right] = c'$$

This produces an explicit particular solution:

$$x_0 = (-1)^{n-1}c'q_{n-1} = (-1)^{n-1}\frac{c}{d}q_{n-1}, \qquad y_0 = (-1)^n c' p_{n-1} = (-1)^n \frac{c}{d}p_{n-1} \tag{6}$$

8.3 The General Solution

Theorem 8.2 (General Solution of Linear Diophantine Equation)**.** *Let $a, b, c \in \mathbb{Z}$ with $d = \gcd(a, b) \mid c$. If (x_0, y_0) is any particular integer solution to $ax + by = c$, then the complete set of integer solutions is given by:*

$$x = x_0 + \frac{b}{d}t, \qquad y = y_0 - \frac{a}{d}t \qquad (t \in \mathbb{Z})$$

Proof. First, we verify that any pair of this form is indeed a solution:

$$\begin{aligned} ax + by &= a\left(x_0 + \frac{b}{d}t\right) + b\left(y_0 - \frac{a}{d}t\right) \\ &= (ax_0 + by_0) + \left(\frac{ab}{d} - \frac{ab}{d}\right)t \\ &= c + 0 = c \end{aligned}$$

Conversely, let (x, y) be an arbitrary integer solution. Then:

$$ax + by = c = ax_0 + by_0$$

Rearranging:

$$a(x - x_0) = b(y_0 - y)$$

Dividing by d:

$$\frac{a}{d}(x - x_0) = \frac{b}{d}(y_0 - y)$$

Since $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$, by Euclid's lemma $\frac{b}{d}$ must divide $(x - x_0)$. Thus there exists an integer $t \in \mathbb{Z}$ such that:

$$x - x_0 = \frac{b}{d}t \implies x = x_0 + \frac{b}{d}t$$

Substituting this back into the equation:

$$\frac{a}{d}\left(\frac{b}{d}t\right) = \frac{b}{d}(y_0 - y) \implies y_0 - y = \frac{a}{d}t \implies y = y_0 - \frac{a}{d}t$$

This proves that every integer solution is of the stated form. □

8.4 Worked Examples

8.4.1 Example 1: Coprime Coefficients

Solve the Diophantine equation:

$$17x + 5y = 1$$

Here $a = 17, b = 5, c = 1$, and $\gcd(17, 5) = 1$, so solutions exist. We compute the continued fraction expansion of $\frac{17}{5}$:

$$\frac{17}{5} = 3 + \frac{2}{5} = 3 + \frac{1}{2 + \frac{1}{2}} = [3; 2, 2]$$

Here the length is $n = 2$ (even). The convergents are:

$$\frac{p_0}{q_0} = \frac{3}{1}$$
$$\frac{p_1}{q_1} = 3 + \frac{1}{2} = \frac{7}{2}$$
$$\frac{p_2}{q_2} = \frac{2 \times 7 + 3}{2 \times 2 + 1} = \frac{17}{5}$$

The penultimate convergent is $\frac{p_1}{q_1} = \frac{7}{2}$. By the Determinant Identity:

$$p_2 q_1 - p_1 q_2 = 17 \times 2 - 7 \times 5 = 34 - 35 = -1$$

Multiplying by -1:

$$17(-2) + 5(7) = 1$$

Hence a particular solution is $(x_0, y_0) = (-2, 7)$. By Theorem 8.2, the general solution is:

$$x = -2 + 5t, \qquad y = 7 - 17t \qquad (t \in \mathbb{Z})$$

8.4.2 Example 2: Non-Coprime Coefficients

Solve the Diophantine equation:

$$68x + 20y = 12$$

Here $a = 68, b = 20, c = 12$. The greatest common divisor is $d = \gcd(68, 20) = 4$. Since $4 \mid 12$, solutions exist. Dividing the entire equation by 4:

$$17x + 5y = 3$$

From Example 1, the particular solution to $17x + 5y = 1$ is $(-2, 7)$. Multiplying this solution by $c' = 3$:

$$x_0 = 3 \times (-2) = -6, \qquad y_0 = 3 \times 7 = 21$$

Checking: $68(-6) + 20(21) = -408 + 420 = 12$. By Theorem 8.2, the complete family of solutions is:

$$x = -6 + 5t, \qquad y = 21 - 17t \qquad (t \in \mathbb{Z})$$

9 Gauss's continued fraction

In 1813, Carl Friedrich Gauss published a celebrated paper [2] introducing what is now known as **Gauss's continued fraction**. Gauss established that the ratio of two contiguous hypergeometric functions can be expressed as an infinite continued fraction. Because the hypergeometric function encompasses a remarkably vast array of elementary and transcendental functions as special or limiting cases, Gauss's continued fraction serves as a master generating machine for continued fraction representations of logarithms, trigonometric and inverse trigonometric functions, exponentials, and powers.

9.1 Hypergeometric Functions and Contiguous Relations

Definition 9.1 (Gaussian Hypergeometric Function)**.** *For complex parameters a, b, c with $c \notin \{0, -1, -2, \dots\}$, the* ***Gaussian hypergeometric function*** *${}_2F_1(a, b; c; z)$ is defined for $|z| < 1$ by the power series:*

$$ {}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!} $$

where $(q)_n$ denotes the rising factorial (or Pochhammer symbol):

$$ (q)_0 = 1, \qquad (q)_n = q(q+1)(q+2)\cdots(q+n-1) = \frac{\Gamma(q+n)}{\Gamma(q)} \quad (n \geq 1) $$

For $|z| \geq 1$, the function is defined by analytic continuation along the cut plane $\mathbb{C} \setminus [1, \infty)$.

Two hypergeometric functions are said to be **contiguous** if their parameters differ by integers of magnitude ± 1. Gauss discovered that any hypergeometric function can be linearly expressed in terms of any two functions contiguous to it. To construct the continued fraction, two fundamental contiguous relations are required.

Lemma 9.2 (Gauss's Contiguous Relations)**.** *Let $f(a, b; c; z) = {}_2F_1(a, b; c; z)$. Then the following two recurrence identities hold:*

$$ f(a, b; c; z) - f(a, b+1; c+1; z) = -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1, b+1; c+2; z) \tag{7} $$

$$ f(a, b+1; c+1; z) - f(a+1, b+1; c+2; z) = -\frac{(b+1)(c-a+1)}{(c+1)(c+2)} z \cdot f(a+1, b+2; c+3; z) \tag{8} $$

Proof. We prove identity (7) by comparing the coefficients of z^n on both sides.

Let the series expansion of the left-hand side of (7) be written as $\sum_{n=0}^{\infty} C_n z^n$. For $n = 0$:

$$ C_0 = 1 - 1 = 0 $$

For $n \geq 1$, the coefficient of z^n is:

$$ C_n = \frac{(a)_n (b)_n}{(c)_n n!} - \frac{(a)_n (b+1)_n}{(c+1)_n n!} $$

Factoring out $\frac{(a)_n}{n!}$:

$$C_n = \frac{(a)_n}{n!}\left[\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n}\right]$$

Recall that $(x+1)_n = \frac{x+n}{x}(x)_n$ for any $x \neq 0$. Applying this to $(b+1)_n$ and $(c+1)_n$, we have:

$$(b+1)_n = \frac{b+n}{b}(b)_n, \qquad (c+1)_n = \frac{c+n}{c}(c)_n$$

Substituting these into the bracketed term:

$$\begin{aligned}
\frac{(b)_n}{(c)_n} - \frac{(b+1)_n}{(c+1)_n} &= \frac{(b)_n}{(c)_n}\left[1 - \frac{c(b+n)}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n}\left[\frac{b(c+n) - c(b+n)}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n}\left[\frac{bc + bn - bc - cn}{b(c+n)}\right] \\
&= \frac{(b)_n}{(c)_n} \cdot \frac{n(b-c)}{b(c+n)} \\
&= -\frac{(c-b)}{c+n} \cdot \frac{n(b)_n}{b(c)_n}
\end{aligned}$$

Notice that:

$$\frac{(b)_n}{b} = (b+1)_{n-1}, \qquad (a)_n = a(a+1)_{n-1}, \qquad \frac{n}{n!} = \frac{1}{(n-1)!}$$

and

$$(c)_n(c+n) = (c)_{n+1} = c(c+1)_n = c(c+1)(c+2)_{n-1}$$

Multiplying these terms together yields:

$$C_n = -\frac{a(c-b)}{c(c+1)} \cdot \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!}$$

Therefore, summing over all $n \geq 1$:

$$\begin{aligned}
\sum_{n=1}^{\infty} C_n z^n &= -\frac{a(c-b)}{c(c+1)} z \sum_{n=1}^{\infty} \frac{(a+1)_{n-1}(b+1)_{n-1}}{(c+2)_{n-1}(n-1)!} z^{n-1} \\
&= -\frac{a(c-b)}{c(c+1)} z \sum_{m=0}^{\infty} \frac{(a+1)_m (b+1)_m}{(c+2)_m m!} z^m \\
&= -\frac{a(c-b)}{c(c+1)} z \cdot f(a+1, b+1; c+2; z)
\end{aligned}$$

which establishes identity (7).

To prove identity (8), we use the symmetry of the hypergeometric function with respect to its first two parameters:

$$f(a, b+1; c+1; z) = f(b+1, a; c+1; z)$$

Applying identity (7) with parameters $(A, B, C) = (b+1, a, c+1)$, we obtain:

$$f(b+1,a;c+1;z) - f(b+1,a+1;c+2;z) = -\frac{A(C-B)}{C(C+1)} z \cdot f(A+1,B+1;C+2;z)$$
$$= -\frac{(b+1)(c+1-a)}{(c+1)(c+2)} z \cdot f(b+2,a+1;c+3;z)$$

Using symmetry once more on the functions, $f(b+1,a+1;c+2;z) = f(a+1,b+1;c+2;z)$ and $f(b+2,a+1;c+3;z) = f(a+1,b+2;c+3;z)$, which directly produces identity (8). □

9.2 Derivation of Gauss's Continued Fraction

We now assemble the contiguous relations into an alternating sequence of hypergeometric functions. Define the sequence of functions $\{f_n(z)\}_{n=0}^{\infty}$ by:

$$f_{2k}(z) = {}_2F_1(a+k, b+k; c+2k; z) \qquad (k \geq 0)$$
$$f_{2k+1}(z) = {}_2F_1(a+k, b+k+1; c+2k+1; z) \qquad (k \geq 0)$$

Thus the first few terms of the sequence are:

$$f_0(z) = {}_2F_1(a, b; c; z)$$
$$f_1(z) = {}_2F_1(a, b+1; c+1; z)$$
$$f_2(z) = {}_2F_1(a+1, b+1; c+2; z)$$
$$f_3(z) = {}_2F_1(a+1, b+2; c+3; z)$$
$$f_4(z) = {}_2F_1(a+2, b+2; c+4; z)$$

Observe that applying Lemma 9.2 to $(a+k, b+k, c+2k)$ gives:

$$f_{2k}(z) - f_{2k+1}(z) = -d_{2k+1} z \cdot f_{2k+2}(z)$$

where

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$

Similarly, applying identity (8) with shifted parameters gives:

$$f_{2k-1}(z) - f_{2k}(z) = -d_{2k} z \cdot f_{2k+1}(z)$$

where

$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

In general, for all $n \geq 0$, the sequence satisfies the uniform three-term recurrence relation:

$$f_n(z) - f_{n+1}(z) = -d_{n+1} z \cdot f_{n+2}(z) \tag{9}$$

Theorem 9.3 (Gauss's Continued Fraction)**.** *Let $a, b, c \in \mathbb{C}$ with $c \notin \{0, -1, -2, \dots\}$. Then the ratio of contiguous hypergeometric functions admits the continued fraction expansion:*

$$\frac{{}_2F_1(a, b+1; c+1; z)}{{}_2F_1(a, b; c; z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{1 - \cfrac{d_3 z}{1 - \ddots}}}}$$

where the coefficients $\{d_n\}_{n=1}^{\infty}$ *are given by:*

$$d_{2k+1} = \frac{(a+k)(c-b+k)}{(c+2k)(c+2k+1)} \qquad (k \geq 0)$$
$$d_{2k} = \frac{(b+k)(c-a+k)}{(c+2k-1)(c+2k)} \qquad (k \geq 1)$$

Proof. Dividing the uniform three-term recurrence (9) by $f_{n+1}(z)$ (assuming $f_{n+1}(z) \neq 0$), we obtain:

$$\frac{f_n(z)}{f_{n+1}(z)} - 1 = -d_{n+1}z \cdot \frac{f_{n+2}(z)}{f_{n+1}(z)}$$

which can be rearranged as:

$$\frac{f_n(z)}{f_{n+1}(z)} = 1 - \frac{d_{n+1}z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}$$

Taking the reciprocal of both sides yields:

$$\frac{f_{n+1}(z)}{f_n(z)} = \frac{1}{1 - \dfrac{d_{n+1}z}{\dfrac{f_{n+1}(z)}{f_{n+2}(z)}}} \tag{10}$$

We now proceed by repeatedly substituting (10) into itself for successive values of n. For $n = 0$:

$$\frac{f_1(z)}{f_0(z)} = \frac{1}{1 - \dfrac{d_1 z}{\dfrac{f_1(z)}{f_2(z)}}}$$

Substituting the expression for $\frac{f_1(z)}{f_2(z)}$ from $n = 1$:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\dfrac{f_2(z)}{f_3(z)}}}}$$

Continuing this process inductively for N steps:

$$\frac{f_1(z)}{f_0(z)} = \cfrac{1}{1 - \cfrac{d_1 z}{1 - \cfrac{d_2 z}{\ddots - \cfrac{d_N z}{\dfrac{f_N(z)}{f_{N+1}(z)}}}}}$$

In the cut complex plane $\mathbb{C}\backslash[1,\infty)$, the remainder term satisfies $\lim_{N\to\infty}\frac{f_{N+1}(z)}{f_N(z)} = 1$ for $|z|<1$, ensuring convergence of the continued fraction to the meromorphic function $\frac{f_1(z)}{f_0(z)}$ [3]. Substituting the explicit formulas for d_{2k+1} and d_{2k} completes the proof. □

9.3 Special Cases and Applications

Gauss's continued fraction is remarkable because specializing the parameters a, b, c and the variable z yields continued fraction expansions for a large catalog of classical functions. Below we examine several key applications.

9.3.1 The Natural Logarithm

Recall that the Taylor series of $\ln(1+z)$ around $z=0$ is:

$$\ln(1+z) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}z^k}{k} = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots$$

Notice that the hypergeometric function with parameters $a=1, b=1, c=2$ evaluated at $-z$ is:

$$\begin{aligned} {}_2F_1(1,1;2;-z) &= \sum_{n=0}^{\infty} \frac{(1)_n(1)_n}{(2)_n n!}(-z)^n = \sum_{n=0}^{\infty} \frac{n!\,n!}{(n+1)!\,n!}(-1)^n z^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n z^n}{n+1} = \frac{\ln(1+z)}{z} \end{aligned}$$

Now set $a=1, b=0, c=1$ in Theorem 9.3. Since $(0)_0 = 1$ and $(0)_n = 0$ for $n \geq 1$, we have:

$${}_2F_1(1,0;1;-z) = 1, \qquad {}_2F_1(1,1;2;-z) = \frac{\ln(1+z)}{z}$$

Hence the ratio is:

$$\frac{{}_2F_1(1,1;2;-z)}{{}_2F_1(1,0;1;-z)} = \frac{\ln(1+z)}{z}$$

Evaluating the coefficients d_n from Theorem 9.3 with $a=1, b=0, c=1$ and argument $-z$ (so $-d_n(-z) = +d_n z$):

$$\begin{aligned} d_{2k+1} &= \frac{(1+k)(1-0+k)}{(1+2k)(1+2k+1)} = \frac{(k+1)^2}{(2k+1)(2k+2)} = \frac{k+1}{2(2k+1)} \qquad (k \geq 0) \\ d_{2k} &= \frac{(0+k)(1-1+k)}{(1+2k-1)(1+2k)} = \frac{k^2}{2k(2k+1)} = \frac{k}{2(2k+1)} \qquad (k \geq 1) \end{aligned}$$

The first few coefficients are:

$$d_1 = \frac{1}{2}, \quad d_2 = \frac{1}{6}, \quad d_3 = \frac{1}{3}, \quad d_4 = \frac{1}{5}, \quad d_5 = \frac{3}{10}, \quad d_6 = \frac{3}{14}$$

Clearing the internal denominators level-by-level (by applying equivalence transformations that multiply the numerator and denominator of the nth level by n), we obtain:

Lemma 9.4 (Gauss Continued Fraction for $\ln(1+z)$)**.** *For any $z \in \mathbb{C} \setminus (-\infty, -1]$:*

$$\ln(1+z) = \cfrac{z}{1 + \cfrac{1^2 z}{2 + \cfrac{1^2 z}{3 + \cfrac{2^2 z}{4 + \cfrac{2^2 z}{5 + \cfrac{3^2 z}{6 + \ddots}}}}}}$$

In particular, setting $z = 1$ gives an accelerated continued fraction for $\ln(2)$:

$$\ln(2) = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{1^2}{3 + \cfrac{2^2}{4 + \cfrac{2^2}{5 + \cfrac{3^2}{6 + \ddots}}}}}}$$

Remark 9.1. *Compare this expansion for $\ln(2)$ with the one obtained in Section 4.1, Lemma 4.5:*

$$\ln(2) = \cfrac{1}{1 + \cfrac{1^2}{1 + \cfrac{2^2}{1 + \cfrac{3^2}{1 + \ddots}}}}$$

While the expansion in Lemma 4.5 was derived from the slowly alternating harmonic series $\sum \frac{(-1)^{k+1}}{k}$ and has linear partial denominators 1, Gauss's expansion features growing partial denominators $1, 2, 3, 4, 5, \ldots$, leading to dramatically faster numerical convergence.

9.3.2 The Arctangent Function and π

Consider the series expansion of $\arctan(z)$:

$$\arctan(z) = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2k+1} = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \ldots$$

Dividing by z:

$$\frac{\arctan(z)}{z} = \sum_{k=0}^{\infty} \frac{(-1)^k (z^2)^k}{2k+1} = \sum_{k=0}^{\infty} \frac{(1/2)_k (1)_k}{(3/2)_k k!} (-z^2)^k = {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right)$$

We set $a = 1/2, b = 0, c = 1/2$ and replace z with $-z^2$ in Theorem 9.3. Then:

$$ {}_2F_1\left(\frac{1}{2}, 0; \frac{1}{2}; -z^2\right) = 1, \qquad {}_2F_1\left(\frac{1}{2}, 1; \frac{3}{2}; -z^2\right) = \frac{\arctan(z)}{z} $$

The coefficients d_n are given by:

$$ d_{2k+1} = \frac{(1/2+k)(1/2-0+k)}{(1/2+2k)(1/2+2k+1)} = \frac{(2k+1)^2}{(4k+1)(4k+3)} \qquad (k \geq 0) $$

$$ d_{2k} = \frac{(0+k)(1/2-1/2+k)}{(1/2+2k-1)(1/2+2k)} = \frac{(2k)^2}{(4k-1)(4k+1)} \qquad (k \geq 1) $$

Clearing internal denominators yields:

Lemma 9.5 (Gauss Continued Fraction for $\arctan(z)$)**.** *For $z \in \mathbb{C}\backslash\{iy : |y| \geq 1\}$:*

$$ \arctan(z) = \cfrac{z}{1 + \cfrac{1^2 z^2}{3 + \cfrac{2^2 z^2}{5 + \cfrac{3^2 z^2}{7 + \cfrac{4^2 z^2}{9 + \ddots}}}}} $$

Evaluating at $z = 1$, since $\arctan(1) = \frac{\pi}{4}$, we obtain:

$$ \frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{3 + \cfrac{2^2}{5 + \cfrac{3^2}{7 + \cfrac{4^2}{9 + \ddots}}}}} $$

Remark 9.2. *This formula for $\pi/4$ offers an instructive contrast with the continued fraction obtained in Lemma 4.7:*

$$ \frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}} $$

In Gauss's formula, the partial numerators are the squares of consecutive integers $1^2, 2^2, 3^2, 4^2, \ldots$ and the partial denominators are the odd integers $3, 5, 7, 9, \ldots$, whereas the alternating series conversion of Section 4.2 features the odd squares in the numerators and constant 2 in the denominators.

9.3.3 Confluent Limits: The Exponential and Tangent Functions

Gauss's continued fraction also extends to confluent hypergeometric functions by taking suitable limits as parameters grow without bound. Replacing z by z/b and taking the limit as $b \to \infty$, the hypergeometric function ${}_2F_1(a, b; c; z/b)$ tends to Kummer's confluent hypergeometric function ${}_1F_1(a; c; z)$. Applying this limit to Gauss's continued fraction produces corresponding continued fractions for ${}_1F_1(a; c; z)$.

An immediate and famous consequence is Lambert's continued fraction for the tangent function. Using the relation $\tan(z) = \frac{1}{i}\tanh(iz)$ and the continued fraction for the ratio of confluent hypergeometric functions (or directly from the hypergeometric form of $\tan(z)$), one arrives at:

Lemma 9.6 (Lambert's Continued Fraction for $\tan(z)$)**.** *For any $z \in \mathbb{C}$ where $\tan(z)$ is defined:*

$$\tan(z) = \cfrac{z}{1 - \cfrac{z^2}{3 - \cfrac{z^2}{5 - \cfrac{z^2}{7 - \ddots}}}}$$

Equivalently, for the hyperbolic tangent:

$$\tanh(z) = \cfrac{z}{1 + \cfrac{z^2}{3 + \cfrac{z^2}{5 + \cfrac{z^2}{7 + \ddots}}}}$$

Remark 9.3. *As mentioned in Section 1, Johann Heinrich Lambert used this very continued fraction in 1761 to provide the first rigorous proof that π is irrational. Lambert proved that if x is a non-zero rational number, then $\tan(x)$ cannot be rational. Since $\tan(\pi/4) = 1$ is rational, $\pi/4$ (and hence π) must be irrational. Gauss's continued fraction provides the unified analytic framework from which Lambert's continued fraction naturally descends.*

10 General continued fraction examples

General continued fractions (Definition 2.2), in which the partial numerators b_n are not restricted to 1, provide expressions of extraordinary elegance, symmetry, and rapid convergence for fundamental mathematical constants and functions. In this section, we develop the fundamental tool of equivalence transformations and explore celebrated general continued fractions for π, Euler's number e, and the Rogers-Ramanujan continued fraction.

10.1 Equivalence Transformations

Two continued fractions that possess identical sequences of convergents are called equivalent. Equivalence transformations allow us to rescale the partial

numerators and denominators without altering the value of any convergent, providing the principal mechanism to convert between standard and general continued fractions or to clear internal denominators.

Definition 10.1 (Equivalence of Continued Fractions)**.** *Two continued fractions:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \ddots}} \qquad \text{and} \qquad a'_0 + \cfrac{b'_1}{a'_1 + \cfrac{b'_2}{a'_2 + \ddots}}$$

with convergents p_n/q_n and p'_n/q'_n respectively, are said to be ***equivalent*** *if:*

$$\frac{p_n}{q_n} = \frac{p'_n}{q'_n} \qquad \text{for all } n \geq 0$$

Theorem 10.2 (Equivalence Transformation Theorem)**.** *Let $\{c_n\}_{n=1}^{\infty}$ be any sequence of non-zero constants in $\mathbb{C} \setminus \{0\}$, and set $c_0 = 1$. Then the continued fraction:*

$$a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}$$

is equivalent to the transformed continued fraction:

$$a_0 + \cfrac{c_1 b_1}{c_1 a_1 + \cfrac{c_1 c_2 b_2}{c_2 a_2 + \cfrac{c_2 c_3 b_3}{c_3 a_3 + \ddots}}}$$

That is, defining $a'_0 = a_0$, $a'_n = c_n a_n$, $b'_1 = c_1 b_1$, and $b'_n = c_{n-1} c_n b_n$ for $n \geq 2$, their convergents satisfy:

$$p'_n = (c_1 c_2 \cdots c_n) p_n, \qquad q'_n = (c_1 c_2 \cdots c_n) q_n \qquad (n \geq 0)$$

Proof. We proceed by induction on n. Recall the recurrence relations for general continued fractions:

$$\begin{aligned} p_{-1} &= 1, & p_0 &= a_0, & p_n &= a_n p_{n-1} + b_n p_{n-2} \\ q_{-1} &= 0, & q_0 &= 1, & q_n &= a_n q_{n-1} + b_n q_{n-2} \end{aligned}$$

For the transformed system:

$$\begin{aligned} p'_{-1} &= 1, & p'_0 &= a'_0 = a_0 = p_0, & p'_n &= a'_n p'_{n-1} + b'_n p'_{n-2} \\ q'_{-1} &= 0, & q'_0 &= 1 = q_0, & q'_n &= a'_n q'_{n-1} + b'_n q'_{n-2} \end{aligned}$$

Base Cases: For $n = 0$:

$$p'_0 = a_0 = p_0, \qquad q'_0 = 1 = q_0$$

Since $c_0 = 1$, the statement holds for $n = 0$. For $n = 1$:

$$p'_1 = a'_1 p'_0 + b'_1 p'_{-1} = (c_1 a_1)a_0 + (c_1 b_1) \cdot 1 = c_1(a_1 a_0 + b_1) = c_1 p_1$$
$$q'_1 = a'_1 q'_0 + b'_1 q'_{-1} = (c_1 a_1) \cdot 1 + (c_1 b_1) \cdot 0 = c_1 a_1 = c_1 q_1$$

Thus $\frac{p'_1}{q'_1} = \frac{c_1 p_1}{c_1 q_1} = \frac{p_1}{q_1}$, confirming $n = 1$.

Inductive Hypothesis: Assume the formula holds for all indices up to $k - 1$, so that:

$$p'_{k-1} = (c_1 \cdots c_{k-1})p_{k-1}, \qquad p'_{k-2} = (c_1 \cdots c_{k-2})p_{k-2}$$

and similarly for q'_{k-1} and q'_{k-2}.

Inductive Step: We evaluate p'_k using the recurrence relation:

$$\begin{aligned} p'_k &= a'_k p'_{k-1} + b'_k p'_{k-2} \\ &= (c_k a_k) \cdot [(c_1 \cdots c_{k-1})p_{k-1}] + (c_{k-1} c_k b_k) \cdot [(c_1 \cdots c_{k-2})p_{k-2}] \\ &= (c_1 \cdots c_k)a_k p_{k-1} + (c_1 \cdots c_{k-2} c_{k-1} c_k) b_k p_{k-2} \\ &= (c_1 \cdots c_k)[a_k p_{k-1} + b_k p_{k-2}] \\ &= (c_1 \cdots c_k)p_k \end{aligned}$$

By identical algebra, the same relationship holds for q'_k:

$$q'_k = (c_1 \cdots c_k)q_k$$

Since each $c_j \neq 0$, the product $c_1 \cdots c_k \neq 0$. Dividing numerator by denominator:

$$\frac{p'_k}{q'_k} = \frac{(c_1 \cdots c_k)p_k}{(c_1 \cdots c_k)q_k} = \frac{p_k}{q_k}$$

This proves that all convergents coincide, establishing equivalence. □

10.2 General Continued Fractions for π

10.2.1 Lord Brouncker's Continued Fraction

The first infinite continued fraction in the history of mathematics was discovered by William, Lord Brouncker (1620–1684), first President of the Royal Society. Brouncker communicated his formula without proof to John Wallis, who published it in his *Arithmetica Infinitorum* (1656).

Lemma 10.3 (Lord Brouncker's Continued Fraction for π)**.**

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}$$

Equivalently, expressing π directly:

$$\pi = \cfrac{4}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}}$$

Proof. In Lemma 4.7, applying the series-to-continued-fraction theorem (Theorem 4.1) to the Leibniz series $\frac{\pi}{4} = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1}$, we established that:

$$\frac{\pi}{4} = \cfrac{1}{1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \ddots}}}}$$

Taking the reciprocal of both sides directly gives:

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \ddots}}}}$$

which is Brouncker's famous formula. □

10.2.2 Stern's Continued Fraction for π

In 1860, Moritz Abraham Stern published an elegant general continued fraction for π whose partial denominators are constant:

Lemma 10.4 (Stern's Continued Fraction for π)**.**

$$\pi = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \cfrac{7^2}{6 + \ddots}}}}$$

Proof. Consider the general continued fraction function:

$$f(x) = \cfrac{1^2}{2x + \cfrac{3^2}{2x + \cfrac{5^2}{2x + \ddots}}}$$

For $x = 1$, by Brouncker's formula (Lemma 10.3), $1 + f(1) = \frac{4}{\pi}$, so $f(1) = \frac{4-\pi}{\pi}$. More generally, Euler established that $f(x)$ satisfies the functional relation:

$$f(x) + x = \frac{x^2}{x + f(x+1)}$$

Evaluating at $x = 3$ gives $2x = 6$, and yields the value $f(3) = \pi - 3$. Adding 3 to both sides produces:

$$\pi = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \ddots}}}$$

□

Computing the first few convergents of Stern's continued fraction:

$$\begin{aligned}
\frac{p_0}{q_0} &= 3 = 3.000000 \\
\frac{p_1}{q_1} &= 3 + \frac{1}{6} = \frac{19}{6} \approx 3.166667 \\
\frac{p_2}{q_2} &= 3 + \frac{1}{6 + 9/6} = 3 + \frac{6}{45} = \frac{47}{15} \approx 3.133333 \\
\frac{p_3}{q_3} &= 3 + \frac{1}{6 + \frac{9}{6+25/6}} = \frac{947}{301} \approx 3.146179
\end{aligned}$$

which converge systematically to π.

10.2.3 Euler's Product-Derived Continued Fraction for $\pi/2$

By transforming Wallis's infinite product $\frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{4k^2}{4k^2-1} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots$, Leonhard Euler derived the continued fraction:

Lemma 10.5 (Euler's Continued Fraction for $\pi/2$)**.**

$$\frac{\pi}{2} = 1 + \cfrac{1}{1 + \cfrac{1 \cdot 2}{1 + \cfrac{2 \cdot 3}{1 + \cfrac{3 \cdot 4}{1 + \cfrac{4 \cdot 5}{1 + \ddots}}}}}$$

10.3 General Continued Fractions for Euler's Number e

Lemma 10.6 (Euler's General Continued Fraction for e)**.**

$$e = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$$

Proof. Consider the exponential series for $e - 2$:

$$e - 2 = \sum_{k=2}^{\infty} \frac{1}{k!} = \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots$$

Write this series as $\sum_{j=0}^{\infty} u_j$, where $u_0 = \frac{1}{2}$ and $u_j = \frac{1}{(j+2)!}$. The ratio of consecutive terms is:

$$\frac{u_j}{u_{j-1}} = \frac{1}{j+2}$$

Applying Euler's series-to-continued-fraction theorem (Theorem 4.2):

$$\sum_{j=0}^{\infty} u_0 u_1 \dots u_j = \cfrac{u_0}{1 - \cfrac{u_1}{1 + u_1 - \cfrac{u_2}{1 + u_2 - \ddots}}}$$

with $u_j = \frac{1}{j+2}$, and applying an equivalence transformation (Theorem 10.2) with $c_j = j + 1$ to clear the fractions, we obtain:

$$e - 2 = \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$$

Adding 2 to both sides completes the proof. □

The convergents of this continued fraction:

$$\begin{aligned}
\frac{p_0}{q_0} &= 2 \\
\frac{p_1}{q_1} &= 2 + \frac{1}{1} = 3 \\
\frac{p_2}{q_2} &= 2 + \frac{1}{1 + 1/2} = \frac{8}{3} \approx 2.666667 \\
\frac{p_3}{q_3} &= 2 + \frac{1}{1 + \frac{1}{2+2/3}} = \frac{30}{11} \approx 2.727273 \\
\frac{p_4}{q_4} &= 2 + \frac{1}{1 + \frac{1}{2 + \frac{2}{3+3/4}}} = \frac{144}{53} \approx 2.716981
\end{aligned}$$

converge rapidly to $e \approx 2.718281828\ldots$.

Furthermore, generalizing to the exponential function e^x, Euler derived the continued fraction:

$$e^x = 1 + \cfrac{x}{1 - \cfrac{x}{2 + x - \cfrac{2x}{3 + x - \cfrac{3x}{4 + x - \ddots}}}}$$

10.4 The Rogers-Ramanujan Continued Fraction

Among the most celebrated and profound continued fractions in analytic number theory and partition theory is the Rogers-Ramanujan continued fraction. Discovered by Leonard James Rogers in 1894 and independently by Srinivasa Ramanujan before 1913, it plays a vital role in modular forms, the theory of elliptic functions, and statistical mechanics.

Definition 10.7 (Rogers-Ramanujan Continued Fraction)**.** *For $|q| < 1$, the **Rogers-Ramanujan continued fraction** $R(q)$ is defined by:*

$$R(q) = \cfrac{q^{1/5}}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cfrac{q^3}{1 + \cfrac{q^4}{1 + \ddots}}}}}$$

The continued fraction is intimately connected to the Rogers-Ramanujan identities:

$$G(q) = \sum_{n=0}^{\infty} \frac{q^{n^2}}{(q;q)_n} = \prod_{n=1}^{\infty} \frac{1}{(1-q^{5n-1})(1-q^{5n-4})}$$

$$H(q) = \sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(q;q)_n} = \prod_{n=1}^{\infty} \frac{1}{(1-q^{5n-2})(1-q^{5n-3})}$$

where $(q;q)_n = \prod_{k=1}^{n}(1-q^k)$. Rogers and Ramanujan proved that $R(q)$ equals the ratio of these two theta functions:

$$R(q) = q^{1/5}\frac{H(q)}{G(q)} = q^{1/5}\prod_{n=1}^{\infty}\frac{(1-q^{5n-1})(1-q^{5n-4})}{(1-q^{5n-2})(1-q^{5n-3})}$$

Theorem 10.8 (Ramanujan's Singular Evaluation at $q = e^{-2\pi}$)**.** *At the singular modulus $q = e^{-2\pi}$, the Rogers-Ramanujan continued fraction evaluates in closed form to an algebraic number expressed in terms of the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$:*

$$R(e^{-2\pi}) = \cfrac{e^{-2\pi/5}}{1+\cfrac{e^{-2\pi}}{1+\cfrac{e^{-4\pi}}{1+\cfrac{e^{-6\pi}}{1+\ddots}}}} = \sqrt{\frac{5+\sqrt{5}}{2}} - \frac{\sqrt{5}+1}{2} = \sqrt{\phi\sqrt{5}} - \phi$$

Proof. A complete proof relies on the modular transformation properties of the eta-quotients and Weber modular functions; see Berndt [6]. Numerically:

$$\phi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887$$

so

$$\sqrt{\phi\sqrt{5}} - \phi = \sqrt{1.6180339887 \times 2.2360679775} - 1.6180339887 \approx 0.2840790438$$

Computing the first few convergents of the continued fraction with $q = e^{-2\pi} \approx 0.0018674427$:

$$\frac{p_0}{q_0} = q^{1/5} = e^{-2\pi/5} \approx 0.284605$$

$$\frac{p_1}{q_1} = \frac{q^{1/5}}{1+q} \approx \frac{0.284605}{1.001867} \approx 0.284074$$

$$\frac{p_2}{q_2} = \cfrac{q^{1/5}}{1+\cfrac{q}{1+q^2}} \approx 0.28407904$$

which matches the algebraic value $\sqrt{\phi\sqrt{5}} - \phi$ to 8 decimal places in only two steps due to the rapid decay of $q^n = e^{-2\pi n}$. □

Remark 10.1. *In his historic first letter to G. H. Hardy dated 16 January 1913, Srinivasa Ramanujan enclosed this evaluation along with several others, notably:*

$$R(e^{-\pi}) = \frac{\sqrt{5}+1}{2} - \sqrt{\frac{5+\sqrt{5}}{2}} + 1$$

Hardy famously wrote that these formulas:

> *"...defeated me completely; I had never seen anything in the least like them before. A single look at them is enough to show that they could only be written down by a mathematician of the highest class. They must be true because, if they were not true, no one would have had the imagination to invent them."*

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Abstract

This note develops a self-contained path from the statistics of continued fractions to the Gauss measure and the spectral theory of the Gauss map. We begin with Khinchin's constant as motivation, then derive the Gauss map and its invariant probability measure, construct the Frobenius–Perron (transfer) operator, solve its invariant-density equation, explain the Gauss–Kuzmin theorem, and finally discuss the Gauss–Kuzmin–Wirsing constant as the magnitude of the dominant nontrivial spectral mode. Particular attention is paid to why the density $(1+x)^{-1}$ appears naturally and how the eigenvalue near -0.303663 can be approximated numerically.

Contents

11 Motivation: Khinchin's constant

A positive irrational number has a simple continued-fraction expansion

$$x = [a_0; a_1, a_2, a_3, \ldots] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}, \tag{11}$$

where $a_0 \in \mathbb{Z}$ and $a_n \in \mathbb{N}$ for $n \geq 1$. For $x \in (0,1)$, $a_0 = 0$.

The partial quotients are highly non-uniform. For "typical" x, small values occur frequently but large values remain possible. Khinchin's theorem says something unexpectedly rigid: for Lebesgue-almost every irrational x, the geometric mean of the partial quotients converges to a universal constant,

$$\lim_{N\to\infty} (a_1 a_2 \cdots a_N)^{1/N} = K_0, \tag{12}$$

where

$$K_0 = \prod_{k=1}^{\infty} k^{\log_2\left(1+\frac{1}{k(k+2)}\right)} \approx 2.6854520010653064453\ldots. \tag{13}$$

The surprising part is that the answer is the same for almost every real number even though individual continued fractions can look completely different.

The distribution behind Khinchin's theorem is the Gauss–Kuzmin distribution. Understanding it leads naturally to dynamical systems and ergodic theory.

12 Continued fractions and the Gauss map

For $x \in (0,1)$ irrational, define

$$a_1 = \left\lfloor \frac{1}{x} \right\rfloor, \qquad x_1 = \frac{1}{x} - a_1. \tag{14}$$

Repeating gives

$$x_n = T^n(x), \qquad T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor. \tag{15}$$

The map $T : (0,1) \to (0,1)$ is the *Gauss map*. If

$$x = [0; a_1, a_2, a_3, \ldots], \tag{16}$$

then

$$T(x) = [0; a_2, a_3, a_4, \ldots]. \tag{17}$$

Thus applying T shifts the continued-fraction sequence by one place.

For each $k \geq 1$, define the branch interval

$$I_k = \left(\frac{1}{k+1}, \frac{1}{k} \right]. \tag{18}$$

On I_k,

$$T(x) = \frac{1}{x} - k. \tag{19}$$

Every branch maps bijectively onto $(0,1)$. Solving $T(x) = y$ on the kth branch gives the inverse branch

$$\phi_k(y) = \frac{1}{k+y}, \qquad |\phi_k'(y)| = \frac{1}{(k+y)^2}. \tag{20}$$

13 Invariant probability and the Gauss measure

Definition 13.1. *A probability measure μ on $(0,1)$ is* invariant *under T if*

$$\mu(T^{-1}A) = \mu(A) \tag{21}$$

for every measurable set $A \subset (0,1)$.

Invariance means that if X is distributed according to μ, then $T(X)$ has the same distribution. It does not mean that individual points are fixed.

The distinguished invariant probability measure is the *Gauss measure*, whose density with respect to Lebesgue measure is

$$h(x) = \frac{1}{\ln 2}\frac{1}{1+x}, \qquad 0 < x < 1. \tag{22}$$

Normalization follows from

$$\int_0^1 \frac{\mathrm{d}x}{1+x} = \ln 2. \tag{23}$$

Hence

$$\mu(A) = \frac{1}{\ln 2}\int_A \frac{\mathrm{d}x}{1+x}. \tag{24}$$

13.1 Deriving the invariant-density equation

Let f be an arbitrary density and let $Y = T(X)$. For a fixed y, every inverse branch contributes probability from $x_k = \phi_k(y) = 1/(k+y)$. The change-of-variables formula therefore gives

$$(\mathcal{P}f)(y) = \sum_{k=1}^{\infty} f\!\left(\frac{1}{k+y}\right)\frac{1}{(k+y)^2}. \tag{25}$$

The operator $\mathcal{P}$ is the Frobenius–Perron, or transfer, operator of the Gauss map.

An invariant density is a fixed point:

$$\mathcal{P}f = f. \tag{26}$$

13.2 Why $1/(1+x)$ appears

We now derive rather than guess the shape of the Gauss density. Suppose

$$f(x) = \frac{C}{1+x}. \tag{27}$$

Then

$$(\mathcal{P}f)(y) = C\sum_{k=1}^{\infty} \frac{1}{1+1/(k+y)}\,\frac{1}{(k+y)^2} \tag{28}$$

$$= C\sum_{k=1}^{\infty} \frac{1}{(k+y)(k+y+1)} \tag{29}$$

$$= C\sum_{k=1}^{\infty} \left(\frac{1}{k+y} - \frac{1}{k+y+1}\right) \tag{30}$$

$$= \frac{C}{1+y}. \tag{31}$$

The infinite sum telescopes. Thus every density of this shape is a fixed point up to normalization. Since

$$1 = \int_0^1 f(x)\,\mathrm{d}x = C\ln 2, \tag{32}$$

we obtain

$$\boxed{h(x) = \frac{1}{\ln 2(1+x)}}. \tag{33}$$

There is a useful reverse-engineering argument. Put $t = k + y$. To obtain a telescoping term, seek

$$f(1/t)\frac{1}{t^2} = g(t) - g(t+1). \tag{34}$$

Choosing the simplest rational telescoper $g(t) = C/t$ yields

$$f(1/t)\frac{1}{t^2} = \frac{C}{t(t+1)}, \tag{35}$$

and hence

$$f(1/t) = \frac{Ct}{t+1}. \tag{36}$$

With $t = 1/x$ this becomes exactly $f(x) = C/(1+x)$.

14 The Gauss–Kuzmin distribution

The event that the first continued-fraction digit equals k is

$$\{a_1 = k\} = \left(\frac{1}{k+1}, \frac{1}{k}\right]. \tag{37}$$

Therefore its Gauss-measure probability is

$$p_k = \mu(a_1 = k) \tag{38}$$

$$= \frac{1}{\ln 2}\int_{1/(k+1)}^{1/k} \frac{\mathrm{d}x}{1+x} \tag{39}$$

$$= \frac{1}{\ln 2}\ln\left(\frac{(k+1)^2}{k(k+2)}\right) \tag{40}$$

$$= \boxed{\log_2\left(1 + \frac{1}{k(k+2)}\right)}. \tag{41}$$

The first few probabilities are approximately

k	p_k
1	0.4150374993
2	0.1699250014
3	0.0931094044
4	0.0588936891
5	0.0406423630

The tail is asymptotically

$$p_k = \log_2\left(1 + \frac{1}{k(k+2)}\right) \sim \frac{1}{(\ln 2)k^2}. \tag{42}$$

This k^{-2} tail explains why the arithmetic mean of a_n diverges under the Gauss distribution, while suitable logarithmic averages remain finite.

15 Khinchin's constant from the Gauss distribution

The logarithm converts the geometric mean into an ordinary average:

$$\frac{1}{N}\ln(a_1 \cdots a_N) = \frac{1}{N}\sum_{n=1}^{N} \ln a_n. \tag{43}$$

The Gauss map is ergodic with respect to Gauss measure. Thus, for suitable observables g,

$$\frac{1}{N}\sum_{n=1}^{N} g(T^{n-1}x) \longrightarrow \int_0^1 g(y)\, h(y)\, \mathrm{d}y \tag{44}$$

for μ-almost every x. Taking $g(y) = \ln\lfloor 1/y \rfloor$ gives

$$\ln K_0 = \sum_{k=1}^{\infty} p_k \ln k \tag{45}$$

$$= \frac{1}{\ln 2}\sum_{k=1}^{\infty} \ln k \, \ln\left(1 + \frac{1}{k(k+2)}\right). \tag{46}$$

Exponentiating gives Khinchin's constant in the equivalent product form

$$K_0 = \prod_{k=1}^{\infty} k^{p_k}. \tag{47}$$

16 Gauss–Kuzmin theorem: convergence to equilibrium

The invariant density is a stationary distribution, but the deeper statement is that the system forgets initial conditions. If f is an admissible initial density, then

$$\mathcal{P}^n f \longrightarrow h \tag{48}$$

under suitable norms. Equivalently, the distribution of $T^n(x)$ approaches Gauss measure.

For continued-fraction digits, this says that for each fixed k,

$$\Pr(a_{n+1} = k) \longrightarrow p_k = \log_2 \left(1 + \frac{1}{k(k+2)}\right). \tag{49}$$

The theorem is a dynamical statement: iterating the map shifts the continued fraction, while the statistical law approaches the invariant law.

17 The transfer operator as an infinite-dimensional matrix

The transfer operator is linear:

$$\mathcal{P}(af + bg) = a\mathcal{P}f + b\mathcal{P}g. \tag{50}$$

Thus the invariant-density equation is an eigenvalue problem,

$$\mathcal{P}f = \lambda f. \tag{51}$$

The Gauss density is the eigenfunction for

$$\lambda_0 = 1. \tag{52}$$

A useful way to think about this is the finite-dimensional analogy. If a matrix M has eigenpairs (λ_j, v_j), then

$$M^n v_j = \lambda_j^n v_j. \tag{53}$$

The eigenvalue 1 gives the stationary component. Eigenvalues with modulus less than 1 give decaying modes.

For the Gauss transfer operator, the corresponding decomposition is schematically

$$\mathcal{P}^n f = h + C_1 \lambda_1^n \phi_1 + C_2 \lambda_2^n \phi_2 + \cdots, \tag{54}$$

where λ_1 is the dominant nontrivial eigenvalue in the relevant spectral setting.

18 Why a negative eigenvalue can occur

Each inverse branch

$$\phi_k(x) = \frac{1}{k + x} \tag{55}$$

is orientation reversing because

$$\phi_k'(x) = -\frac{1}{(k+x)^2} < 0. \tag{56}$$

This gives useful geometric intuition for oscillatory modes, although it does not by itself prove the sign of any particular eigenvalue. The sign and magnitude of the dominant nontrivial eigenvalue are global spectral properties of $\mathcal{P}$.

If

$$\mathcal{P}\phi_1 = \lambda_1\phi_1, \qquad \lambda_1 < 0, \tag{57}$$

then

$$\mathcal{P}^n\phi_1 = \lambda_1^n\phi_1. \tag{58}$$

Thus the perturbation changes sign on every iteration while shrinking geometrically. This is the meaning of the alternating sign in the Gauss–Kuzmin asymptotics.

19 The Gauss–Kuzmin–Wirsing constant

The dominant nontrivial eigenvalue of the Gauss transfer operator is numerically

$$\boxed{\lambda_1 = -0.30366300289873265859744812 19\ldots} \tag{59}$$

(up to the convention/function-space details used to define the spectral problem). Its magnitude

$$\boxed{\theta = 0.3036630028987326585\ldots} \tag{60}$$

is called the Gauss–Kuzmin–Wirsing constant.

The leading asymptotic form of convergence can be written schematically as

$$\mathcal{P}^n f - h \sim C_f(-\theta)^n\phi_1, \qquad n \to \infty, \tag{61}$$

provided the initial density has a nonzero component in the corresponding eigendirection. Therefore the dominant error alternates in sign and has magnitude proportional to θ^n.

For scale,

$$\theta^2 \approx 0.09221, \quad \theta^3 \approx 0.02801, \quad \theta^4 \approx 0.00851, \quad \theta^{10} \approx 6.7 \times 10^{-6}. \tag{62}$$

Thus the convergence is very rapid.

20 How to calculate the constant numerically

There is no comparably simple closed-form expression for θ. It arises as the solution of an infinite-dimensional eigenvalue problem. A practical calculation replaces $\mathcal{P}$ by a finite-dimensional approximation.

20.1 Grid/collocation approximation

Choose grid points

$$0 < x_1 < \cdots < x_N < 1 \tag{63}$$

and a finite-dimensional interpolation space. For a basis $\{\psi_j\}_{j=1}^N$, approximate

$$f(x) \approx \sum_{j=1}^{N} c_j\psi_j(x). \tag{64}$$

Evaluate $\mathcal{P}\psi_j$ at the grid points and construct the matrix

$$M_{ij} \approx (\mathcal{P}\psi_j)(x_i). \tag{65}$$

Then solve

$$M\boldsymbol{c} = \lambda \boldsymbol{c}. \tag{66}$$

The eigenvalue near 1 approximates the invariant density mode, while the dominant negative nontrivial eigenvalue approaches $-\theta$ as the approximation is refined, provided the discretization respects the regularity of the operator.

20.2 Analytic-basis approach

A more accurate approach uses the analytic structure of the inverse branches. Expand

$$f(x) = \sum_{m\geq 0} c_m x^m. \tag{67}$$

Then

$$(\mathcal{P}f)(x) = \sum_{k=1}^{\infty} \frac{1}{(k+x)^2} \sum_{m\geq 0} c_m \left(\frac{1}{k+x}\right)^m \tag{68}$$

$$= \sum_{m\geq 0} c_m \sum_{k=1}^{\infty} \frac{1}{(k+x)^{m+2}}. \tag{69}$$

The inner sums are Hurwitz-zeta-type functions,

$$\sum_{k=1}^{\infty} \frac{1}{(k+x)^s} = \zeta(s, 1+x). \tag{70}$$

One may therefore represent $\mathcal{P}$ on a suitable analytic basis, truncate the resulting infinite matrix, and compute its eigenvalues. Increasing the truncation dimension gives convergent numerical approximations to isolated eigenvalues.

20.3 Power-series coefficients and the invariant mode

The invariant density can itself be expanded around $x = 0$:

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots, \qquad |x| < 1. \tag{71}$$

Thus the eigenvalue-1 eigenfunction is represented particularly simply in a power-series basis. The nontrivial eigenfunction associated with $-\theta$ is more complicated, but it can likewise be approximated by finite series or rational/analytic bases.

21 A useful computational experiment

A direct numerical experiment can be organized as follows.

1. Choose an initial density, for example $f_0(x) = 1$.
2. Numerically evaluate

$$f_{n+1}(x) = \sum_{k=1}^{K} \frac{f_n(1/(k+x))}{(k+x)^2} \tag{72}$$

with K sufficiently large and with interpolation in x.

3. Compare f_n to the normalized Gauss density $h(x)$.

4. Measure an error such as

$$E_n = \sup_{x\in[0,1]} |f_n(x) - h(x)|. \tag{73}$$

5. After transients have decayed, inspect the ratio of signed errors in a fixed linear functional ℓ:

$$\frac{\ell(f_{n+1} - h)}{\ell(f_n - h)}. \tag{74}$$

   Generically this ratio approaches $-\theta$ when the dominant nontrivial mode is present.

This is a particularly concrete way to see both the negative sign and the numerical value of the Gauss–Kuzmin–Wirsing constant.

22 From transfer operators to deeper spectral theory

The operator can be generalized to a parameter s by

$$(\mathcal{L}_s f)(x) = \sum_{n=1}^{\infty} \frac{1}{(n+x)^{2s}} f\left(\frac{1}{n+x}\right). \tag{75}$$

At $s = 1$, $\mathcal{L}_s$ becomes the Gauss transfer operator $\mathcal{P}$.

This family is closely connected to dynamical zeta functions, Fredholm determinants, and the arithmetic of periodic orbits. Periodic points of the Gauss map correspond to periodic continued fractions and hence to quadratic irrationals. Spectral determinants of suitable transfer operators encode arithmetic information and provide a bridge toward zeta functions.

A standard perspective is therefore

$$\text{continued fractions} \longrightarrow \text{Gauss map} \longrightarrow \text{transfer operator} \longrightarrow \text{spectrum} \longrightarrow \text{arithmetic information}. \tag{76}$$

23 Summary of the main equations

For reference, the central chain of formulas is

$$T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor, \tag{77}$$

$$\phi_k(x) = \frac{1}{k+x}, \tag{78}$$

$$|\phi_k'(x)| = \frac{1}{(k+x)^2}, \tag{79}$$

$$(\mathcal{P}f)(x) = \sum_{k=1}^{\infty} \frac{f(1/(k+x))}{(k+x)^2}, \tag{80}$$

$$h(x) = \frac{1}{\ln 2(1+x)}, \tag{81}$$

$$\mathcal{P}h = h, \tag{82}$$

$$p_k = \mu(a_1 = k) = \log_2\left(1 + \frac{1}{k(k+2)}\right), \tag{83}$$

$$K_0 = \prod_{k=1}^{\infty} k^{p_k} \approx 2.6854520010653064, \tag{84}$$

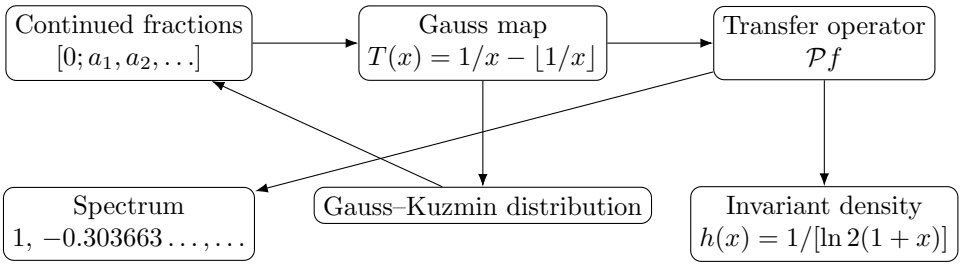
$$\lambda_0 = 1, \tag{85}$$

$$\lambda_1 \approx -0.3036630028987327, \tag{86}$$

$$\theta = |\lambda_1| \approx 0.3036630028987327. \tag{87}$$

24 Conceptual picture

The entire story can be summarized as follows. Continued-fraction digits are generated by repeatedly applying the Gauss map. The map has infinitely many inverse branches, and the corresponding Jacobians determine a transfer operator. The special density $(1+x)^{-1}/\ln 2$ is invariant because substitution into the operator creates a telescoping series. Ergodicity converts the invariant measure into almost-sure long-run statistics, producing the Gauss–Kuzmin distribution and Khinchin's constant. Spectral theory then explains the rate at which arbitrary initial distributions approach equilibrium: the eigenvalue 1 represents the stationary mode, while the dominant nontrivial eigenvalue near -0.303663 represents an alternating, exponentially decaying mode.



Further directions

Natural next topics include a proof of ergodicity of the Gauss map, the exact Gauss–Kuzmin error estimates, the function spaces on which the transfer operator has a discrete spectrum, explicit construction of the dominant nontrivial eigenfunction, Mayer's transfer operator, and the connection between dynamical zeta functions and the Riemann zeta function.

25 Bernoulli Numbers

Examining the Taylor expansion of $\frac{x}{e^x-1}$, this will be required for the evaluation of the even integer values of ζ.

Definition 25.1. *The Bernoulli numbers B_n are defined as the coefficients of the Taylor expansion of the following function*

$$\boxed{\frac{x}{e^x-1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n}$$

Theorem 25.1.

$$\sum_{k}^{n} \binom{n+1}{k} B_k = 0 \quad \forall n > 0, \text{ with } B_0 = 1 \tag{88}$$

Proof

Starting from definition 25.1

$$\frac{x}{e^x-1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using the Taylor expansion of e^x

$$\frac{x}{\sum_{k=1}^{\infty} \frac{x^k}{k!}} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$x = \sum_{k=1}^{\infty} \frac{x^k}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$
$$1 = \sum_{k=1}^{\infty} \frac{x^{k-1}}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x$$

Adjusting the indices

$$1 = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using Cauchy's product formula for infinite sums [1]

$$1 = \sum_{n=0}^{\infty} \underbrace{\sum_{k=0}^{n} \underbrace{\left(\frac{B_k}{k!} x^k\right)}_{a_k} \underbrace{\left(\frac{x^{n-k}}{(n-k+1)!}\right)}_{b_{n-k}}}_{c_n}$$

Simplifying

$$1 = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k x^k x^{n-k}$$
$$1 = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

Comparing coefficients of powers of x of both sides we get

$$n = 0, \quad 1 = B_0$$
$$n \neq 0, \quad 0 = \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

25.1 First few Bernoulli numbers

Using equation **??** with $n = 1$ implies $\binom{2}{0}B_0 + \binom{2}{1}B_1 = 0 = 1 + 2B_1$ which implies

$$B_1 = -\frac{1}{2} \tag{89}$$

Using equation **??** with $n = 2$ implies $\binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2 = 0 = 1 - \frac{3}{2} + 3B_2$ which implies

$$B_2 = \frac{1}{6} \tag{90}$$

Similarly when we apply this equation for increasing values of n

$$B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0, B_6 = \frac{1}{42}, \ldots \tag{91}$$

Theorem 25.2. *In fact we can show that all odd Bernoulli numbers after n=1 are 0*

$$B_n = 0 \quad \forall \text{ odd } n > 1 \tag{92}$$

Proof

Starting from definition 25.1

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Removing the first two terms of the sum

$$\frac{x}{e^x - 1} = B_0 + B_1 x + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

[1] $\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n$ where $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. See Theorem 29.1 for the proof.

Rearranging

$$\frac{x}{e^x - 1} - 1 + \frac{x}{2} = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$
$$\frac{x + xe^x}{2e^x - 2} - 1 = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

If we can show that the RHS is an even function then we have established the proof

$$y(x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = \frac{-x + -xe^{-x}}{2e^{-x} - 2} - 1$$

Simplifying

$$y(-x) = \frac{-x - xe^{-x}}{2e^{-x} - 2} \cdot \frac{e^x}{e^x} - 1$$
$$y(-x) = \frac{-xe^x - x}{2 - 2e^x} - 1$$
$$y(-x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = y(x)$$
$$B_n = 0 \quad \forall \ odd \ n > 1$$

26 Bernoulli Polynomials

Definition 26.1. *The Bernoulli polynomials $B_n(x)$ are defined as*

$$\boxed{B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}}$$

26.1 Examples

$$B_0(x) = B_0 = 1$$
$$B_1(x) = B_0 x + B_1 = x - \frac{1}{2}$$
$$B_2(x) = B_0 x^2 + \binom{2}{1} B_1 x + B_2 = x^2 - x + \frac{1}{6}$$
$$B_3(x) = B_0 x^3 + \binom{3}{1} B_1 x^2 + \binom{3}{2} B_2 x + B_3 = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x$$

Note that $B_n(x)$ is a polynomial of degree n.

26.2 Useful Properties

Theorem 26.1. $B_n(0)$

$$\boxed{B_n(0) = B_n} \tag{93}$$

Proof

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}$$
$$B_n(x) = \sum_{k=0}^{n-1} \binom{n}{k} B_k x^{n-k} + B_n$$
$$B_n(0) = \sum_{k=0}^{n-1} \binom{n}{k} B_k \cdot 0 + B_n = B_n$$

Theorem 26.2. $B_n(1)$

$$\boxed{B_n(1) = B_n \quad n \neq 1 \quad B_1(1) = -B_1} \tag{94}$$

Proof

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}$$
$$B_n(1) = \sum_{k=0}^{n} \binom{n}{k} B_k$$

Using equation **??**, $\sum_k^n \binom{n+1}{k} B_k = 0$

$$\sum_k^{n-1} \binom{n}{k} B_k = 0$$

Adding B_n *to both sides*

$$\sum_k^{n} \binom{n}{k} B_k = B_n$$

Theorem 26.3. *Derivative*

$$\boxed{B_n^{'}(x) = nB_{n-1}(x)} \tag{95}$$

Proof

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}$$

Taking derivatives of both sides

$$B_n^{'}(x) = \sum_{k=0}^{n-1} \binom{n}{k} B_k (n-k) x^{n-k-1}$$

$$B_n^{'}(x) = \sum_{k=0}^{n-1} \frac{n!(n-k)}{k!(n-k)!} B_k x^{n-1-k}$$

$$B_n^{'}(x) = \sum_{k=0}^{n-1} \frac{(n-1)! \cdot n}{k!(n-k-1)!} B_k x^{n-1-k}$$

$$B_n^{'}(x) = n \sum_{k=0}^{n-1} \binom{n-1}{k} B_k x^{n-1-k}$$

$$B_n^{'}(x) = nB_{n-1}(x)$$

Theorem 26.4. *Integral*

$$\boxed{\int_{x=0}^{1} B_n(x) dx = 0} \tag{96}$$

Proof

Starting from the derivative equation, using $m = n - 1$

$$B_{m+1}^{'}(x) = (m+1) \cdot B_m(x)$$

Integrating both sides

$$\int_{x=0}^{1} B_{m+1}^{'}(x) = B_{m+1}(1) - B_{m+1}(0) = (m+1) \cdot \int_{x=0}^{1} B_m(x)$$

Using the Bernoulli polynomial properties

$$B_{m+1}(1) - B_{m+1}(0) = B_{m+1} - B_{m+1} = 0$$

Therefore for $m \neq -1$

$$\int_{x=0}^{1} B_m(x) = 0$$

27 Euler-Maclaurin formula

Let $\{x\} = x - \lfloor x \rfloor$ then

Theorem 27.1.

$$\sum_{k=1}^{n-1} f(k) = \int_{x=1}^{n} f(x)dx - \sum_{k=1}^{m} \frac{B_k}{k!} \cdot \left(f^{(k-1)}(n) - f^{(k-1)}(1)\right) + R_{mn} \tag{97a}$$

$$R_{mn} = \frac{(-1)^{m+1}}{m!} \cdot \int_{x=1}^{n} B_m(\{x\}) f^{(m)}(x)dx \tag{97b}$$

Proof

See the Darboux Formula paper for proof

27.1 Stirling's Approximation

Theorem 27.2.

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n} \cdot exp\left(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} + O(\frac{1}{n^m})\right) \tag{98}$$

Proof

See the Gamma function paper for proof

28 Connection to Zeta(2n)

See the Zeta function document.

Theorem 28.1. *We can express all even integer values of ζ using Bernoulli numbers*

$$\zeta(2n) = (-1)^{n+1} \frac{(2\pi)^{2n} B_{2n}}{2 \cdot (2n)!} \tag{99}$$

Proof: See the Zeta function document.

28.1 The first few even integer values of Zeta

$$\zeta(2) = \frac{\pi^2}{6}, \zeta(4) = \frac{\pi^4}{90}, \zeta(6) = \frac{\pi^6}{945}, \zeta(8) = \frac{\pi^8}{9450}, \zeta(10) = \frac{\pi^{10}}{93555},$$
$$\zeta(12) = \frac{691}{638512875}\pi^{12}, \zeta(14) = \frac{2}{18243225}\pi^{14}, \zeta(16) = \frac{3617}{325641566250}\pi^{16},$$
$$\zeta(18) = \frac{43867}{38979295480125}\pi^{18}, \zeta(20) = \frac{174611}{1531329465290625}\pi^{20}, \zeta(22) = \frac{155366}{13447856940643125}\pi^{22},$$
$$\zeta(24) = \frac{236364091}{2019195719637565218 75}\pi^{24}, \ldots$$

29 Cauchy Product of Series (Mertens' Theorem)

When multiplying two infinite series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, the resulting series arranged by total degree is known as the *Cauchy product*:

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} \tag{100}$$

Theorem 29.1 (Mertens' Theorem on Cauchy Products)**.** *Let $\sum_{n=0}^{\infty} a_n = A$ be an absolutely convergent series, and let $\sum_{n=0}^{\infty} b_n = B$ be a convergent series. Then their Cauchy product series converges to their product:*

$$\sum_{n=0}^{\infty} c_n = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) = AB = \left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) \tag{101}$$

Proof. Let $A_N = \sum_{n=0}^{N} a_n$, $B_N = \sum_{n=0}^{N} b_n$, and $C_N = \sum_{n=0}^{N} c_n$. Let $\beta_N = B_N - B$, so that $\lim_{N\to\infty} \beta_N = 0$.

Expanding the partial sum C_N:

$$\begin{aligned} C_N &= \sum_{n=0}^{N} \sum_{k=0}^{n} a_k b_{n-k} \\ &= \sum_{k=0}^{N} a_k \sum_{n=k}^{N} b_{n-k} \\ &= \sum_{k=0}^{N} a_k B_{N-k} \\ &= \sum_{k=0}^{N} a_k (B + \beta_{N-k}) \\ &= B A_N + \sum_{k=0}^{N} a_k \beta_{N-k} \end{aligned}$$

Let $\gamma_N = \sum_{k=0}^{N} a_k \beta_{N-k}$. We want to show that $\lim_{N\to\infty} \gamma_N = 0$.

Since $\sum_{n=0}^{\infty} a_n$ is absolutely convergent, let $M = \sum_{n=0}^{\infty} |a_n| < \infty$. If $M = 0$, all $a_n = 0$ and the result is trivial. Otherwise, given any $\varepsilon > 0$:

1. Since $\beta_N \to 0$, there exists $N_0 \in \mathbb{N}$ such that $|\beta_m| < \frac{\varepsilon}{2M}$ for all $m > N_0$.

2. Let $K = \sup_{m \geq 0} |\beta_m| < \infty$. Since the tail of the absolutely convergent series $\sum |a_k|$ vanishes, there exists $N_1 > N_0$ such that for all $N > N_1$:

$$\sum_{k=N-N_0}^{N} |a_k| < \frac{\varepsilon}{2K+1}$$

Then for any $N > N_1$:

$$\begin{aligned}
|\gamma_N| &\leq \sum_{k=0}^{N-N_0-1} |a_k||\beta_{N-k}| + \sum_{k=N-N_0}^{N} |a_k||\beta_{N-k}| \\
&< \frac{\varepsilon}{2M} \sum_{k=0}^{N-N_0-1} |a_k| + K \sum_{k=N-N_0}^{N} |a_k| \\
&< \frac{\varepsilon}{2M}(M) + K\left(\frac{\varepsilon}{2K+1}\right) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon
\end{aligned}$$

Therefore $\lim_{N\to\infty} \gamma_N = 0$, which yields:

$$\lim_{N\to\infty} C_N = \lim_{N\to\infty} (BA_N + \gamma_N) = BA + 0 = AB$$

□

30 Definition

Definition 30.1. *The Gamma Function is defined as:*

$$\Gamma(z) = \int_0^\infty e^{-t}t^{z-1}dt$$

The first few evaluations of the Gamma function are:

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty e^{-t}\ dt = -e^{-t}\Big|_0^\infty = 0-(-1) = 1 \\
\Gamma(2) &= \int_0^\infty e^{-t}t\ dt = -e^{-t}t\Big|_0^\infty + \int_0^\infty e^{-t}\ dt = 1 \\
\Gamma(3) &= \int_0^\infty e^{-t}t^2\ dt = -e^{-t}t^2\Big|_0^\infty + 2\int_0^\infty e^{-t}t\ dt = 2 \\
\Gamma(4) &= \int_0^\infty e^{-t}t^3\ dt = -e^{-t}t^3\Big|_0^\infty + 3\int_0^\infty e^{-t}t\ dt = 6
\end{aligned} \tag{102}$$

31 Relation to the Factorial Function

Expanding on the pattern we found by evaluating the first few integer values of the Gamma function through integration by parts we obtain:

Theorem 31.1. $\Gamma(z+1) = z\Gamma(z)$
Proof by induction

Assume true for k

$$\Gamma(k+1) = k\Gamma(k)$$

Proving for k+1

$$\begin{aligned}
\Gamma(k+1) &= \int_0^\infty e^{-t}t^k\ dt \\
&= -e^{-t}t^k\Big|_0^\infty + k\int_0^\infty e^{-t}t^{k-1}\ dt \\
&= 0-0+k\Gamma(k-1) \\
&= k\Gamma(k-1)
\end{aligned}$$

31.1 Integer Values

Therefore since $\Gamma(1) = 1$, then for positive integer values

$$\Gamma(n) = (n-1)! \tag{103}$$

32 Equivalent Integral Forms

$$\Gamma(z) = \int_0^\infty e^{-t}t^{z-1}dt \tag{104}$$

Theorem 32.1. *Starting from Equation 30.1, we can use the substitution* $t = -ln(u)$ *to obtain*

$$\Gamma(z) = \int_0^1 (-lnu)^{z-1} dt \tag{105}$$

Proof

$$t = -ln(u)$$
$$dt = -\frac{du}{u}$$
$$\Gamma(z) = \int_1^0 e^{ln(u)}(-ln(u))^{z-1}\frac{-du}{u}$$
$$\Gamma(z) = -\int_1^0 u(-ln(u))^{z-1}\frac{du}{u}$$
$$\Gamma(z) = -\int_1^0 (-ln(u))^{z-1} du$$
$$\Gamma(z) = \int_0^1 (-ln(u))^{z-1} du$$

33 Finite Gamma Expression

Define the finite gamma function (find out its real name)

$$\Gamma_p(x) = \frac{p!p^x}{x(x+1)(x+2)...(x+p)} = \frac{p^x}{x(\frac{x}{1}+1)(\frac{x}{2}+1)...(\frac{x}{p}+1)} \quad x > 0 \tag{106}$$

Note that this function has the following properties:

$$\Gamma_p(1) = \frac{p!p}{1(1+1)(1+2)...(1+p)} = \frac{p!p}{p!(1+p)} = \frac{p}{1+p} \tag{107}$$

$$\begin{aligned}\Gamma_p(x+1) &= \frac{p!pp^x}{(x+1)(x+2)...(x+p)(x+p+1)} \\ &= \frac{xp!pp^x}{x(x+1)(x+2)...(x+p)(x+p+1)} \\ &= \frac{xp}{x+p+1}\Gamma_p(x)\end{aligned} \tag{108}$$

33.1 Taking the limit to obtain Gamma

Using Equation 108 and assuming $\Gamma_\infty(x)$ converges we get:

$$\lim_{p\to\infty}\Gamma_p(x+1) = \lim_{p\to\infty}\frac{xp}{x+p+1}\Gamma_p(x) = x\lim_{p\to\infty}\Gamma_p(x) \tag{109}$$

Therefore using the above properties we see that if $\Gamma_\infty(x)$ converges then it must be equal to $\Gamma(x)$.

34 Euler Mascheroni constant

The Euler Mascheroni constant is defined as:

$$\gamma = \lim_{p\to\infty} \sum_{i=1}^{p} 1/i - ln(p) \approx 0.5772 \tag{110}$$

This constant can be interpreted as the difference between the sum of $1/x$ and the integral of $1/x$ from 1 to ∞. It is unknown whether this number is transcendental or even irrational! One can show numerically that if it is rational then the denominator must be greater than 10^{242080}.

35 Weierstrass Formula

Theorem 35.1. *Let γ be the Euler Mascheroni constant as defined in section 34, and Γ be the Gamma function, then*

$$\frac{1}{\Gamma(x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k} \tag{111}$$

Proof
Re-writing p^x

$$p^x = e^{xlnp} = e^{x[lnp - 1 - 1/2 - 1/3 - \cdots - 1/p]} e^{x + x/2 + x/3 + \cdots + x/p}$$

Plugging p^x into equation 106

$$\Gamma_p(x) = \frac{e^{x[lnp - 1 - 1/2 - 1/3 - \cdots - 1/p]} e^{x + x/2 + x/3 + \cdots + x/p}}{x(\frac{x}{1} + 1)(\frac{x}{2} + 1)...(\frac{x}{p} + 1)}$$

$$\Gamma_p(x) = e^{x[lnp - 1 - 1/2 - 1/3 - \cdots - 1/p]} \cdot \frac{1}{x} \cdot \frac{e^x}{x+1} \cdot \frac{e^{x/2}}{x/2+1} \cdot \frac{e^{x/3}}{x/3+1} \cdot \cdots \frac{e^{x/p}}{x/p+1}$$

$$\Gamma_p(x) = e^{x[lnp - 1 - 1/2 - 1/3 - \cdots - 1/p]} \cdot \frac{1}{x} \cdot \prod_{k=1}^{p} \frac{e^{x/k}}{1 + x/k}$$

Taking the limit as $p \to \infty$

$$\Gamma(x) = \lim_{p\to\infty} \Gamma_p(x) = \lim_{p\to\infty} e^{x[lnp - 1 - 1/2 - 1/3 - \cdots - 1/p]} \cdot \frac{1}{x} \cdot \prod_{k=1}^{p} \frac{e^{x/k}}{1 + x/k}$$

$$\Gamma(x) = e^{-x\gamma} \cdot \frac{1}{x} \cdot \prod_{k=1}^{\infty} \frac{e^{x/k}}{1 + x/k}$$

Equivalently

$$\frac{1}{\Gamma(x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k}$$

35.1 Symmetry around 1/2 relation

Using 35 we derive the following functional form.

Theorem 35.2.

$$\boxed{\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = \frac{sin(\pi x)}{\pi}} \tag{112}$$

Proof
Starting from 35 and considering $\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)}$ *we obtain*

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)} = x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty}(1 + x/k) \cdot e^{-x/k} \cdot -x \cdot e^{-x\gamma} \cdot \prod_{k=1}^{\infty}(1 - x/k) \cdot e^{x/k}$$

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(-x)} = -x^2 \cdot \prod_{k=1}^{\infty}(1 - x^2/k^2)$$

Using the functional form of Γ*, we get* $\Gamma(1-x) = \Gamma(-x+1) = -x\Gamma(-x)$

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = x \cdot \prod_{k=1}^{\infty}(1 - x^2/k^2)$$

Using the product form of $\frac{\sin(\pi x)}{\pi x} = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2}\right)$ *, See Euler's Sine Product Form document*

$$\frac{1}{\Gamma(x)} \cdot \frac{1}{\Gamma(1-x)} = \frac{sin(\pi x)}{\pi}$$

35.2 Special Examples

Plugging in various values for x we obtain certain values and relations for $\Gamma(x)$

$x = \frac{1}{2}$

$$\Gamma(\frac{1}{2}) = \sqrt{\frac{\pi}{sin(\frac{\pi}{2})}} = \sqrt{\pi} \tag{113}$$

$x = \frac{1}{3}$

$$\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = \frac{\pi}{sin(\frac{\pi}{3})} = \frac{2\pi}{\sqrt{3}} \tag{114}$$

$x = \frac{1}{4}$

$$\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \frac{\pi}{sin(\frac{\pi}{4})} = \sqrt{2}\pi \tag{115}$$

Theorem 35.3. $x = -\frac{1}{2}$ *with equation 113*

$$\Gamma(-\frac{1}{2}) = -2\sqrt{\pi} \tag{116}$$

Proof

$$\Gamma(-\frac{1}{2}) \cdot \Gamma(1-(-\frac{1}{2})) = \frac{\pi}{sin(-\frac{\pi}{2})}$$
$$\Gamma(-\frac{1}{2}) = -\frac{\pi}{\Gamma(\frac{3}{2})}$$
$$\Gamma(-\frac{1}{2}) = -\frac{\pi}{\frac{1}{2}\Gamma(\frac{1}{2})}$$
$$\Gamma(-\frac{1}{2}) = -\frac{\pi}{\frac{1}{2}\sqrt{\pi}}$$
$$\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$$

Note that equation 113 implies that

$$\Gamma(\frac{1}{2}) = \int_0^\infty \frac{1}{e^t\sqrt{t}}dt = \sqrt{\pi} \tag{117}$$

Since $\Gamma(x) = (x-1)!$ for integer values of x, this is where the idea of $(-\frac{1}{2})! = \sqrt{\pi}$ comes from.

36 Other relations

36.1 Gauss's Multiplication Formula

The multiplication formula, discovered by Carl Friedrich Gauss in 1812, generalizes Legendre's duplication formula (Section 39.5) to an arbitrary positive integer multiplier $n \geq 1$.

Theorem 36.1. *For any integer $n \geq 1$ and any complex number $x \notin \{0, -1/n, -2/n, \dots\}$:*

$$\boxed{\Gamma(x) \cdot \Gamma\left(x+\frac{1}{n}\right) \cdot \Gamma\left(x+\frac{2}{n}\right) \cdots \Gamma\left(x+\frac{n-1}{n}\right) = (2\pi)^{\frac{n-1}{2}} n^{\frac{1}{2}-nx}\Gamma(nx)} \tag{118}$$

Proof
We prove the multiplication formula using Euler's limit product representation of the Gamma function from Section 33:

$$\Gamma(z) = \lim_{p\to\infty} \frac{p!\, p^z}{z(z+1)(z+2)\cdots(z+p)} = \lim_{p\to\infty} \frac{p!\, p^z}{\prod_{j=0}^{p}(z+j)}$$

For each $k \in \{0, 1, \dots, n-1\}$, evaluating at $z = x + \frac{k}{n}$:

$$\begin{aligned}\Gamma\left(x+\frac{k}{n}\right) &= \lim_{p\to\infty} \frac{p!\, p^{x+k/n}}{\prod_{j=0}^{p}\left(x+\frac{k}{n}+j\right)} \\ &= \lim_{p\to\infty} \frac{p!\, p^{x+k/n}}{\prod_{j=0}^{p}\frac{nx+jn+k}{n}} \\ &= \lim_{p\to\infty} \frac{n^{p+1}p!\, p^{x+k/n}}{\prod_{j=0}^{p}(nx+jn+k)}\end{aligned}$$

Multiplying these n limits together for $k = 0, 1, \ldots, n-1$:

$$\prod_{k=0}^{n-1} \Gamma\left(x + \frac{k}{n}\right) = \lim_{p\to\infty} \frac{\prod_{k=0}^{n-1} \left(n^{p+1} p!\, p^{x+k/n}\right)}{\prod_{k=0}^{n-1} \prod_{j=0}^{p} (nx + jn + k)}$$

Notice that as k ranges from 0 to $n-1$ and j ranges from 0 to p, the index $jn + k$ covers every integer from 0 to $n(p+1) - 1$ exactly once. Therefore, the denominator is:

$$\prod_{k=0}^{n-1} \prod_{j=0}^{p} (nx + jn + k) = \prod_{m=0}^{n(p+1)-1} (nx + m)$$

For the numerator:

$$\begin{aligned}\prod_{k=0}^{n-1} \left(n^{p+1} p!\, p^{x+k/n}\right) &= n^{n(p+1)} (p!)^n p^{nx + \sum_{k=0}^{n-1} \frac{k}{n}} \\ &= n^{n(p+1)} (p!)^n p^{nx + \frac{n-1}{2}}\end{aligned}$$

Now compare this with Euler's limit formula for $\Gamma(nx)$ with parameter np:

$$\Gamma(nx) = \lim_{p\to\infty} \frac{(np)!\,(np)^{nx}}{\prod_{m=0}^{np} (nx + m)}$$

Taking the ratio of the two expressions:

$$\begin{aligned}\frac{\prod_{k=0}^{n-1} \Gamma\left(x + \frac{k}{n}\right)}{\Gamma(nx)} &= \lim_{p\to\infty} \frac{n^{n(p+1)} (p!)^n p^{nx + \frac{n-1}{2}}}{(np)!\,(np)^{nx}} \cdot \frac{\prod_{m=0}^{np} (nx + m)}{\prod_{m=0}^{n(p+1)-1} (nx + m)} \\ &= n^{-nx} \lim_{p\to\infty} \frac{n^{np+n} (p!)^n p^{\frac{n-1}{2}}}{(np)!} \cdot \frac{1}{\prod_{m=np+1}^{n(p+1)-1} (nx + m)}\end{aligned}$$

The remaining product has $(n(p+1) - 1) - np = n - 1$ factors:

$$\prod_{m=np+1}^{n(p+1)-1} (nx + m) = (nx + np + 1)(nx + np + 2) \cdots (nx + np + n - 1)$$

As $p \to \infty$, this product is asymptotic to $(np)^{n-1} = n^{n-1} p^{n-1}$. Substituting this asymptotic behavior:

$$\begin{aligned}\frac{\prod_{k=0}^{n-1} \Gamma\left(x + \frac{k}{n}\right)}{\Gamma(nx)} &= n^{-nx} \lim_{p\to\infty} \frac{n^{np+n} (p!)^n p^{\frac{n-1}{2}}}{(np)! \cdot n^{n-1} p^{n-1}} \\ &= n^{1-nx} \lim_{p\to\infty} \frac{n^{np} (p!)^n p^{\frac{1-n}{2}}}{(np)!}\end{aligned}$$

Using Stirling's asymptotic formula for factorials (Section 40):

$$\begin{aligned}p! &\sim \sqrt{2\pi p} \left(\frac{p}{e}\right)^p \implies (p!)^n \sim (2\pi)^{n/2} p^{n/2} \left(\frac{p}{e}\right)^{np} \\ (np)! &\sim \sqrt{2\pi np} \left(\frac{np}{e}\right)^{np} = \sqrt{2\pi n}\, n^{np} p^{1/2} \left(\frac{p}{e}\right)^{np}\end{aligned}$$

Substituting these into the limit:

$$\begin{aligned}\lim_{p\to\infty}\frac{n^{np}(p!)^n p^{\frac{1-n}{2}}}{(np)!} &= \lim_{p\to\infty}\frac{n^{np}\cdot(2\pi)^{n/2}p^{n/2}(p/e)^{np}\cdot p^{\frac{1-n}{2}}}{\sqrt{2\pi n}\,n^{np}p^{1/2}(p/e)^{np}}\\ &= \frac{(2\pi)^{n/2}}{\sqrt{2\pi n}}\lim_{p\to\infty}\frac{p^{n/2+(1-n)/2}}{p^{1/2}}\\ &= (2\pi)^{\frac{n-1}{2}}n^{-1/2}\end{aligned}$$

Multiplying by $n^{1-nx}\Gamma(nx)$:

$$\prod_{k=0}^{n-1}\Gamma\left(x+\frac{k}{n}\right) = n^{1-nx}\cdot(2\pi)^{\frac{n-1}{2}}n^{-1/2}\Gamma(nx) = (2\pi)^{\frac{n-1}{2}}n^{\frac{1}{2}-nx}\Gamma(nx)$$

which completes the proof.

Corollary 36.1.1. *Multiplying both sides of Equation (118) by x and taking the limit as $x\to 0$, we have $\lim_{x\to 0}x\Gamma(x)=\Gamma(1)=1$, and $\lim_{x\to 0}x\Gamma(nx)=\frac{1}{n}\lim_{x\to 0}(nx)\Gamma(nx)=\frac{1}{n}$. Therefore:*

$$\boxed{\Gamma\left(\frac{1}{n}\right)\cdot\Gamma\left(\frac{2}{n}\right)\cdots\Gamma\left(\frac{n-1}{n}\right)=\frac{(2\pi)^{\frac{n-1}{2}}}{\sqrt{n}}} \tag{119}$$

Example 36.2. *For $n=2$, Gauss's multiplication formula reduces directly to Legendre's duplication formula (Theorem 39.5):*

$$\Gamma(x)\Gamma\left(x+\frac{1}{2}\right) = (2\pi)^{1/2}2^{\frac{1}{2}-2x}\Gamma(2x) = \frac{\sqrt{\pi}}{2^{2x-1}}\Gamma(2x)$$

For $n=3$, Equation (119) gives:

$$\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right) = \frac{(2\pi)^{2/2}}{\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

which perfectly agrees with Euler's reflection formula (112):

$$\Gamma\left(\frac{1}{3}\right)\Gamma\left(1-\frac{1}{3}\right) = \frac{\pi}{\sin(\pi/3)} = \frac{\pi}{\sqrt{3}/2} = \frac{2\pi}{\sqrt{3}}$$

36.2 Relation to the Riemann Zeta Function

Theorem 36.3. *For $\mathrm{Re}(s)>1$, the Gamma function is related to the Riemann Zeta function $\zeta(s)=\sum_{n=1}^{\infty}\frac{1}{n^s}$ by:*

$$\boxed{\Gamma(s)\zeta(s)=\int_0^\infty\frac{u^{s-1}}{e^u-1}\,\mathrm{d}u} \tag{120}$$

Proof
Starting from the integral definition of the Gamma function (Equation 30.1):

$$\Gamma(s)=\int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t \qquad (\mathrm{Re}(s)>0)$$

For any positive integer $n \geq 1$, substitute $t = nu$, so that $\mathrm{d}t = n\,\mathrm{d}u$:

$$\Gamma(s) = \int_0^\infty (nu)^{s-1} e^{-nu} n\,\mathrm{d}u = n^s \int_0^\infty u^{s-1} e^{-nu}\,\mathrm{d}u$$

Dividing by n^s:

$$\frac{\Gamma(s)}{n^s} = \int_0^\infty u^{s-1} e^{-nu}\,\mathrm{d}u$$

Summing both sides over all positive integers $n \geq 1$:

$$\Gamma(s) \sum_{n=1}^\infty \frac{1}{n^s} = \sum_{n=1}^\infty \int_0^\infty u^{s-1} e^{-nu}\,\mathrm{d}u$$

For $\mathrm{Re}(s) > 1$, we can interchange the summation and the integral by the monotone convergence theorem (or dominated convergence):

$$\begin{aligned}
\Gamma(s)\zeta(s) &= \int_0^\infty u^{s-1} \left(\sum_{n=1}^\infty e^{-nu} \right) \mathrm{d}u \\
&= \int_0^\infty u^{s-1} \left(\frac{e^{-u}}{1 - e^{-u}} \right) \mathrm{d}u \\
&= \int_0^\infty \frac{u^{s-1}}{e^u - 1}\,\mathrm{d}u
\end{aligned}$$

which proves the identity.

37 Digamma function

37.1 Definition

Define the Digamma function

$$\Psi(x) = \frac{\partial}{\partial x} ln(\Gamma(x)) = \frac{\Gamma'(x)}{\Gamma(x)} \tag{121}$$

37.2 Series Representation

Theorem 37.1. *Let γ be the Euler Mascheroni constant as defined in section 34 then,*

$$\boxed{\Psi(x) = -\gamma - \frac{1}{x} + \sum_{k=1}^\infty \left(\frac{1}{k} - \frac{1}{x+k} \right)} \tag{122}$$

Proof

$$\Psi(x) = \frac{\partial}{\partial x} ln(\Gamma(x))$$

Using equation 111

$$\Psi(x) = -\frac{\partial}{\partial x} ln(x \cdot e^{x\gamma} \cdot \prod_{k=1}^{\infty} (1 + x/k) \cdot e^{-x/k})$$

$$\Psi(x) = -\frac{\partial}{\partial x} \left[ln(x) + x\gamma + \sum_{k=1}^{\infty} ln(1 + x/k) - \frac{x}{k} \right]$$

$$\Psi(x) = -\frac{1}{x} - \gamma - \sum_{k=1}^{\infty} \frac{1}{k} \cdot \frac{1}{1 + x/k} - \frac{1}{k}$$

$$\Psi(x) = -\frac{1}{x} - \gamma + \sum_{k=1}^{\infty} \frac{1}{k} - \frac{1}{x + k}$$

37.3 Derivatives

Theorem 37.2.

$$\boxed{\Psi'(x) = \sum_{k=1}^{\infty} \frac{1}{(k + x - 1)^2}} \tag{123}$$

Proof

Using equation 122

$$\Psi(x) = -\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{x + k} \right)$$

$$\Psi'(x) = \frac{1}{x^2} + \sum_{k=1}^{\infty} \frac{1}{(x + k)^2}$$

Including the first term in the sum

$$\Psi'(x) = \sum_{k=1}^{\infty} \frac{1}{(x + k - 1)^2}$$

Theorem 37.3.

$$\boxed{\Psi^n(x) = \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k + x - 1)^{n+1}}} \tag{124}$$

Proof
Take repeated derivatives of equation 123

37.4 Special values

Theorem 37.4.

$$\Psi^n(x + 1) = \Psi^n(x) + \frac{(-1)^n n!}{x^{n+1}} \tag{125}$$

Proof

$$\Psi^n(x+1) = \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k+x)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \sum_{k=2}^{\infty} \frac{(-1)^{n+1} n!}{(k+x)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \sum_{k=1}^{\infty} \frac{(-1)^{n+1} n!}{(k+x-1)^{n+1}}$$

$$\Psi^n(x+1) = \frac{(-1)^n n!}{x^{n+1}} + \Psi^n(x)$$

Theorem 37.5. *Reflection formula*

$$\Psi(1-x) = \Psi(x) + \pi cot(\pi x) \tag{126}$$

Proof

$$\Psi(1-x) = -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \Big(\frac{1}{k} - \frac{1}{k+1-x} \Big)$$

Adding and subtracting $\Psi(x)$

$$\Psi(1-x) = -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \Big(\frac{1}{k} - \frac{1}{k+1-x} \Big) + \Psi(x) - \Psi(x)$$

$$\Psi(1-x) = -\gamma - \frac{1}{1-x} + \sum_{k=1}^{\infty} \Big(\frac{1}{k} - \frac{1}{k+1-x} \Big) + \Psi(x) - \left(-\gamma - \frac{1}{x} + \sum_{k=1}^{\infty} \Big(\frac{1}{k} - \frac{1}{x+k} \Big) \right)$$

Simplifying

$$\Psi(1-x) = \Psi(x) + \frac{1}{x} - \frac{1}{1-x} + \sum_{k=1}^{\infty} \Big(\frac{1}{k} - \frac{1}{k+1-x} - \frac{1}{k} + \frac{1}{x+k} \Big)$$

$$\Psi(1-x) = \Psi(x) + \frac{1}{x} - \frac{1}{1-x} + \sum_{k=1}^{\infty} \Big(\frac{1}{x+k} - \frac{1}{k+1-x} \Big)$$

$$\Psi(1-x) = \Psi(x) + \sum_{k=0}^{\infty} \Big(\frac{1}{x+k} + \frac{1}{x-k-1} \Big)$$

$$\Psi(1-x) = \Psi(x) + \sum_{k=-\infty}^{\infty} \Big(\frac{1}{x+k} \Big)$$

Using the series [2] *for* $\pi cot(\pi x) = \sum_{k=-\infty}^{\infty} \frac{1}{x+k}$

$$\Psi(1-x) = \Psi(x) + \pi cot(\pi x)$$

[2] See the Euler's Sine Product formula document.

Theorem 37.6 (Duplication Formula for the Digamma Function)**.** *For any complex number $x \notin \{0, -1/2, -1, -3/2, \dots\}$:*

$$\boxed{\Psi(2x) = \frac{1}{2}\Psi(x) + \frac{1}{2}\Psi\left(x + \frac{1}{2}\right) + \ln(2)} \tag{127}$$

Proof
We prove the duplication formula by taking the logarithmic derivative of Legendre's duplication formula (Theorem 39.5):

$$\Gamma(2x) = \frac{2^{2x-1}}{\sqrt{\pi}}\Gamma(x)\Gamma\left(x + \frac{1}{2}\right)$$

Taking the natural logarithm of both sides:

$$\begin{aligned}\ln\Gamma(2x) &= \ln\left(\frac{2^{2x-1}}{\sqrt{\pi}}\Gamma(x)\Gamma\left(x + \frac{1}{2}\right)\right) \\ &= (2x-1)\ln(2) - \frac{1}{2}\ln(\pi) + \ln\Gamma(x) + \ln\Gamma\left(x + \frac{1}{2}\right)\end{aligned}$$

Differentiating both sides with respect to x using the chain rule and the definition of the Digamma function $\Psi(z) = \frac{\mathrm{d}}{\mathrm{d}z}\ln\Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}$ (Equation 121):

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln\Gamma(2x) = 2\frac{\Gamma'(2x)}{\Gamma(2x)} = 2\Psi(2x)$$

Differentiating the right-hand side term by term:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left[(2x-1)\ln(2) - \frac{1}{2}\ln(\pi) + \ln\Gamma(x) + \ln\Gamma\left(x + \frac{1}{2}\right)\right] &= 2\ln(2) + \frac{\Gamma'(x)}{\Gamma(x)} + \frac{\Gamma'\left(x + \frac{1}{2}\right)}{\Gamma\left(x + \frac{1}{2}\right)} \\ &= 2\ln(2) + \Psi(x) + \Psi\left(x + \frac{1}{2}\right)\end{aligned}$$

Equating the derivatives gives:

$$2\Psi(2x) = \Psi(x) + \Psi\left(x + \frac{1}{2}\right) + 2\ln(2)$$

Dividing both sides by 2:

$$\Psi(2x) = \frac{1}{2}\Psi(x) + \frac{1}{2}\Psi\left(x + \frac{1}{2}\right) + \ln(2)$$

which completes the proof.

Remark 37.7. *More generally, taking the logarithmic derivative of Gauss's multiplication formula (Theorem 118):*

$$\prod_{k=0}^{n-1}\Gamma\left(x + \frac{k}{n}\right) = (2\pi)^{\frac{n-1}{2}}n^{\frac{1}{2}-nx}\Gamma(nx)$$

yields Gauss's multiplication formula for the Digamma function:

$$\boxed{\Psi(nx) = \frac{1}{n}\sum_{k=0}^{n-1}\Psi\left(x+\frac{k}{n}\right)+\ln(n)} \tag{128}$$

for any integer $n \geq 1$. The duplication formula is the special case $n = 2$.

Example 37.8. *Setting $x = 1/2$ in the duplication formula (127):*

$$\begin{aligned}\Psi(1) &= \frac{1}{2}\Psi\left(\frac{1}{2}\right)+\frac{1}{2}\Psi(1)+\ln(2)\\ \frac{1}{2}\Psi(1) &= \frac{1}{2}\Psi\left(\frac{1}{2}\right)+\ln(2)\\ \Psi\left(\frac{1}{2}\right) &= \Psi(1)-2\ln(2)\end{aligned}$$

Since $\Psi(1) = -\gamma$ (Equation 130), we obtain the special value:

$$\boxed{\Psi\left(\frac{1}{2}\right) = -\gamma-2\ln(2)} \tag{129}$$

$$\Psi(1) = \Gamma'(1) = -\gamma \tag{130}$$

$$\Psi^{(n)}(1) = (-1)^{n+1}n!\zeta(n+1) \tag{131}$$

Where $\zeta(z)$ is the Riemann Zeta function defined as $\zeta(z) = \sum_{k=1}^{\infty}\frac{1}{k^z}$. This follows directly from evaluating the n-th derivative formula (Equation 124) at $x = 1$:

$$\Psi^{(n)}(1) = \sum_{k=1}^{\infty}\frac{(-1)^{n+1}n!}{(k+1-1)^{n+1}} = (-1)^{n+1}n!\sum_{k=1}^{\infty}\frac{1}{k^{n+1}} = (-1)^{n+1}n!\zeta(n+1)$$

$$\Psi(n) = \frac{\Gamma'(n)}{\Gamma(n)} = -\gamma+\sum_{k=1}^{n-1}\frac{1}{k} = -\gamma+H_{n-1} \tag{132}$$

Where H_n is the sum of the n first harmonic numbers. Note that this implies that the derivative of the Γ function is diverges to ∞.

38 Using The Gamma Function for Euler Integrals

Theorem 38.1. *Let $p = x + yi$ and $\alpha = tan^{-1}\left(\frac{y}{x}\right)$ then*

$$\boxed{\frac{\Gamma(s)}{n|p|^s}sin(\alpha s) = \int_0^{\infty}u^{ns-1}e^{-xu^n}sin(yu^n)du} \tag{133}$$

$$\boxed{\frac{\Gamma(s)}{n|p|^s}cos(\alpha s) = \int_0^{\infty}u^{ns-1}e^{-xu^n}cos(yu^n)du} \tag{134}$$

Proof

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$$

Using substitution $t = p \cdot u^n$ *hence* $dt = p \cdot n \cdot u^{n-1} \cdot du$

$$\Gamma(s) = \int_0^\infty p^{n-1} u^{ns-n} e^{-pu} npu^{n-1} du = n \int_0^\infty p^s u^{ns-1} e^{-pu^n} du$$

$$\frac{\Gamma(s)}{np^s} = \int_0^\infty u^{ns-1} e^{-pu^n} du$$

Assuming p *is a complex number with real part* x*, imaginary part* y

Taking the conjugate of both sides, we obtain a second equation.

$$\frac{\Gamma(s)}{n\bar{p}^s} = \int_0^\infty u^{ns-1} e^{-\bar{p}u^n} du$$

Note that $p = x + yi = |p|e^{i\alpha}$ *and* $\bar{p} = x - yi = |p|e^{-i\alpha}$ *where* $\alpha = tan^{-1}\left(\frac{y}{x}\right)$

Adding and subtracting both equations we get

$$\frac{\Gamma(s)}{n}\Big[\frac{1}{\bar{p}^s} \pm \frac{1}{p^s}\Big] = \int_0^\infty u^{ns-1}\Big[e^{-\bar{p}u^n} \pm e^{-pu^n}\Big] du$$

$$\frac{\Gamma(s)}{n|p|^s}\Big[\frac{1}{e^{-i\alpha s}} \pm \frac{1}{e^{i\alpha s}}\Big] = \int_0^\infty u^{ns-1}\Big[e^{-xu^n+yu^n i} \pm e^{-xu^n-yu^n i}\Big] du$$

$$\frac{\Gamma(s)}{n|p|^s}\Big[e^{i\alpha s} \pm e^{-i\alpha s}\Big] = \int_0^\infty u^{ns-1} e^{-xu^n}\Big[e^{yu^n i} \pm e^{-yu^n i}\Big] du$$

With addition we get

$$\frac{\Gamma(s)}{n|p|^s}\Big[2cos(\alpha s)\Big] = \int_0^\infty u^{ns-1} e^{-xu^n}\Big[2cos(yu^n)\Big] du$$

$$\frac{\Gamma(s)}{n|p|^s} cos(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} cos(yu^n)\Big] du$$

With subtraction we get

$$\frac{\Gamma(s)}{n|p|^s}\Big[2isin(\alpha s)\Big] = \int_0^\infty u^{ns-1} e^{-xu^n}\Big[2isin(yu^n)\Big] du$$

$$\frac{\Gamma(s)}{n|p|^s} sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} sin(yu^n)\Big] du$$

38.1 Examples

38.1.1 Sinc Integral $\int \frac{\sin(x)}{x}$

Theorem 38.2.

$$\boxed{\int_0^\infty \frac{\sin(x)}{x} = \frac{\pi}{2}} \tag{135}$$

Proof
Starting from equation 133

$$\frac{\Gamma(s)}{n|p|^s}sin(\alpha s) = \int_0^\infty u^{ns-1}e^{-xu^n}sin(yu^n)du$$

setting $y = 1, n = 1, x = 0$ *and taking the limit as* $s \to 0$ *we obtain*

$$|p| = 1, \alpha = tan^{-1}(1/0) = \pi/2$$

$$\lim_{s\to 0}\Gamma(s)sin(s\frac{\pi}{2}) = \lim_{s\to 0}\int_0^\infty u^{s-1}sin(u)du = \int_0^\infty \frac{sin(u)}{u}du$$

To evaluate the RHS limit we use equation 112 $\Gamma(s) = \dfrac{\pi}{\Gamma(1-s)sin(\pi s)}$

$$\lim_{s\to 0}\Gamma(s)sin(s\frac{\pi}{2}) = \lim_{s\to 0}\frac{\pi}{\Gamma(1-s)sin(\pi s)}sin(s\frac{\pi}{2})$$

We can split this limit into the product of 3 limits since all of these limits exist

$$\lim_{s\to 0}\Gamma(s)sin(s\frac{\pi}{2}) = \lim_{s\to 0}\frac{\frac{\pi}{2}}{\Gamma(1-s)}\cdot\lim_{s\to 0}\frac{\pi}{sin(\pi s)}\cdot\lim_{s\to 0}\frac{sin(s\frac{\pi}{2})}{\frac{\pi}{2}}$$

$$\lim_{s\to 0}\Gamma(s)sin(s\frac{\pi}{2}) = \frac{\pi}{2}$$

$$\int_0^\infty \frac{sin(u)}{u}du = \frac{\pi}{2}$$

since $\dfrac{sin(u)}{u}$ *is an even function*

$$\int_{-\infty}^\infty \frac{sin(u)}{u}du = \pi$$

38.1.2 $\int \sin(x^2)$

Theorem 38.3.

$$\boxed{\int_0^\infty \sin(x^2) = \frac{\sqrt{2\pi}}{4}} \tag{136}$$

Proof
Starting from equation 133

$$\frac{\Gamma(s)}{n|p|^s}sin(\alpha s) = \int_0^\infty u^{ns-1}e^{-xu^n}sin(yu^n)du$$

setting $y = 1, n = 2, x = 0$ *and* $ns - 1 = 0$ *we obtain*

$$|p| = 1, \alpha = tan^{-1}(1/0) = \pi/2, s = 1/2$$

$$\frac{\Gamma(\frac{1}{2})sin(\frac{\pi}{4})}{2} = \int_0^\infty sin(u^2)du$$

using equation 113: $\Gamma(\dfrac{1}{2}) = \sqrt{\pi}$

$$\int_0^\infty sin(u^2)du = \frac{\sqrt{\pi}\frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2\pi}}{4}$$

38.1.3 $\int \cos(x^2)$

Theorem 38.4.

$$\boxed{\int_0^\infty \cos(x^2) = \frac{\sqrt{2\pi}}{4}} \tag{137}$$

Proof
Starting from equation 134

$$\frac{\Gamma(s)}{n|p|^s} cos(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} cos(yu^n) du$$

setting $y = 1, n = 2, x = 0$ *and* $ns - 1 = 0$ *we obtain*

$$|p| = 1, \alpha = tan^{-1}(1/0) = \pi/2, s = 1/2$$

$$\frac{\Gamma(\frac{1}{2}) cos(\frac{\pi}{4})}{2} = \int_0^\infty cos(u^2) du$$

using equation 113: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

$$\int_0^\infty cos(u^2) du = \frac{\sqrt{\pi}\frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2\pi}}{4}$$

38.1.4 $\int \frac{\sin(x^2)}{x^2}$

Theorem 38.5.

$$\boxed{\int_0^\infty \frac{\sin(x^2)}{x^2} = \sqrt{\frac{\pi}{2}}} \tag{138}$$

Proof
Starting from equation 133

$$\frac{\Gamma(s)}{n|p|^s} sin(\alpha s) = \int_0^\infty u^{ns-1} e^{-xu^n} sin(yu^n) du$$

setting $y = 1, n = 2, x = 0$ *and* $ns - 1 = -2$ *we obtain*

$$|p| = 1, \alpha = tan^{-1}(1/0) = \pi/2, s = -1/2$$

$$\frac{\Gamma(\frac{-1}{2}) sin(-\frac{\pi}{4})}{2} = \int_0^\infty \frac{sin(u^2)}{u^2} du$$

using equation 116: $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$

$$\int_0^\infty \frac{sin(u^2)}{u^2} du = -\frac{\Gamma(\frac{-1}{2})}{2\sqrt{2}} = \sqrt{\frac{\pi}{2}}$$

38.1.5 $\int \frac{\sin(x^n)}{x^n}$

Theorem 38.6.

$$\boxed{\int_0^\infty \frac{\sin(x^n)}{x^n} = \frac{\Gamma(\frac{1}{n})}{n-1} \cdot cos(\frac{\pi}{2n}) \quad , n > 1} \tag{139}$$

Proof
Starting from equation 133

$$\frac{\Gamma(s)}{n|p|^s}sin(\alpha s) = \int_0^\infty u^{ns-1}e^{-xu^n}sin(yu^n)du$$

setting $y = 1, x = 0$ *and* $ns - 1 = -n$ *we obtain*

$$|p| = 1, \alpha = tan^{-1}(1/0) = \pi/2, s = 1/n - 1$$

$$\frac{\Gamma(\frac{1}{n} - 1)sin(\frac{\pi}{2n} - \frac{\pi}{2})}{n} = \int_0^\infty \frac{sin(u^n)}{u^n}du$$

using equation 31.1 and assuming $n \neq 1$*:* $\Gamma(k+1) = k\Gamma(k)$

$$\int_0^\infty \frac{sin(u^n)}{u^n}du = \frac{\Gamma(\frac{1}{n})}{n(\frac{1}{n} - 1)}sin(\frac{\pi}{2n} - \frac{\pi}{2})$$

$$\int_0^\infty \frac{sin(u^n)}{u^n}du = \frac{\Gamma(\frac{1}{n})}{1 - n)}sin(\frac{\pi}{2n} - \frac{\pi}{2})$$

$$using\ \sin\left(x - \frac{\pi}{2}\right) = -\cos(x)$$

$$\int_0^\infty \frac{sin(u^n)}{u^n}du = \frac{\Gamma(\frac{1}{n})}{n - 1} \cdot cos(\frac{\pi}{2n})$$

39 Relation to the Beta Function

39.1 Definition

The Beta Function is defined as:

Definition 39.1.

$$\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$$

39.2 Symmetry

Theorem 39.1.

$$\beta(x,y) = \beta(y,x) \tag{140}$$

Proof
Starting from the definition 39.1

$$\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$$

setting $u = 1-t$, *hence* $du = -dt$ *we obtain*

$$\beta(x,y) = \int_1^0 (1-u)^{x-1}u^{y-1} - du$$

Switching the bounds of integration we get

$$\beta(x,y) = \int_0^1 (1-u)^{x-1}u^{y-1}du = \beta(y,x)$$

39.3 Relation to Gamma

Theorem 39.2.

$$\boxed{\beta(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}} \tag{141}$$

Proof

Starting from the definition 39.1

$$\beta(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$$

setting $t = \frac{1}{1+u}$, *hence* $dt = -\frac{1}{(1+u)^2}du$ *we obtain*

$$\beta(x,y) = \int_\infty^0 (\frac{1}{1+u})^{x-1}(1-\frac{1}{1+u})^{y-1}\frac{1}{(1+u)^2}(-du)$$

Switching the bounds of integration and simplifying we get

$$\beta(x,y) = \int_0^\infty (\frac{1}{1+u})^{x-1}(\frac{u}{1+u})^{y-1}\frac{1}{(1+u)^2}du$$

collecting powers of $(1+u)$

$$\beta(x,y) = \int_0^\infty \frac{u^{y-1}}{(1+u)^{x+y}}du$$

Using the Gamma definition 30.1 and applying a change of variables $\alpha w = t$, *hence* $dt = \alpha dw$

$$\Gamma(z) = \alpha^z \int_0^\infty e^{-\alpha w}w^{z-1}dw$$

setting $z = x+y$ *and* $\alpha = 1+u$, *we obtain*

$$\Gamma(x+y) = (1+u)^{x+y}\int_0^\infty e^{-(1+u)w}w^{x+y-1}dw$$

Combining with the Beta function integral we obtained

$$\beta(x,y)\Gamma(x+y) = \int_0^\infty \Gamma(x+y)\frac{u^{y-1}}{(1+u)^{x+y}}du$$

$$\beta(x,y)\Gamma(x+y) = \int_0^\infty \Big((1+u)^{x+y}\int_0^\infty e^{-(1+u)w}w^{x+y-1}dw\Big)\frac{u^{y-1}}{(1+u)^{x+y}}du$$

Simplfying

$$\beta(x,y)\Gamma(x+y) = \int_0^\infty\int_0^\infty e^{-(1+u)w}w^{x+y-1}u^{y-1}dwdu$$

Rearranging

$$\beta(x,y)\Gamma(x+y) = \int_0^\infty \Big(\int_0^\infty e^{-uw}u^{y-1}du\Big)e^{-w}w^{x+y-1}dw$$

Using $\Gamma(z) = \alpha^z\int_0^\infty e^{-\alpha u}u^{z-1}du$

$$\beta(x,y)\Gamma(x+y) = \int_0^\infty \frac{\Gamma(y)}{w^y}e^{-w}w^{x+y-1}dw$$

Simplifying and moving $\Gamma(y)$ *outside the integral*

$$\beta(x,y)\Gamma(x+y) = \Gamma(y)\int_0^\infty e^{-w}w^{x-1}dw$$

Using the definition of the $\Gamma(z)$

$$\beta(x,y)\Gamma(x+y) = \Gamma(y)\Gamma(x)$$

$$\beta(x,y) = \frac{\Gamma(y)\Gamma(x)}{\Gamma(x+y)}$$

39.4 Relation to integral of powers of sin(x) and cos(x)

Theorem 39.3.

$$\boxed{\int_0^{\pi/2} sin^n(t)cos^m(t)dt = \frac{1}{2}\beta\Big(\frac{n+1}{2}, \frac{m+1}{2}\Big) = \frac{\Gamma(\frac{n+1}{2})\Gamma(\frac{m+1}{2})}{\Gamma(\frac{n+m}{2}+1)}} \tag{142}$$

Proof

$$A(m,n) = \int_0^{\pi/2} sin^n(t)cos^m(t)dt$$

setting $sin^2(t) = u$, *hence* $2\sin(t)\cos(t)dt = du$ *we obtain*

note $cos^2(t) = 1-u$, *and* $sin^2(0) = 0$ *and* $sin^2\left(\frac{\pi}{2}\right) = 1$

$$A(m,n) = \int_0^1 u^{\frac{n}{2}}(1-u)^{\frac{m}{2}}\frac{du}{2\sqrt{u}\sqrt{1-u}}$$

Simplifying the powers we get

$$A(m,n) = \frac{1}{2}\int_0^1 u^{\frac{n}{2}-\frac{1}{2}}(1-u)^{\frac{m}{2}-\frac{1}{2}}du$$

Writing it in the Beta function form we get

$$A(m,n) = \frac{1}{2}\int_0^1 u^{\frac{n+1}{2}-1}(1-u)^{\frac{m+1}{2}-1}du = \frac{1}{2}\beta(\frac{n+1}{2}, \frac{m+1}{2})$$

39.5 Proof of the Legendre duplication formula

Theorem 39.4.

$$\boxed{\Gamma(x)\cdot\Gamma(x+\frac{1}{2}) = \frac{\sqrt{\pi}}{2^{2x-1}}\Gamma(2x)} \tag{143}$$

Proof

Using the relation to the Beta function, and the definition of the Beta function

$$\frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 t^{x-1}(1-t)^{y-1}dt$$

let $x = y = s$

$$\frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = \int_0^1 t^{s-1}(1-t)^{s-1}dt$$

$$\text{let } t = \frac{1+x}{2}, \text{ hence } dt = \frac{dx}{2}$$

note when $t = 0$ *then* $x = -1$, *and when* $t = 1$ *then* $x = 1$

$$\frac{\Gamma(s)\Gamma(s)}{\Gamma(2s)} = \int_{-1}^1 (\frac{1+x}{2})^{s-1}(\frac{1-x}{2})^{s-1}\frac{dx}{2}$$

Simplifying the powers of 2 and rearranging

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s)\int_{-1}^1 (1+x)^{s-1}(1-x)^{s-1}dx$$

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s)\int_{-1}^1 (1-x^2)^{s-1}dx$$

Notice that the integral is even around 0, hence

$$2^{2s-1}\Gamma(s)\Gamma(s) = 2\Gamma(2s)\int_0^1 (1-x^2)^{s-1}dx$$

To evaluate the integral, notice that after a substitution of $u = x^2$ *the integral is just* $\beta(1/2, s)$

$$\beta(1/2, s) = \int_0^1 t^{-\frac{1}{2}}(1-t)^{s-1}dt$$

Let $t = x^2$, *hence* $dt = 2xdx$

$$\beta(1/2, s) = \int_0^1 \frac{1}{x}(1-x^2)^{s-1}2xdx = 2\int_0^1 (1-x^2)^{s-1}dx$$

Hence

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s)\beta(1/2, s)$$

Using theorem 39.2

$$2^{2s-1}\Gamma(s)\Gamma(s) = \Gamma(2s)\frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)}$$

Rearranging and using equation 113

$$\Gamma(s)\Gamma(s+1/2) = \Gamma(2s)\frac{\sqrt{\pi}}{2^{2s-1}}$$

40 Approximations

40.1 Stirling's Approximation

Theorem 40.1. *The basic approximation of the factorial function for large n is*

$$\boxed{n! \approx (\frac{n}{e})^n \sqrt{2\pi n}} \tag{144}$$

Proof
Starting from the definition of the Gamma function 30.1

$$\Gamma(n+1) = n! = \int_0^\infty e^{-t} t^n dt$$

the idea is to take the log of the integrand and then slightly perturb it

$$ln(e^{-t}t^n) = nln(t) - t$$

$$let\ t = n + \epsilon$$

$$\ln(e^{-t}t^n) = nln(n+\epsilon) - (n+\epsilon) = nln(n) + nln(1+\frac{\epsilon}{n}) - (n+\epsilon)$$

$$for\ large\ n,\ \frac{\epsilon}{n} \ll 1$$

Using the Taylor-Expansion of $\ln(1+x)$

$$\ln(e^{-t}t^n) = n\Big(ln(n) + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}(\frac{\epsilon}{n})^k\Big) - (n+\epsilon)$$

$$\ln(e^{-t}t^n) = nln(n) - n + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}(\frac{\epsilon^k}{n^{k-1}}) - \epsilon$$

$$\ln(e^{-t}t^n) = nln(n) - n - \frac{\epsilon^2}{2n} + \frac{\epsilon^3}{3n^2} - \frac{\epsilon^4}{4n^3} + \dots$$

$$\ln(e^{-t}t^n) \approx nln(n) - n - \frac{\epsilon^2}{2n}$$

$$e^{-t}t^n \approx \frac{n^n}{e^n} e^{-\frac{\epsilon^2}{2n}}$$

$$n! \approx \int_{-n}^{\infty} \frac{n^n}{e^n} e^{-\frac{\epsilon^2}{2n}} d\epsilon$$

$$n! \approx \frac{n^n}{e^n} \int_{-n}^{\infty} e^{\frac{-\epsilon^2}{2n}} d\epsilon$$

Assuming n is large enough

$$n! \approx \frac{n^n}{e^n} \int_{-\infty}^{\infty} e^{-\frac{\epsilon^2}{2n}} d\epsilon$$

$$Using \int_{-\infty}^{\infty} e^{-px^2} dx = \sqrt{\frac{\pi}{p}}$$

$$n! \approx \frac{n^n}{e^n} \sqrt{2n\pi}$$

40.1.1 Limit as n approaches infinity

The approximation in the this result can be stated more correctly as follows

$$\boxed{\lim_{n\to\infty} \frac{n!}{(\frac{n}{e})^n\sqrt{2\pi n}} = 1} \tag{145}$$

We can state this equivalently by taking logarithms of both sides. We will use this form to help derive the full Stirling approximation.

$$\boxed{\lim_{n\to\infty} ln(n!) - nln(n) + n - \frac{ln(n)}{2} = ln(\sqrt{2\pi})} \tag{146}$$

40.2 Stirling's Full Approximation

Theorem 40.2. *Expanding on Stirling's approximation to include as many terms as needed we need to use the Euler-Maclaurin formula*[3].

$$\boxed{n! = (\frac{n}{e})^n\sqrt{2\pi n}\cdot exp\Big(\sum_{k=1}^{\lfloor m/2\rfloor} \frac{B_{2k}}{2k\cdot(2k-1)}\frac{1}{n^{2k-1}} - \frac{1}{m}\int_{n-1}^{\infty}\frac{B_m(\{t\})}{t^m}dt\Big)} \tag{147}$$

Proof
Starting from the definition of the simple version of the Euler-Maclaurin formula

$$\sum_{j=0}^{r} g(j) - \int_{x=0}^{r} g(x)dx = \frac{g(0)+g(r)}{2} + \sum_{k=1}^{\lfloor m/2\rfloor}\frac{B_{2k}}{(2k)!}\Big(g^{(2k-1)}(r) - g^{(2k-1)}(0)\Big) + R_{r,m}$$

$$R_{r,m} = \frac{(-1)^{m+1}}{m!}\int_0^r B_m(\{t\})g^{(m)}(t)dt$$

Define $f(x) = g(x+1)$ *and* $n = r+1$, *so that* $f(n) = f(r+1) = g(r)$. *we get*

$$\sum_{j=1}^{n} f(j) - \int_{x=1}^{n} f(x)dx = \frac{f(1)+f(n)}{2} + \sum_{k=1}^{\lfloor m/2\rfloor}\frac{B_{2k}}{(2k)!}\Big(f^{(2k-1)}(n) - f^{(2k-1)}(1)\Big) + R_{n,m}$$

$$R_{n,m} = \frac{(-1)^{m+1}}{m!}\int_1^{n-1} B_m(\{t\})f^{(m)}(t)dt$$

[3] See the Darboux Formula document

Let $f(x) = \ln(x)$, therefore

$$f^{(m)}(x) = \frac{(-1)^{m-1} \cdot (m-1)!}{(x)^m} \text{ or } f^{(2k-1)}(x) = \frac{(-1)^{2k-2} \cdot (2k-2)!}{(x)^{2k-1}}$$

Plugging in $f(x)$ and $f^{(m)}(x)$ into the Euler-Maclaurin formula we get

$$\sum_{j=1}^{n} ln(j) = ln(n!) = \int_{x=1}^{n} ln(x)dx + \frac{ln(1) + ln(n)}{2}$$

$$+ \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{(2k)!}\Big(\frac{(-1)^{2k-2} \cdot (2k-2)!}{n^{2k-1}} - \frac{(-1)^{2k-2} \cdot (2k-2)!}{1^{2k-1}}\Big) + R_{n,m}$$

$$R_{n,m} = \frac{(-1)^{m+1}}{m!} \int_{1}^{n-1} B_m(\{t\}) \frac{(-1)^{m-1} \cdot (m-1)!}{t^m} dt$$

Evaluating the integral and simplifying

$$ln(n!) = \big(n \cdot ln(n) - n\big) - \big(1 \cdot ln(1) - 1\big) + \frac{ln(n)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k} \cdot (2k-2)!}{(2k)!}\Big(\frac{1}{n^{2k-1}} - 1\Big) + R_{n,m}$$

$$R_{n,m} = \frac{1}{m} \int_{1}^{n-1} \frac{B_m(\{t\})}{t^m} dt$$

Further simplifying

$$ln(n!) = n \cdot ln(n) - n + 1 + \frac{ln(n)}{2} + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)}\Big(\frac{1}{n^{2k-1}} - 1\Big) + R_{n,m}$$

To eliminate some terms we can rearranging and take limits of both sides. This would allow us to use the simple Stirling Approximation to skip the evaluation of one of the sums.

$$\lim_{n\to\infty} ln(n!) - n \cdot ln(n) + n - \frac{ln(n)}{2}$$

$$= \lim_{n\to\infty} 1 + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + R_{n,m}$$

$$= 1 - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + \lim_{n\to\infty} R_{n,m}$$

Using equation 146

$$1 - \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} + \lim_{n\to\infty} R_{n,m} = ln(\sqrt{2\pi})$$

Plugging this back in, we get:

$$ln(n!) - n \cdot ln(n) + n - \frac{ln(n)}{2} = ln(\sqrt{2\pi}) + \sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} + R_{n,m} - \lim_{n\to\infty} R_{n,m}$$

Using the integral form of $R_{n,m}$ we can then re-write the difference $D = R_{n,m} - \lim_{n\to\infty} R_{n,m}$

$$D = R_{n,m} - \lim_{n\to\infty} R_{n,m} = \frac{1}{m} \int_1^{n-1} \frac{B_m(\{t\})}{t^m} dt - \frac{1}{m} \int_1^{\infty} \frac{B_m(\{t\})}{t^m} dt = -\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt$$

Now raising e to the value of both sides, we get

$$n! = (\frac{n}{e})^n \sqrt{2\pi n} \cdot exp\Big(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} - \frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt \Big)$$

40.2.1 Approximation big-O notation

Theorem 40.3. *The integral in the exponent can be bound by a polynomial power with respect to n.*

$$\boxed{n! = (\frac{n}{e})^n \sqrt{2\pi n} \cdot exp\Big(\sum_{k=1}^{\lfloor m/2 \rfloor} \frac{B_{2k}}{2k \cdot (2k-1)} \frac{1}{n^{2k-1}} + O(\frac{1}{n^m}) \Big)} \tag{148}$$

Proof

We need to show that $\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt = O\left(\frac{1}{n^m}\right)$

$$D = -\frac{1}{m} \int_{n-1}^{\infty} \frac{B_m(\{t\})}{t^m} dt$$

$$D \leq |D| \leq \frac{1}{|m|} \int_{n-1}^{\infty} \frac{|B_m(\{t\})|}{t^m} dt \leq \frac{\max_{x\in[0,1]} |B_m(x)|}{|m|} \int_{n-1}^{\infty} \frac{1}{t^m} dt$$

$$= \underbrace{\frac{\max_{x\in[0,1]} |B_m(x)|}{m \cdot (m+1)}}_{\neq f(n)} \underbrace{\frac{1}{(n-1)^m}}_{f(n)} = O(\frac{1}{n^m})$$

40.3 Plugging in values

Generating an approximation with $m = 7$ and using $B_2 = \frac{1}{6}, B_4 = -\frac{1}{30}, B_6 = \frac{1}{42}$ we obtain:

$$n! = (\frac{n}{e})^n \sqrt{2\pi n} \cdot exp\Big(\frac{1}{12n} - \frac{1}{360n^3} + \frac{1}{1260n^5} + O(\frac{1}{n^7}) \Big) \tag{149}$$

Since the factorial function grows very fast, it's useful to consider the logarithm of the factorial function. This would result in the following approximation.

$$ln(n!) = \frac{ln(2\pi)}{2} + nln(n) - n + + \frac{ln(n)}{2} + \frac{1}{12n} - \frac{1}{360n^3} + \frac{1}{1260n^5} + O(\frac{1}{n^7}) \tag{150}$$

41 Definition

Definition 41.1. *The Riemann Zeta function is defined as*

$$\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$$

We have already shown that [4]

$$\zeta(2) = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \tag{151}$$

42 Bernoulli Numbers

Examining the Taylor expansion of $\frac{x}{e^x-1}$, this will be required for the evaluation of the even integer values of ζ.

Definition 42.1. *The Bernoulli numbers B_n are defined as the coefficients of the Taylor expansion of the following function*

$$\boxed{\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n}$$

Theorem 42.1.

$$\sum_{k}^{n} \binom{n+1}{k} B_k = 0 \quad \forall n > 0, \text{ with } B_0 = 1 \tag{152}$$

Proof

Starting from definition 42.1

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using the Taylor expansion of e^x

$$\frac{x}{\sum_{k=1}^{\infty} \frac{x^k}{k!}} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$x = \sum_{k=1}^{\infty} \frac{x^k}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

$$1 = \sum_{k=1}^{\infty} \frac{x^{k-1}}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x$$

[4] See Gamma function document

Adjusting the indicies

$$1 = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using Cauchy's product formula for infinite sums [5]

$$1 = \sum_{n=0}^{\infty} \underbrace{\sum_{k=0}^{n} \underbrace{\left(\frac{B_k}{k!} x^k\right)}_{a_k} \underbrace{\left(\frac{x^{n-k}}{(n-k+1)!}\right)}_{b_{n-k}}}_{c_n}$$

Simplifying

$$1 = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k x^k x^{n-k}$$

$$1 = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

Comparing coefficients of powers of x of both sides we get

$$n = 0, \quad 1 = B_0$$

$$n \neq 0, \quad 0 = \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

42.1 First few Bernoulli numbers

Using equation 152 with $n = 1$ implies $\binom{2}{0}B_0 + \binom{2}{1}B_1 = 0 = 1 + 2B_1$ which implies

$$B_1 = -\frac{1}{2} \tag{153}$$

Using equation 152 with $n = 2$ implies $\binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2 = 0 = 1 - \frac{3}{2} + 3B_2$ which implies

$$B_2 = \frac{1}{6} \tag{154}$$

Similarly when we apply this equation for increasing values of n

$$B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0, B_6 = \frac{1}{42}, \ldots \tag{155}$$

Theorem 42.2. *In fact we can show that all odd Bernoulli numbers after n=1 are 0*

$$B_n = 0 \quad \forall \text{ odd } n > 1 \tag{156}$$

Proof

[5] $\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n$ where $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. See Mertens' Theorem (proof in Bernoulli Numbers document).

Starting from definition 42.1

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Removing the first two terms of the sum

$$\frac{x}{e^x - 1} = B_0 + B_1 x + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$\frac{x}{e^x - 1} - 1 + \frac{x}{2} = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$
$$\frac{x + xe^x}{2e^x - 2} - 1 = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

If we can show that the RHS is an even function then we have established the proof

$$y(x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = \frac{-x + -xe^{-x}}{2e^{-x} - 2} - 1$$

Simplfying

$$y(-x) = \frac{-x - xe^{-x}}{2e^{-x} - 2} \cdot \frac{e^x}{e^x} - 1$$
$$y(-x) = \frac{-xe^x - x}{2 - 2e^x} - 1$$
$$y(-x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = y(x)$$
$$B_n = 0 \quad \forall \ odd \ n > 1$$

43 Even integer values $\zeta(2n)$

While the odd integer values of $\zeta(n)$ do not have a closed form expression, the even values do.

Theorem 43.1. *We can express all even integer values of ζ using Bernoulli numbers*

$$\zeta(2n) = (-1)^{n+1}\frac{(2\pi)^{2n}B_{2n}}{2\cdot(2n)!} \tag{157}$$

Proof

Starting from the series representation [6] *of cotangent*

$$\pi x cot(\pi x) = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the complex exponential representation [7] *of cotangent*

$$i\pi x + \frac{2i\pi x}{e^{2i\pi x}-1} = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the Bernoulli numbers definition 42.1

$$i\pi x + \sum_{n=0}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Rearranging and using the first two values of the Bernoulli numbers

$$i\pi x + B_0 + B_1(2i\pi x) + \sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$
$$i\pi x + 1 + -i\pi x + \sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$
$$\sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi)^n x^n = -2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the fact the the odd Bernoulli numbers after B_1 are 0

$$\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^k(2\pi)^{2k}x^{2k} = \sum_{k=1}^{\infty}-2\zeta(2k)x^{2k}$$

Comparing coefficients

$$\zeta(2k) = \frac{B_{2k}}{2\cdot(2k)!}(-1)^{k+1}(2\pi)^{2k} \quad \forall k \in \mathbb{N}_{>0}$$

[6] See Euler's Sine Product Formula document

[7] $\cot(x) = i + \frac{2i}{e^{2ix}-1}$. See Basic trig functions document

43.1 The first few even integer values of ζ

$$\zeta(2) = \frac{\pi^2}{6}, \zeta(4) = \frac{\pi^4}{90}, \zeta(6) = \frac{\pi^6}{945}, \zeta(8) = \frac{\pi^8}{9450}, \zeta(10) = \frac{\pi^{10}}{93555},$$
$$\zeta(12) = \frac{691}{638512875}\pi^{12}, \zeta(14) = \frac{2}{18243225}\pi^{14}, \zeta(16) = \frac{3617}{325641566250}\pi^{16},$$
$$\zeta(18) = \frac{43867}{38979295480125}\pi^{18}, \zeta(20) = \frac{174611}{1531329465290625}\pi^{20}, \zeta(22) = \frac{155366}{13447856940643125}\pi^{22},$$
$$\zeta(24) = \frac{236364091}{2019195719637565218 75}\pi^{24}, \ldots$$

44 Related functions

44.1 Dirichlet Eta Function

Definition 44.1. *The Dirichlet Eta function is defined as*

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = 1 - \frac{1}{2^z} + \frac{1}{3^z} - \frac{1}{4^z} + ...$$

Theorem 44.1.

$$\boxed{\eta(z) = (1 - 2^{1-s})\zeta(z)} \tag{158}$$

Proof

Starting from the definition of $\eta(z)$

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - 2\sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\eta(z) = \zeta(z) - \frac{2}{2^z}\zeta(z)$$

$$\eta(z) = (1 - 2^{1-z})\zeta(z)$$

44.1.1 Values

$$\eta(1) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = ln(2) \tag{159}$$ [8]

$$\eta(2) = (1 - 2^{1-2}) \cdot \zeta(2) = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12} \tag{160}$$

[8] This can be seen by taking the Taylor-Expansion of $\ln(x+1)$ and evaluating at $x = 1$

44.2 Dirichlet Lambda Function

Definition 44.2. *The Dirichlet Lambda function is defined as*

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1))^z} = 1 + \frac{1}{3^z} + \frac{1}{5^z} + \frac{1}{7^z} + ...$$

Theorem 44.2.

$$\boxed{\lambda(z) = (1 - 2^{-s})\zeta(z)} \tag{161}$$

Proof

Starting from the definition of $\lambda(z)$

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1))^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - \sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\lambda(z) = \zeta(z) - \frac{1}{2^z}\zeta(z)$$

$$\lambda(z) = (1 - 2^{-z})\zeta(z)$$

44.2.1 Values

$$\lambda(2) = (1 - 2^{-2}) \cdot \zeta(2) = \frac{3}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{8} \tag{162}$$

45 Integral Forms

45.1 Gamma with Zeta functions

Theorem 45.1. *See the Gamma function document for the definition and properties of the Gamma function.*

$$\boxed{\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du} \tag{163}$$

Proof

Using $\Gamma(z) = n^z \int_0^\infty e^{-nu} u^{z-1} du$ *from the Gamma Function document and rearranging*

$$\Gamma(z)\frac{1}{n^z} = \int_0^\infty e^{-nu} u^{z-1} du$$

Let n *be all integers and take the sum of all equalities*

$$\sum_{n=1}^\infty \Gamma(z)\frac{1}{n^z} = \Gamma(z)\zeta(z) = \sum_{n=1}^\infty \int_0^\infty e^{-nu} u^{z-1} du$$

Bringing the summation into the integral since this integral converges uniformly

$$\Gamma(z)\zeta(z) = \int_0^\infty \Big(\sum_{n=1}^\infty e^{-nu}\Big) u^{z-1} du$$

Notice that $\sum_{n=1}^\infty e^{-nu}$ *is a geometric series if* $u > 0$

$$\Gamma(z)\zeta(z) = \int_0^\infty \Big(\frac{e^{-u}}{1 - e^{-u}}\Big) u^{z-1} du$$

Simplfiying by multiplying top and bottom by e^u

$$\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du$$

46 Jacobi Theta Function

46.1 Definition

Definition 46.1. *The Jacobi Theta function is defined as*

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$$

Theorem 46.1. *The Jacobi Theta function has the following functional form*

$$\Theta(x) = \frac{1}{\sqrt{x}}\Theta(\frac{1}{x}) \tag{164}$$

Proof

Using the Poisson Summation Formula [9]

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x} e^{-2\pi i k y} dy = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x - 2\pi i k y} dy$$

Re-writting the exponent

$$-\pi y^2 x - 2\pi i k y = -\pi x(y^2 - 2\pi i \frac{k}{x} y) = -\pi x(y + i\frac{k}{x})^2 - \pi \frac{k^2}{x}$$

Plugging in the factored exponent, we get

$$\Theta(x) = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi x(y+i\frac{k}{x})^2 - \pi \frac{k^2}{x}} dy = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \int_{-\infty}^{\infty} e^{-\pi x(y+i\frac{k}{x})^2} dy$$

After a change of variables $z = y + i\frac{k}{x}$ we can use the Guassian integral identity as proved before, refer to the Guassian integral document, $\int_{-\infty}^{\infty} e^{-az^2} dz = \sqrt{\frac{\pi}{a}}$, we get:

$$\Theta(x) = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \sqrt{\frac{\pi}{x\pi}} = \frac{1}{\sqrt{x}}\Theta(\frac{1}{x})$$

47 Functional Equations

47.1 First Symmetric Form

Theorem 47.1 (Riemann's Symmetric Functional Equation)**.** *The completed zeta function $\xi(s)$ satisfies the symmetric functional equation:*

$$\boxed{\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s)} \tag{165}$$

[9] $\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n)$ is the Poisson summation formula with Fourier transform $\hat{f}(k) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i k x} dx$. See Fourier Series and Integrals notes.

Proof. Recall the integral definition of the Gamma function:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty t^{\frac{s}{2}-1} e^{-t}\, dt$$

Making the substitution $t = \pi n^2 x$ (so $dt = \pi n^2 dx$) for $n \geq 1$:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty (\pi n^2 x)^{\frac{s}{2}-1} e^{-\pi n^2 x} \pi n^2\, dx = \pi^{\frac{s}{2}} n^s \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

Rearranging yields:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} = \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

Summing both sides over all positive integers $n = 1, 2, 3, \ldots$ for $\mathrm{Re}(s) > 1$:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \sum_{n=1}^\infty n^{-s} = \sum_{n=1}^\infty \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

By Fubini-Tonelli theorem, we can interchange the sum and integral:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \int_0^\infty x^{\frac{s}{2}-1} \omega(x)\, dx$$

where $\omega(x) := \sum_{n=1}^\infty e^{-\pi n^2 x} = \frac{\Theta(x)-1}{2}$.

From Jacobi's theta transformation $\Theta(x) = \frac{1}{\sqrt{x}}\Theta\left(\frac{1}{x}\right)$, the function $\omega(x)$ satisfies:

$$\omega(x) = \frac{1}{\sqrt{x}}\omega\left(\frac{1}{x}\right) + \frac{1}{2\sqrt{x}} - \frac{1}{2}$$

Now split the integral into two regions $[0, 1]$ and $[1, \infty)$:

$$\int_0^\infty x^{\frac{s}{2}-1} \omega(x)\, dx = \int_0^1 x^{\frac{s}{2}-1} \omega(x)\, dx + \int_1^\infty x^{\frac{s}{2}-1} \omega(x)\, dx$$

Substitute the modular transformation of $\omega(x)$ into the integral over $[0, 1]$:

$$\begin{aligned}
\int_0^1 x^{\frac{s}{2}-1} \omega(x)\, dx &= \int_0^1 x^{\frac{s}{2}-1} \left(\frac{1}{\sqrt{x}}\omega\left(\frac{1}{x}\right) + \frac{1}{2x^{1/2}} - \frac{1}{2}\right) dx \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{2}\left[\frac{x^{s/2-1/2}}{s/2-1/2}\right]_0^1 - \frac{1}{2}\left[\frac{x^{s/2}}{s/2}\right]_0^1 \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s-1} - \frac{1}{s} \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s(s-1)}
\end{aligned}$$

In the integral $\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right)\, dx$, substitute $u = 1/x$ ($dx = -u^{-2} du$):

$$\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx = \int_\infty^1 u^{-\left(\frac{s-1}{2}-1\right)} \omega(u) \left(-\frac{du}{u^2}\right) = \int_1^\infty u^{\frac{1-s}{2}-1} \omega(u)\, du$$

Combining both parts, we obtain the fundamental representation:

$$\xi(s) := \pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \left(x^{\frac{s}{2}-1} + x^{\frac{1-s}{2}-1}\right)\omega(x)\,dx \tag{166}$$

Because $\omega(x) \sim e^{-\pi x}$ as $x \to \infty$, the integral converges absolutely and uniformly for all $s \in \mathbb{C}$, providing the analytic continuation of $\zeta(s)$ to a meromorphic function on $\mathbb{C}$ with a single simple pole at $s = 1$.

Moreover, replacing s with $1 - s$ leaves the right-hand side completely invariant:

$$\xi(1-s) = \frac{1}{(1-s)(-s)} + \int_1^\infty \left(x^{\frac{1-s}{2}-1} + x^{\frac{s}{2}-1}\right)\omega(x)\,dx = \xi(s)$$

which proves the theorem. □

47.2 Second Asymmetric Form

Theorem 47.2 (Reflection Form of the Functional Equation)**.** *For all* $s \in \mathbb{C} \setminus \{0, 1\}$*:*

$$\boxed{\zeta(s) = 2^s\pi^{s-1}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s)} \tag{167}$$

Proof. From the first functional equation:

$$\zeta(s) = \pi^{s-\frac{1}{2}}\frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)}\zeta(1-s)$$

We simplify the ratio $\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}$ using two classical identities of the Gamma function:

1. **Legendre's Duplication Formula**:

$$\Gamma\left(\frac{s}{2}\right)\Gamma\left(\frac{s+1}{2}\right) = 2^{1-s}\sqrt{\pi}\,\Gamma(s) \implies \frac{1}{\Gamma(s/2)} = \frac{2^{s-1}}{\sqrt{\pi}}\frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)}$$

2. **Euler's Reflection Formula**:

$$\Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{1+s}{2}\right) = \frac{\pi}{\sin\left(\pi\frac{1-s}{2}\right)} = \frac{\pi}{\cos\left(\frac{\pi s}{2}\right)}$$

and for $\Gamma(s)$:

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)} \implies \frac{1}{\Gamma(s)} = \frac{\sin(\pi s)}{\pi}\Gamma(1-s) = \frac{2\sin\left(\frac{\pi s}{2}\right)\cos\left(\frac{\pi s}{2}\right)}{\pi}\Gamma(1-s)$$

Combining these gives:

$$\begin{aligned}
\frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)} &= \Gamma\left(\frac{1-s}{2}\right)\cdot\frac{2^{s-1}}{\sqrt{\pi}}\frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)} \\
&= \frac{2^{s-1}}{\sqrt{\pi}}\left[\Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{1+s}{2}\right)\right]\cdot\frac{1}{\Gamma(s)} \\
&= \frac{2^{s-1}}{\sqrt{\pi}}\left[\frac{\pi}{\cos\left(\frac{\pi s}{2}\right)}\right]\cdot\left[\frac{2\sin\left(\frac{\pi s}{2}\right)\cos\left(\frac{\pi s}{2}\right)}{\pi}\Gamma(1-s)\right] \\
&= 2^s\pi^{-\frac{1}{2}}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)
\end{aligned}$$

Multiplying by $\pi^{s-1/2}\zeta(1-s)$ yields:

$$\zeta(s) = \pi^{s-\frac{1}{2}} \cdot 2^s \pi^{-\frac{1}{2}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s)$$

□

48 Zeros of the Riemann Zeta Function and the Prime Numbers

48.1 The Euler Product: The Bridge to Prime Numbers

The profound connection between the Riemann zeta function and prime numbers was discovered by Leonhard Euler in 1737. It shows that the analytical behavior of $\zeta(s)$ directly encodes the distribution of prime numbers.

Theorem 48.1 (Euler Product Formula)**.** *For any complex number s with $\mathrm{Re}(s) > 1$, the zeta function can be expressed as an absolutely convergent infinite product over all prime numbers:*

$$\boxed{\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}} \tag{168}$$

Proof. Consider the series definition $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ for $\mathrm{Re}(s) > 1$. For each prime p, the term p^{-s} satisfies $|p^{-s}| = p^{-\mathrm{Re}(s)} < 1$. Expanding each geometric factor:

$$\frac{1}{1-p^{-s}} = 1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \cdots = \sum_{k=0}^{\infty} \frac{1}{p^{ks}}$$

Now consider the finite product over the first N primes $p_1, p_2, \ldots, p_N$:

$$\begin{aligned}\prod_{j=1}^{N} \frac{1}{1-p_j^{-s}} &= \left(\sum_{k_1=0}^{\infty} \frac{1}{p_1^{k_1 s}}\right)\left(\sum_{k_2=0}^{\infty} \frac{1}{p_2^{k_2 s}}\right) \cdots \left(\sum_{k_N=0}^{\infty} \frac{1}{p_N^{k_N s}}\right) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} \cdots \sum_{k_N=0}^{\infty} \frac{1}{(p_1^{k_1} p_2^{k_2} \cdots p_N^{k_N})^s}\end{aligned}$$

By the **Fundamental Theorem of Arithmetic**, every positive integer $n \geq 1$ has a unique factorization as a product of prime powers $n = p_1^{k_1} p_2^{k_2} \cdots$. Therefore, multiplying out the terms produces the sum $\sum \frac{1}{n^s}$ over all positive integers n whose prime factors are all among $\{p_1, \ldots, p_N\}$:

$$\zeta(s) - \prod_{j=1}^{N} \frac{1}{1-p_j^{-s}} = \sum_{\substack{n > p_N \\ \exists q > p_N \text{ s.t. } q \mid n}} \frac{1}{n^s}$$

Because $\sum_{n=1}^{\infty} |n^{-s}| = \sum_{n=1}^{\infty} n^{-\mathrm{Re}(s)} < \infty$, the tail of the series vanishes as $N \to \infty$:

$$\left|\zeta(s) - \prod_{j=1}^{N} \frac{1}{1-p_j^{-s}}\right| \leq \sum_{n=p_N+1}^{\infty} \frac{1}{n^{\mathrm{Re}(s)}} \xrightarrow[N\to\infty]{} 0$$

which completes the proof. □

Corollary 48.1.1 (No Zeros in the Half-Plane $\mathrm{Re}(s) > 1$)**.** *The Riemann zeta function has no zeros in the region* $\mathrm{Re}(s) > 1$.

Proof. Taking the complex logarithm of the Euler product for $\mathrm{Re}(s) > 1$:

$$\ln \zeta(s) = -\sum_p \ln(1 - p^{-s}) = \sum_p \sum_{m=1}^{\infty} \frac{1}{m\, p^{ms}}$$

Since $\sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\,\mathrm{Re}(s)}} \leq \sum_p \frac{2}{p^{\mathrm{Re}(s)}} < \infty$, this series converges absolutely. Consequently, $\ln \zeta(s)$ is finite everywhere on $\mathrm{Re}(s) > 1$. Therefore:

$$|\zeta(s)| = \exp(\mathrm{Re}(\ln \zeta(s))) > 0$$

meaning $\zeta(s) \neq 0$ for all $\mathrm{Re}(s) > 1$. □

48.2 Trivial Zeros and the Critical Strip

With the half-plane $\mathrm{Re}(s) > 1$ established as zero-free, we examine the remaining regions using the reflection functional equation (Theorem 47.2):

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1 - s)\zeta(1 - s) \tag{169}$$

48.2.1 Trivial Zeros

Consider negative even integers $s = -2m$ for $m \in \{1, 2, 3, \dots\}$:

(i) The trigonometric factor vanishes: $\sin\left(\frac{\pi(-2m)}{2}\right) = \sin(-\pi m) = 0$.

(ii) The gamma factor $\Gamma(1 - (-2m)) = \Gamma(1 + 2m) = (2m)!$ is a positive real number, hence finite and non-zero.

(iii) The zeta factor $\zeta(1 - (-2m)) = \zeta(2m + 1)$ is evaluated at an odd integer $2m + 1 \geq 3 > 1$, where the Euler product guarantees $\zeta(2m + 1) > 1 \neq 0$.

Therefore, every negative even integer is a zero of $\zeta(s)$:

$$\boxed{\zeta(-2m) = 0 \quad \text{for all } m \in \{1, 2, 3, \dots\}} \tag{170}$$

These are the **trivial zeros** of the Riemann zeta function.

Remark 48.2 (Values at Negative Odd Integers)**.** *At negative odd integers* $s = -(2m-1)$*, the sine factor does not vanish:* $\sin\left(-\frac{(2m-1)\pi}{2}\right) = (-1)^m \neq 0$*. Using the functional equation together with Euler's formula for* $\zeta(2m)$ *(Theorem* **??***), we obtain:*

$$\zeta(-(2m - 1)) = -\frac{B_{2m}}{2m} \neq 0 \tag{171}$$

In particular:

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}, \quad \zeta(-5) = -\frac{1}{252}$$

48.2.2 The Critical Strip

Combining our results:

1. **Region** $\mathrm{Re}(s) > 1$: $\zeta(s) \neq 0$ by the Euler product (Corollary 48.1.1).

2. **Region** $\mathrm{Re}(s) < 0$: By the functional equation (169), the only zeros for $\mathrm{Re}(s) < 0$ are the trivial zeros at $s = -2, -4, -6, \ldots$, since $\zeta(1-s) \neq 0$ for $\mathrm{Re}(1-s) > 1$.

3. **The Critical Strip**: All non-trivial zeros must lie strictly within the strip:

$$\mathcal{S} = \{s \in \mathbb{C} : 0 \leq \mathrm{Re}(s) \leq 1\} \tag{172}$$

4. **Zero-Free Boundary**: In 1896, Jacques Hadamard and Charles Jean de la Vallée Poussin independently proved that $\zeta(s)$ has no zeros on the line $\mathrm{Re}(s) = 1$:

$$\zeta(1+it) \neq 0 \quad \text{for all } t \in \mathbb{R}$$

   By the reflection symmetry $s \leftrightarrow 1-s$, $\zeta(0+it) \neq 0$ as well. This zero-free boundary is mathematically equivalent to the **Prime Number Theorem** ($\pi(x) \sim \frac{x}{\ln x}$).

Consequently, all non-trivial zeros lie strictly in the interior $0 < \mathrm{Re}(s) < 1$.

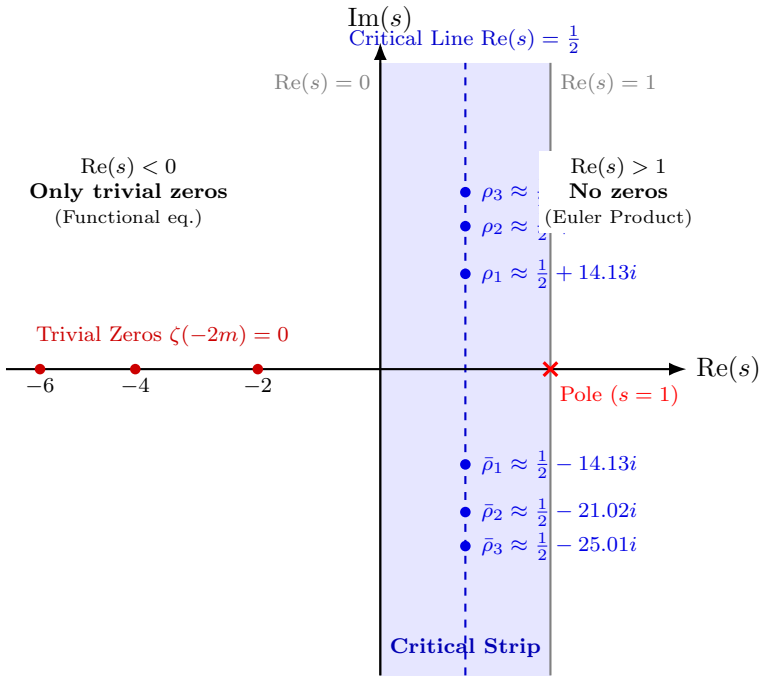


Figure 1: The complex s-plane: trivial zeros on the negative real axis, the pole at $s = 1$, the critical strip $0 < \mathrm{Re}(s) < 1$, and non-trivial zeros lying symmetrically on the critical line $\mathrm{Re}(s) = \frac{1}{2}$.

48.3 The Riemann Xi Function and Critical Line Symmetry

To study the non-trivial zeros on the critical line, Riemann introduced the completed function $\xi(s)$ in his 1859 paper:

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{173}$$

The prefactor $\frac{1}{2}s(s-1)$ cancels the simple pole of $\zeta(s)$ at $s=1$ and the pole of $\Gamma(s/2)$ at $s=0$. Consequently, $\xi(s)$ is an **entire function** of order 1.

From Theorem 47.1, $\xi(s)$ satisfies the perfect reflection symmetry:

$$\xi(s) = \xi(1-s) \tag{174}$$

Notice that the zeros of $\xi(s)$ are precisely the non-trivial zeros of $\zeta(s)$ (the trivial zeros are cancelled by the poles of $\Gamma(s/2)$).

Theorem 48.3 (The Real-Valued Function $\Xi(t)$)**.** *Parameterizing the critical line by $s = \frac{1}{2} + it$ for $t \in \mathbb{R}$, define:*

$$\Xi(t) := \xi\left(\frac{1}{2} + it\right) \tag{175}$$

*Then $\Xi(t)$ is an **even**, **purely real-valued** function for all real t:*

$$\Xi(-t) = \Xi(t) \quad \textit{and} \quad \Xi(t) \in \mathbb{R} \quad (\forall t \in \mathbb{R})$$

Proof. Using the functional equation $\xi(s) = \xi(1-s)$ with $s = \frac{1}{2} + it$:

$$\Xi(-t) = \xi\left(\frac{1}{2} - it\right) = \xi\left(1 - \left(\frac{1}{2} + it\right)\right) = \xi\left(\frac{1}{2} + it\right) = \Xi(t)$$

which proves that $\Xi(t)$ is an even function.

To prove that $\Xi(t)$ is real-valued, observe that for real s, $\zeta(s)$, $\Gamma(s/2)$, and $s(s-1)$ are all real. By the Schwarz reflection principle, $\overline{\xi(s)} = \xi(\bar{s})$ for all $s \in \mathbb{C}$. Evaluating at $s = \frac{1}{2} + it$:

$$\overline{\Xi(t)} = \overline{\xi\left(\frac{1}{2} + it\right)} = \xi\left(\overline{\frac{1}{2} + it}\right) = \xi\left(\frac{1}{2} - it\right) = \Xi(-t) = \Xi(t)$$

Since $\overline{\Xi(t)} = \Xi(t)$, $\Xi(t)$ must be real for all $t \in \mathbb{R}$. □

Remark 48.4 (Significance for Zeros)**.** *Theorem 48.3 converts the two-dimensional problem of locating complex zeros of $\zeta(s)$ on the critical line into the one-dimensional problem of finding the **real roots of a real-valued curve** $\Xi(t)$.*

In 1914, G. H. Hardy used this real-variable formulation to prove that $\Xi(t)$ changes sign infinitely many times, establishing:

> ***Hardy's Theorem (1914):** There are infinitely many non-trivial zeros of $\zeta(s)$ lying exactly on the critical line $\mathrm{Re}(s) = \frac{1}{2}$.*

In 1989, J. B. Conrey proved that at least 40.77% of all non-trivial zeros lie strictly on the critical line.

48.4 The Riemann Hypothesis

The Riemann Hypothesis (Bernhard Riemann, 1859)

All non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the **critical line**:

$$\boxed{\mathrm{Re}(s) = \frac{1}{2}} \tag{176}$$

Equivalently, all roots of the Riemann Ξ-function $\Xi(t)$ are purely real:

$$\Xi(t) = 0 \implies t \in \mathbb{R} \tag{177}$$

The first five non-trivial zeros on the critical line have imaginary parts:

$$\begin{aligned}
\rho_1 &= \frac{1}{2} + i \cdot 14.13472514\ldots \\
\rho_2 &= \frac{1}{2} + i \cdot 21.02203964\ldots \\
\rho_3 &= \frac{1}{2} + i \cdot 25.01085758\ldots \\
\rho_4 &= \frac{1}{2} + i \cdot 30.42487613\ldots \\
\rho_5 &= \frac{1}{2} + i \cdot 32.93506159\ldots
\end{aligned}$$

To date, more than 10^{13} non-trivial zeros have been checked by computer algorithms, and every single one lies precisely on the critical line $\mathrm{Re}(s) = 1/2$. A general mathematical proof remains the most celebrated open problem in mathematics and is designated as a Clay Millennium Prize Problem.

48.5 The "Music of the Primes": Riemann's Explicit Formula

Riemann's ultimate goal in studying $\zeta(s)$ was to understand the distribution of prime numbers with exact precision.

48.5.1 Von Mangoldt and Chebyshev Functions

To link primes with analytic continuation, define the **von Mangoldt function**:

$$\Lambda(n) = \begin{cases} \ln p & \text{if } n = p^k \text{ for some prime } p \text{ and integer } k \geq 1 \\ 0 & \text{otherwise} \end{cases} \tag{178}$$

Taking the logarithmic derivative of the Euler product (Theorem 48.1):

$$\begin{aligned}
\ln \zeta(s) &= -\sum_{p} \ln(1 - p^{-s}) \\
-\frac{\zeta'(s)}{\zeta(s)} &= \sum_{p} \frac{p^{-s} \ln p}{1 - p^{-s}} = \sum_{p} \sum_{k=1}^{\infty} (\ln p)\, p^{-ks} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}
\end{aligned}$$

The cumulative sum $\psi(x) = \sum_{n \le x} \Lambda(n)$ is **Chebyshev's second function**. Counting primes via $\psi(x)$ is weighted by $\ln p$, giving $\psi(x) \approx \sum_{p \le x} \ln p \approx \pi(x) \ln x$.

48.5.2 Riemann's Explicit Formula

By applying Perron's contour inversion formula to $-\frac{\zeta'(s)}{\zeta(s)}$ and evaluating the residues at the simple pole $s = 1$, the trivial zeros, and the non-trivial zeros ρ, Riemann derived his celebrated **Explicit Formula**:

Theorem 48.5 (Riemann's Explicit Formula)**.** *For any non-integer $x > 1$:*

$$\boxed{\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \ln(2\pi) - \frac{1}{2}\ln(1 - x^{-2})} \tag{179}$$

where the sum ranges over all non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$, paired as $\lim_{T\to\infty} \sum_{|\gamma|<T} \frac{x^\rho}{\rho}$.

48.5.3 The Harmonic Interpretation

Writing each non-trivial zero as $\rho = \beta + i\gamma$:

$$x^\rho = x^{\beta+i\gamma} = x^\beta e^{i\gamma \ln x} = x^\beta \left[\cos(\gamma \ln x) + i \sin(\gamma \ln x)\right] \tag{180}$$

Pairing conjugate zeros $\rho = \beta + i\gamma$ and $\bar{\rho} = \beta - i\gamma$ produces real oscillating harmonics:

$$\frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} = 2\,\mathrm{Re}\left(\frac{x^{\beta+i\gamma}}{\beta + i\gamma}\right) = \frac{2x^\beta}{|\rho|}\cos\left(\gamma \ln x - \arg(\rho)\right)$$

Equation (179) decomposes the prime distribution into:

1. **The Baseline Trend** x: Produced by the residue at the pole $s = 1$. This corresponds to the smooth asymptotic distribution of primes $\psi(x) \sim x$ (the Prime Number Theorem).
2. **The Harmonic Oscillations** $-\sum_\rho \frac{x^\rho}{\rho}$: Each non-trivial zero ρ contributes a pure wave with frequency $\frac{\gamma}{2\pi}$ on a logarithmic scale $\ln x$. The primes are literally the **constructive interference** of these infinite waves!
3. **The Smooth Tail** $-\ln(2\pi) - \frac{1}{2}\ln(1 - x^{-2})$: Produced by the residues at the trivial zeros $s = -2, -4, \ldots$ and $s = 0$.

48.5.4 The Optimal Error Bound

The real part $\beta = \mathrm{Re}(\rho)$ controls the **amplitude** of the prime fluctuations (x^β).

- If any zero had $\beta > 1/2$, the fluctuations would blow up with amplitude x^β, creating massive, irregular swings in prime density.
- Under the **Riemann Hypothesis**, all zeros have $\beta = 1/2$, making every oscillation have amplitude exactly $\sqrt{x}$.

This establishes the ultimate equivalence:

Theorem 48.6 (Equivalence with the Prime Error Bound)**.** *The Riemann Hypothesis is mathematically equivalent to the sharpest possible error bound for prime counting:*

$$\boxed{\pi(x) = \mathrm{Li}(x) + O\left(\sqrt{x}\ln x\right)} \tag{181}$$

where $\mathrm{Li}(x) = \int_2^x \frac{dt}{\ln t}$ *is the logarithmic integral.*

The error $O(\sqrt{x}\ln x)$ is identical to the statistical fluctuation of an unbiased random walk of size x. Thus, the Riemann Hypothesis asserts that prime numbers are distributed with the maximum possible regularity permitted by arithmetic.

49 Integral Representation of Zeta(2)

$$\int_0^\infty \frac{t}{e^{\alpha t}-1}dt = \int_0^\infty \frac{te^{-\alpha t}}{1-e^{-\alpha t}}dt = \int_0^\infty t\sum_{n=1}^\infty e^{-n\alpha t}dt$$

By the Dominant Convergence Theorem: (Include Reference)

$$\int_0^\infty t\sum_{n=1}^\infty e^{-n\alpha t}dt = \sum_{n=1}^\infty \int_0^\infty te^{-n\alpha t}dt$$

Using $n\cdot\alpha\cdot t = x \rightarrow n\cdot\alpha\cdot dt = dx$

$$\sum_{n=1}^\infty \int_0^\infty \frac{x}{n\alpha}e^{-x}\frac{dx}{n\alpha} = \sum_{n=1}^\infty \frac{1}{n^2\alpha^2}\int_0^\infty xe^{-x}dx = \frac{1}{\alpha^2}\sum_{n=1}^\infty \frac{1}{n^2}\Gamma(2)$$

Recall $\Gamma(2) = 1! = 1$ (Include Reference)

$$\frac{1}{\alpha^2}\sum_{n=1}^\infty \frac{1}{n^2}\Gamma(2) = \frac{1}{\alpha^2}\sum_{n=1}^\infty \frac{1}{n^2} = \frac{1}{\alpha^2}\zeta(2)$$

Recall $\zeta(2) = \frac{\pi^2}{6}$ (Include Reference)

$$=> \int_0^\infty \frac{t}{e^{\alpha t}-1}dt = \frac{\pi^2}{6\alpha^2}$$

50 Sine Product Formula

Theorem 50.1. $\boxed{\frac{sin(x)}{x} = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{(k\pi)^2}\right)}$

Proof

$$\frac{sin(x)}{x} \text{ has roots } x = \pi, -\pi, 2\pi, -2\pi, 3\pi, -3\pi, ...$$

If we treat it as a normal polynomial then it can be written as a product of it's roots like such

$$\frac{sin(x)}{x} = A\left(1 - \frac{x}{\pi}\right)\left(1 + \frac{x}{\pi}\right)\left(1 - \frac{x}{2\pi}\right)\left(1 + \frac{x}{2\pi}\right)\left(1 - \frac{x}{3\pi}\right)\left(1 + \frac{x}{3\pi}\right)\cdot..$$

For justification of why we can do this, see section 51

$$\frac{sin(x)}{x} = A\left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{2^2\pi^2}\right)\left(1 - \frac{x^2}{3^2\pi^2}\right)\cdot..$$

$$\frac{sin(x)}{x} = A\prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2\pi^2}\right)$$

Solving for A :

From the taylor expansion $sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...$

$$\lim_{x\to 0}\frac{sin(x)}{x} = \lim_{x\to 0}\frac{x}{x} - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + ...$$

$$\lim_{x\to 0}\frac{sin(x)}{x} = 1$$

$$A = 1$$

$$\frac{sin(x)}{x} = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2\pi^2}\right)$$

50.1 Change of variables of the Sine Product Formula

Theorem 50.2. $\boxed{\frac{sin(\pi x)}{\pi x} = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2}\right)}$

Proof

Replace x *with* x'/π *in Theorem 50.1*

50.2 Basel Problem

50.2.1 Context

This product form of sine was originally developed by Euler to solve the Basel problem.

The Basel problem was to determine what value the sum $\sum_{k=1}^{\infty}\frac{1}{k^2}$ converged to. Mathematicians at the time knew that this infinite sum converged due to the various convergence tests, but no one knew the exact value.

Euler's solution to this open problem was one of the first of many solutions for

him to gain a reputation in the international math community.

This problem would later be generalized as a particular value of the zeta function, $\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$. Riemann was the first to consider complex value of this function, and even today there is an open millennium prize problem revolving around the zeros of this function.

50.2.2 Solution

Theorem 50.3. $\boxed{\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}}$

Proof

Expanding the product form

$$\frac{sin(x)}{x} = \prod_{k=1}^{\infty} \Big(1 - \frac{x^2}{(k\pi)^2}\Big)$$

$$\frac{sin(x)}{x} = 1 - \Big(\sum_{k=1}^{\infty} \frac{1}{k^2\pi^2}\Big)x^2 + O\Big(x^4\Big)$$

Expanding using the taylor series

$$sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...$$

Comparing the coefficient of x^2

$$\sum_{k=1}^{\infty} \frac{1}{k^2\pi^2} = \frac{1}{3!}$$

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

50.3 Series representation of $cot(\pi x)$

Theorem 50.4. $\boxed{\pi cot(\pi x) = \sum_{k=-\infty}^{\infty} \frac{1}{x+k}}$

Proof

Starting from Theorem 50.2

$$\frac{sin(\pi x)}{\pi x} = \prod_{k=1}^{\infty} \Big(1 - \frac{x^2}{k^2}\Big)$$

The idea is to take the derivative of the log of both sides. [10]

First taking the log of both sides

$$ln(sin(\pi x)) = ln(\pi x) + \sum_{k=1}^{\infty} ln\Big(1 - \frac{x^2}{k^2}\Big)$$

Now taking the derivative of both sides w.r.t. x we get

$$\pi \frac{cos(\pi x)}{sin(\pi x)} = \frac{1}{x} + \sum_{k=1}^{\infty} \frac{\frac{-2x}{k^2}}{\Big(1 - \frac{x^2}{k^2}\Big)}$$

Simplifying

$$\pi \cdot cot(\pi x) = \frac{1}{x} + 2x \sum_{k=1}^{\infty} \frac{1}{x^2 - k^2}$$

$$Note\ \frac{1}{x+k} + \frac{1}{x-k} = \frac{x - k + x + k}{x^2 - k^2} = \frac{2x}{x^2 - k^2}$$

Plugging this into the series expression

$$\pi \cdot cot(\pi x) = \frac{1}{x} + \sum_{k=1}^{\infty} \frac{1}{x+k} + \frac{1}{x-k}$$

Cleaning up the indicies

$$\pi \cdot cot(\pi x) = \sum_{k=-\infty}^{\infty} \frac{1}{x+k}$$

Theorem 50.5. $\boxed{x\pi \cot(\pi x) = 1 - 2\sum_{k=1}^{\infty} \zeta(2k) x^{2k}}$

Proof

Starting from previous intermediate result

$$\pi cot(\pi x) = \frac{1}{x} + \sum_{k=1}^{\infty} \frac{\frac{-2x}{k^2}}{\Big(1 - \frac{x^2}{k^2}\Big)}$$

$$\pi x \cdot cot(\pi x) = 1 + \sum_{k=1}^{\infty} \frac{\frac{-2x^2}{k^2}}{\Big(1 - \frac{x^2}{k^2}\Big)}$$

Using the infinite geometric series formula $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ *for* $|x| < 1$

$$\pi x \cdot cot(\pi x) = 1 + \sum_{k=1}^{\infty} \left(\sum_{n=0}^{\infty} \Big(\frac{x^2}{k^2}\Big)^n \right) \frac{-2x^2}{k^2}$$

[10] This is similar to the definition of the Di-Gamma function Ψ, see Gamma function document.

Simplfying

$$\pi x \cdot cot(\pi x) = 1 - 2\sum_{k=1}^{\infty}\left(\sum_{n=0}^{\infty}\left(\frac{x^2}{k^2}\right)^{n+1}\right)$$
$$\pi x \cdot cot(\pi x) = 1 - 2\sum_{k=1}^{\infty}\left(\sum_{n=1}^{\infty}\left(\frac{x}{k}\right)^{2n}\right)$$

Interchanging the summations since both series are absolutly convergent [11]

$$\pi x \cdot cot(\pi x) = 1 - 2\sum_{n=1}^{\infty}\left(\sum_{k=1}^{\infty}\left(\frac{1}{k^{2n}}\right)\right)x^{2n}$$

Using the definition of $\zeta(x) = \sum_{k=1}^{\infty}\frac{1}{k^x}$

$$\pi x \cdot cot(\pi x) = 1 - 2\sum_{n=1}^{\infty}\zeta(2n)x^{2n}$$

50.4 Wallis Product

Theorem 50.6. $\boxed{\prod_{n=1}^{\infty}\left(\frac{2n}{2n-1}\right)\left(\frac{2n}{2n+1}\right) = \frac{\pi}{2}}$

Proof

Starting from Theorem 50.2

$$\frac{sin(\pi x)}{\pi x} = \prod_{k=1}^{\infty}\left(1 - \frac{x^2}{k^2}\right)$$

Plugging in $x = \frac{1}{2}$

$$\frac{sin(\frac{\pi}{2})}{\frac{\pi}{2}} = \frac{2}{\pi} = \prod_{k=1}^{\infty}\left(1 - \frac{1}{4k^2}\right)$$
$$\frac{2}{\pi} = \prod_{k=1}^{\infty}\left(\frac{4k^2-1}{4k^2}\right)$$
$$\frac{2}{\pi} = \prod_{k=1}^{\infty}\left(\frac{2k-1}{2k}\right)\cdot\left(\frac{2k+1}{2k}\right)$$
$$\frac{\pi}{2} = \prod_{k=1}^{\infty}\left(\frac{2k}{2k-1}\right)\cdot\left(\frac{2k}{2k+1}\right)$$

51 Rigorous proof of convergence

Euler's original proof did not include proper justification of why we are allowed to write $\sin(x)$ as an infinite product of it's roots. The justification comes from what is known as Weierstrass factorization theorem.

[11] Technically this needs to be shown.

51.1 Weierstrass factorization theorem

Define elementary factors

$$E_n(z) = \begin{cases} (1-z) & \text{if } n = 0 \\ (1-z)e^{z+z^2/2+\ldots+z^n/n} & otherwise \end{cases} \tag{182}$$

Theorem 51.1. *Let f be an entire function, and let $\{a_n\}$ be the non-zero zeros of f repeated according to multiplicity. Suppose also that f has a zero at z=0 of order $m \geq 0$ (a zero of order $m = 0$ at $z = 0$ means $f(0) \neq 0$). Then there exists an entire function g and a sequence of integers $\{p_n\}$ such that*

$$f(z) = z^m e^{g(z)} \prod_{n=1}^{\infty} E_{p_n}(\frac{z}{a_n}) \tag{183}$$

Proof TODO

52 Darboux's formula

Let $\phi(t)$ be a polynomial of degree n and $f(x)$ is an analytic function then,

Theorem 52.1.

$$\sum_{m=0}^{n}(-1)^m(z-a)^m\Big[\phi^{(n-m)}(1)f^{(m)}(z)-\phi^{(n-m)}(0)f^{(m)}(a)\Big]=$$

$$(-1)^n(z-a)^{n+1}\int_0^1\phi(t)f^{(n+1)}\Big[a+t(z-a)\Big]dt \quad (184)$$

Proof

For $n=0$ we have $\phi(t)=c$ then

$$LHS_0=c\Big[f(z)-f(a)\Big]$$

$$RHS_0=(z-a)\int_0^1 cf^{(1)}\Big[a+t(z-a)\Big]dt$$

After a change of variables $u=a+t(z-a)$, $du=dt(z-a)$ we get

$$RHS_0=c\int_a^z f^{(1)}(u)du=c\Big[f(u)\Big]\Big|_{u=a}^{u=z}=c\Big[f(z)-f(a)\Big]=LHS_0$$

For $n>0$

$$RHS_n=(-1)^n(z-a)^{n+1}\int_0^1\phi(t)f^{(n+1)}\Big[a+t(z-a)\Big]dt$$

Using integration by parts $\int_{t=0}^1 uv'=uv\big|_{t=0}^1-\int_{t=0}^1 u'v$, with $u=\phi(t)$, and $v'=f^{(n+1)}(a+t(z-a))$

$$RHS_n=(-1)^n(z-a)^{n+1}\left[\phi(t)\frac{f^{(n)}\Big[a+t(z-a)\Big]}{(z-a)}\Big|_{t=0}^{t=1}-\int_0^1\phi'(t)\frac{f^{(n)}\Big[a+t(z-a)\Big]}{(z-a)}dt\right]$$

Evaluating the first definite integral and multipying by $(z-a)$

$$RHS_n=(-1)^n(z-a)^n\left[\phi(1)f^{(n)}(z)-\phi(0)f^{(n)}(a)-\int_0^1\phi'(t)f^{(n)}\Big[a+t(z-a)\Big]dt\right]$$

Moving the second integral out to it's own term

$$RHS_n=(-1)^n(z-a)^n\Big[\phi(1)f^{(n)}(z)-\phi(0)f^{(n)}(a)\Big]+(-1)^{n-1}(z-a)^n\int_0^1\phi'(t)f^{(n)}\Big[a+t(z-a)\Big]dt$$

Repeating the integration by parts process we get

$$RHS_n=(-1)^n(z-a)^n\Big[\phi(1)f^{(n)}(z)-\phi(0)f^{(n)}(a)\Big]$$

$$+(-1)^{n-1}(z-a)^{n-1}\Big[\phi^{(1)}(1)f^{(n-1)}(z)-\phi^{(1)}(0)f^{(n-1)}(a)\Big]$$

$$+(-1)^{n-2}(z-a)^{n-1}\int_0^1\phi^{(2)}(t)f^{(n-1)}\Big[a+t(z-a)\Big]dt$$

Repeating the integration by parts n-times until $\phi^{(n+1)}(t) = 0$ *we get*

$$RHS_n = \sum_{k=0}^{n} (-1)^{n-k}(z-a)^{n-k}\left[\phi^{(k)}(1)f^{(n-k)}(z) - \phi^{(k)}(0)f^{(n-k)}(a)\right] + 0$$

Changing the index of summation $m = n - k$

$$RHS_n = \sum_{m=0}^{n} (-1)^{m}(z-a)^{m}\left[\phi^{(n-m)}(1)f^{(m)}(z) - \phi^{(n-m)}(0)f^{(m)}(a)\right]$$

Therefore we get

$$\sum_{m=0}^{n} (-1)^{m}(z-a)^{m}\Big[\phi^{(n-m)}(1)f^{(m)}(z) - \phi^{(n-m)}(0)f^{(m)}(a)\Big] =$$
$$(-1)^{n}(z-a)^{n+1}\int_0^1 \phi(t)f^{(n+1)}\Big[a + t(z-a)\Big]dt$$

53 Taylor's formula

Set $\phi(t) = (t-1)^n$ into Darnoux's formula to obtain Taylor's formula.

Theorem 53.1.

$$\boxed{f(z) = \sum_{m=0}^{n} \frac{(z-a)^m f^{(m)}(a)}{m!} + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt} \tag{185}$$

Proof

Starting from Darboux's formula **??**

$$\sum_{m=0}^{n} (-1)^m (z-a)^m \Big[\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a)\Big] =$$
$$(-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

Setting $\phi(t) = (t-1)^n$*, which is an n-degree polynomial*

Note that $\phi^{(k)}(t) = \frac{n!}{(n-k)!}(t-1)^{n-k}$ *for* $k \leq n$*, therefore*

$$\phi^{(k)}(0) = \frac{n!}{(n-k)!}(-1)^{n-k} \quad \textit{for } k \leq n$$
$$\phi^{(k)}(1) = 0 \quad \textit{for } k < n$$
$$\phi^{(n)}(1) = n!$$
$$\sum_{m=1}^{n} (-1)^m (z-a)^m \Big[0 - \frac{n!(-1)^m}{(m)!} f^{(m)}(a)\Big] + (-1)^0 (z-a)^0 \Big[n! f^{(0)}(z) - n! f^{(0)}(a)\Big]$$
$$= (-1)^n (z-a)^{n+1} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

Simplfiying and dividing by $n!$

$$-\sum_{m=1}^{n} \frac{(z-a)^m}{m!} f^{(m)}(a) + \Big[f(z) - f(a)\Big]$$
$$= (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

Rearranging

$$f(z) = f(a) + \sum_{m=1}^{n} \frac{(z-a)^m}{m!} f^{(m)}(a) + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

Adding the extra term into the summation to finish the proof

$$f(z) = \sum_{m=0}^{n} \frac{(z-a)^m}{m!} f^{(m)}(a) + (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

53.1 Infinite Series

Theorem 53.2. *If* $|f^{(k)}(x)| \leq M$ *and* $|z-a| \leq R \quad \forall x$ *then*

$$\boxed{f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!}} \tag{186}$$

Proof

Starting from Taylor's formula 185 and taking the limit as n approaches infinity

$$f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!} + \lim_{n\to\infty} (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

Define the remainder term, and then we will show that it converges to 0

$$R_n = (-1)^n \frac{(z-a)^{n+1}}{n!} \int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt$$

$$|R_n| = \Big|(-1)^n\Big| \cdot \Big|\frac{(z-a)^{n+1}}{n!}\Big| \cdot \Big|\int_0^1 (t-1)^n f^{(n+1)}\Big[a + t(z-a)\Big] dt\Big|$$

$$|R_n| \leq \frac{|z-a|^{n+1}}{n!} \cdot \int_0^1 \Big|(t-1)^n\Big| \cdot \Big|f^{(n+1)}\Big[a + t(z-a)\Big]\Big| dt$$

Using $|f^{(k)}(x)| \leq M$

$$|R_n| \leq \frac{|z-a|^{n+1}}{n!} \cdot \int_0^1 |t-1|^n \cdot M dt$$

Using $|z-a| \leq R$

$$|R_n| \leq \frac{R^{n+1} M}{n!} \cdot \int_0^1 |t-1|^n dt$$

Note that between $0 \leq t \leq 1$, $|t-1| \leq 1$, *therefore*

$$|R_n| \leq \frac{R^{n+1} M}{n!}$$

Therefore since M *and* R *are constant:*

$$\lim_{n\to\infty} R_n \leq \lim_{n\to\infty} |R_n| \leq 0$$

Therefore

$$f(z) = \sum_{m=0}^{\infty} \frac{(z-a)^m f^{(m)}(a)}{m!}$$

54 Euler-Maclaurin formula

Set $\phi(t) = B_n(x)$ into Darnoux's formula to obtain the Euler-Maclaurin formula, where $B_n(x)$ is the Bernoulli polynomial[12].

Theorem 54.1. *For any analytical $g(x)$*

$$\boxed{\begin{aligned} w\sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2}\Big(g(a)-g(a+rw)\Big)+ \\ \sum_{k=1}^{\lfloor n/2\rfloor} \frac{w^{2k}B_{2k}}{(2k)!}\Big(g^{(2k-1)}(a+rw)-g^{(2k-1)}(a)\Big)+R_{n,r} \\ R_{n,r} = \frac{(-1)^{n+1}w^{n+1}}{n!}\int_0^r B_n(\{t\})f^{(n+1)}(a+tw)dt \end{aligned}} \tag{187}$$

Proof

Starting from Darboux's formula **??**

$$\sum_{m=0}^{n}(-1)^m(z-a)^m\Big[\phi^{(n-m)}(1)f^{(m)}(z)-\phi^{(n-m)}(0)f^{(m)}(a)\Big] = (-1)^n(z-a)^{n+1}\int_0^1 \phi(t)f^{(n+1)}\Big[a+t(z-a)\Big]dt$$

Setting $\phi(t) = B_n(t)$, which is an n-degree polynomial

Note that $\phi^{(k)}(t) = \frac{n!}{(n-k)!}B_{n-k}(t)$ for $k \le n$, therefore

$$\phi^{(k)}(0) = \phi^{(k)}(1) = \frac{n!}{(n-k)!}B_{n-k} \quad \text{for } k \le n, k \neq 1$$
$$\phi^{(1)}(0) = -\phi^{(1)}(1) = B_1$$

adjusting the indicies we get

$$\phi^{(n-m)}(0) = \phi^{(n-m)}(1) = \frac{n!}{m!}B_m \quad \text{for } n \ge m \ge 0, m \neq 1$$
$$\phi^{(n-1)}(0) = B_1 = -\phi^{(n-1)}(1)$$

Plugging these values into Darboux's formula, and removing the first two terms we get

$$\sum_{m=2}^{n}(-1)^m(z-a)^m\Big[\frac{n!}{m!}B_m f^{(m)}(z)-\frac{n!}{m!}B_m f^{(m)}(a)\Big]+n!B_0\Big(f(z)-f(a)\Big)$$
$$+(-1)n!(z-a)\Big(-B_1 f'(z)-B_1 f'(a)\Big) = (-1)^n(z-a)^{n+1}\int_0^1 B_n(t)f^{(n+1)}\Big[a+t(z-a)\Big]dt$$

[12] $B_n(x) = \sum_{k=0}^{n}\binom{n}{k}B_k x^{n-k}$. See the Bernoulli number document for the full definition.

simplfying

$$\sum_{m=2}^{n} \frac{(-1)^m (z-a)^m B_m}{m!}(f^{(m)}(z) - f^{(m)}(a)) + B_0\Big(f(z) - f(a)\Big) + (z-a)B_1\Big(f^{'}(z) + f^{'}(a)\Big) = \frac{(-1)^n (z-a)^{n+1}}{n!}\int_0^1 B_n(t) f^{(n+1)}\Big[a + t(z-a)\Big]dt$$

To simplify notation let us define R_n^a

$$R_n^a = \frac{(-1)^n (z-a)^{n+1}}{n!}\int_0^1 B_n(t) f^{(n+1)}\Big[a + t(z-a)\Big]dt$$

Therefore we get

$$\sum_{m=2}^{n} \frac{(-1)^m (z-a)^m B_m}{m!}(f^{(m)}(z) - f^{(m)}(a)) + B_0\Big(f(z) - f(a)\Big) + (z-a)B_1\Big(f^{'}(z) + f^{'}(a)\Big) = R_n^a$$

Plugging in $B_0 = 1$ *and* $B_1 = -\frac{1}{2}$

$$\sum_{m=2}^{n} \frac{(-1)^m (z-a)^m B_m}{m!}\Big(f^{(m)}(z) - f^{(m)}(a)\Big) + \Big(f(z) - f(a)\Big) - (z-a)\frac{1}{2}\Big(f^{'}(z) + f^{'}(a)\Big) = R_n^a$$

Adding $(z-a)f^{'}(a)$ *to both sides and rearranging we get*

$$(z-a)f^{'}(a) = f(z) - f(a) - \frac{(z-a)}{2}\Big(f^{'}(z) - f^{'}(a)\Big) + \sum_{m=2}^{n} \frac{(-1)^m (z-a)^m B_m}{m!}\Big(f^{(m)}(z) - f^{(m)}(a)\Big) - R_n^a$$

Define $w = z - a$ *and* $g(x) = f'(x)$ *therefore we get*

$$wg(a) = \int_{x=a}^{a+w} g(x)dx - \frac{w}{2}\Big(g(a+w) - g(a)\Big) + \sum_{m=2}^{n} \frac{(-1)^m w^m B_m}{m!}\Big(g^{(m-1)}(a+w) - g^{(m-1)}(a)\Big) - R_n^a$$

Allowing a *to be multiples of* w, $a' = a + jw$ *and summing over* i *we get*

$$\sum_{j=0}^{r-1} wg(a+jw) = \sum_{j=0}^{r-1}\int_{x=a+jw}^{a+(j+1)w} g(x)dx - \sum_{j=0}^{r-1}\frac{w}{2}\Big(g(a+(j+1)w) - g(a+jw)\Big)$$
$$+\sum_{j=0}^{r-1}\sum_{m=2}^{n} \frac{(-1)^m w^m B_m}{m!}\Big(g^{(m-1)}(a+(j+1)w) - g^{(m-1)}(a+jw)\Big) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Simplyfying and evaluating the sums we get

$$w\sum_{j=0}^{r-1} g(a+jw) = \int_{x=a}^{a+rw} g(x)dx - \frac{w}{2}\Big(g(a+rw) - g(a)\Big)$$
$$+\sum_{m=2}^{n} \frac{(-1)^m w^m B_m}{m!}\Big(g^{(m-1)}(a+rw) - g^{(m-1)}(a)\Big) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Rearranging

$$w\sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2}\Big(g(a)-g(a+rw)\Big)$$
$$+\sum_{m=2}^{n}\frac{(-1)^m w^m B_m}{m!}\Big(g^{(m-1)}(a+rw)-g^{(m-1)}(a)\Big) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Using the fact that $B_{2n+1}=0, n\geq 1$ we can simplfy the sum using $m=2k$

$$w\sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2}\Big(g(a)-g(a+rw)\Big)$$
$$+\sum_{k=1}^{\lfloor n/2\rfloor}\frac{w^{2k} B_{2k}}{(2k)!}\Big(g^{(2k-1)}(a+rw)-g^{(2k-1)}(a)\Big) - \sum_{j=0}^{r-1} R_n^{a+jw}$$

Recall the remainder term definition

$$R_n^a = \frac{(-1)^n (z-a)^{n+1}}{n!}\int_0^1 B_n(t) f^{(n+1)}\Big[a+t(z-a)\Big]dt$$

Plugging in the definition of $w=(z-a)$ we get

$$R_n^a = \frac{(-1)^n w^{n+1}}{n!}\int_0^1 B_n(t) f^{(n+1)}\Big[a+tw\Big]dt$$

Define the new remainder term

$$R_n = -\sum_{j=0}^{r-1} R_n^{a+jw}$$

Plug in the definition of R_n^{a+jw} and simplfy

$$R_n = -\sum_{j=0}^{r-1}\frac{(-1)^n w^{n+1}}{n!}\int_0^1 B_n(t) f^{(n+1)}(a+jw+tw)dt$$

Simplfy the remainder term using the fractional operator $\{x\}$

$$R_{n,r} = \frac{(-1)^{n+1} w^{n+1}}{n!}\int_0^r B_n(\{t\}) f^{(n+1)}(a+tw)dt$$

54.1 Infinite Series

Note that as n approaches infinity, the remainder term goes to 0.

Theorem 54.2. *If $|g^{(k)}(x)|\leq M \quad \forall x$ and a, w, M finite then*

$$\boxed{\lim_{n\to\infty} R_{n,r} = \lim_{n\to\infty}\frac{(-1)^{n+1} w^{n+1}}{n!}\int_0^r B_n(\{t\}) f^{(n+1)}(a+tw)dt = 0} \qquad (188)$$

Proof

Starting from the definition of $R_{n,r}$

$$\lim_{n\to\infty} R_n = \lim_{n\to\infty} \frac{(-1)^{n+1}w^{n+1}}{n!}\int_0^r B_n(\{t\})f^{(n+1)}(a+tw)dt = 0$$

$$\lim_{n\to\infty} R_n \leq \lim_{n\to\infty} |R_n|$$

$$|R_n| \leq \frac{|w|^{n+1}}{n!}\int_0^r |B_n(\{t\})|Mdt \leq \frac{|w|^{n+1}|B_n(t)|rM}{n!}$$

Using $\lim_{n\to\infty}\frac{B_n(x)}{n!} = 0$ *from BERNOULLI DOCUMENT TODO*

$$\lim_{n\to\infty} R_n \leq \lim_{n\to\infty} |R_n| = 0$$

54.2 Simple version

By specifying $a = 0$ and $w = 1$ we obtain an exact difference between the summation and the integral of any analytic function $g(x)$.

Theorem 54.3. *For any analytical* $g(x)$

$$\boxed{\begin{gathered}\sum_{j=0}^{r} g(j) - \int_{x=0}^{r} g(x)dx = \frac{g(0)+g(r)}{2} + \sum_{k=1}^{\lfloor n/2\rfloor}\frac{B_{2k}}{(2k)!}\Big(g^{(2k-1)}(r) - g^{(2k-1)}(0)\Big) + R_{n,r}\\ R_{n,r} = \frac{(-1)^{n+1}}{n!}\int_0^r B_n(\{t\})g^{(n)}(t)dt\end{gathered}} \tag{189}$$

Proof

Starting from the Euler-Maclaurin formula

$$w\sum_{j=0}^{r-1} g(a+jw) - \int_{x=a}^{a+rw} g(x)dx = \frac{w}{2}\Big(g(a) - g(a+rw)\Big) +$$

$$\sum_{k=1}^{\lfloor n/2\rfloor}\frac{w^{2k}B_{2k}}{(2k)!}\Big(g^{(2k-1)}(a+rw) - g^{(2k-1)}(a)\Big) + R_{n,r}$$

$$R_{n,r} = \frac{(-1)^{n+1}w^{n+1}}{n!}\int_0^r B_n(\{t\})g^{(n)}(a+tw)dt$$

Plugging in $a = 0$ *and* $w = 1$ *we get*

$$\sum_{j=0}^{r-1} g(j) - \int_{x=0}^{r} g(x)dx = \frac{1}{2}\Big(g(0) - g(r)\Big) + \sum_{k=1}^{\lfloor n/2\rfloor}\frac{B_{2k}}{(2k)!}\Big(g^{(2k-1)}(r) - g^{(2k-1)}(0)\Big) + R_{n,r}$$

$$R_{n,r} = \frac{(-1)^{n+1}}{n!}\int_0^r B_n(\{t\})g^{(n)}(t)dt$$

Adding $g(r)$ *to both sides*

$$\sum_{j=0}^{r} g(j) - \int_{x=0}^{r} g(x)dx = \frac{g(0)+g(r)}{2} + \sum_{k=1}^{\lfloor n/2\rfloor}\frac{B_{2k}}{(2k)!}\Big(g^{(2k-1)}(r) - g^{(2k-1)}(0)\Big) + R_{n,r}$$

55 Gaussian Integral

Theorem 55.1.

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

Proof

Defining I

$$I = \int_{-\infty}^{\infty} e^{-ax^2} dx$$

Squaring I

$$I^2 = \int_{-\infty}^{\infty} e^{-ax^2} dx \cdot \int_{-\infty}^{\infty} e^{-ay^2} dy$$

Combining the integrals

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-ax^2 - ay^2} dx dy$$

Converting to Polar coordinates

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-ar^2} r dr d\theta$$

Solving

$$\begin{aligned} I^2 &= \int_0^{2\pi} \tfrac{-1}{2a} e^{-ar^2} \Big|_0^{\infty} d\theta \\ I^2 &= \int_0^{2\pi} \tfrac{1}{2a} d\theta \\ I^2 &= 2\pi \tfrac{1}{2a} \\ I^2 &= \frac{\pi}{a} \\ I &= \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} e^{-ax^2} dx &= \sqrt{\frac{\pi}{a}} \end{aligned}$$

Since e^{-ax^2} is an even function we also get the integral

$$\int_0^{\infty} e^{-ax^2} dx = \frac{\sqrt{\frac{\pi}{a}}}{2} = \sqrt{\frac{\pi}{4a}} \tag{190}$$

56 Gaussian Type Integrals

Using $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ we can take derivatives with respect to (w.r.t.) a

$$\frac{\partial}{\partial a}\int_{-\infty}^{\infty} e^{-ax^2}dx = \frac{\partial}{\partial a}\sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} -x^2 e^{-ax^2}dx = \sqrt{\pi}\frac{-1}{2}a^{-3/2}$$

$$\boxed{\int_{-\infty}^{\infty} x^2 e^{-ax^2}dx = \frac{1}{2a}\sqrt{\frac{\pi}{a}}}$$

This process can be repeated to obtain the general identity

$$\boxed{\int_{-\infty}^{\infty} x^{2n} e^{-ax^2}dx = \frac{\pi}{|\Gamma(\frac{1}{2}-n)|a^n\sqrt{a}}} \tag{191}$$

57 Properties of Fourier Series

A function $f(x)$ can be written as the series shown in **??**, known as a Fourier Series. The following provides derivation of the properties of Fourier Series.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n cos(nx) + b_n sin(nx) \tag{192}$$

57.0.1 Side Note: Trigonometric Identities of addition

Recall equations **????????**

$$cos(a+b) = cos(a)cos(b) - sin(a)sin(b) \tag{193}$$
$$cos(a-b) = cos(a)cos(b) + sin(a)sin(b) \tag{194}$$
$$sin(a+b) = sin(a)cos(b) + sin(b)cos(a) \tag{195}$$
$$sin(a-b) = sin(a)cos(b) - sin(b)cos(b) \tag{196}$$

From equations **????????** we obtain **??????**

$$cos(a)cos(b) = \frac{1}{2}(cos(a+b) + cos(a-b)) \tag{197}$$
$$sin(a)cos(b) = \frac{1}{2}(sin(a+b) + sin(a-b)) \tag{198}$$
$$sin(a)sin(b) = \frac{1}{2}(cos(a-b) - cos(a+b)) \tag{199}$$

57.0.2 Integrals

Since $sin(nx)$ is an odd function, its integral over a symmetric region is zero.

$$\int_{-\pi}^{\pi} sin(nx)dx = 0 \tag{200}$$

Since n is an integer, and assuming $n > 0$, we obtain the following.

$$\int_{-\pi}^{\pi} cos(nx)dx = \frac{2sin(\pi n)}{n} = 0 \tag{201}$$

Since $sin(mx)cos(nx)$ is an odd function, its integral over a symmetric region is zero.

$$\int_{-\pi}^{\pi} sin(mx)cos(nx)dx = 0 \tag{202}$$

Use **??** to write the following.

$$\int_{-\pi}^{\pi} cos(mx)cos(nx)dx = \frac{1}{2}\int_{-\pi}^{\pi} cos((n+m)x)dx + \frac{1}{2}\int_{-\pi}^{\pi} cos((n-m)x)dx$$
$$= \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases} \tag{203}$$

Using **??** we obtain the following.

$$\int_{-\pi}^{\pi} sin(mx)sin(nx)dx = \frac{1}{2}\int_{-\pi}^{\pi} cos((n-m)x)dx - \frac{1}{2}\int_{-\pi}^{\pi} cos((n+m)x)dx$$
$$= \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases} \quad (204)$$

57.0.3 a_0 coefficient

Integrate both sides of **??** to obtain:

$$\int_{-\pi}^{\pi} f(x)dx = \int_{-\pi}^{\pi} a_0 dx + \sum_{n=1}^{\infty}(a_n \int_{-\pi}^{\pi} cos(nx)dx + b_n \int_{-\pi}^{\pi} sin(nx)dx)$$

Using **????** the sum terms become zero.

$$= \int_{-\pi}^{\pi} a_0 + 0 + 0 = a_0 x|_{-\pi}^{\pi} = a_0\pi + a_0\pi = 2\pi a_0$$

And thus we obtain

$$a_0 = \frac{\int_{-\pi}^{\pi} f(x)dx}{2\pi} \quad (205)$$

57.0.4 a_n coefficients

Multiply both sides of **??** by $cos(mx)$ to obtain:

$$f(x)cos(mx) = a_0 cos(mx) + \sum_{n=1}^{\infty} a_n cos(nx)cos(mx) + b_n sin(nx)cos(mx)$$

Integrate both sides.

$$\int_{-\pi}^{\pi} f(x)cos(mx)dx = a_0 \int_{-\pi}^{\pi} cos(mx)dx + \sum_{n=1}^{\infty}(a_n \int_{-\pi}^{\pi} cos(nx)cos(mx)dx + b_n \int_{-\pi}^{\pi} sin(nx)cos(mx)dx)$$

$\rightarrow \int_{-\pi}^{\pi} cos(mx)dx = 0$ (from **??**)

$\rightarrow \sum_{n=1}^{\infty}(a_n \int_{-\pi}^{\pi} cos(nx)cos(mx)dx = \int_{-\pi}^{\pi} a_m cos(mx)cos(mx) + \sum_{n=1,n\neq m}^{\infty} \int_{-\pi}^{\pi} cos(nx)cos(mx),$

which (from **??**) is equal to $a_m\pi + \sum_{n=1,n\neq m}^{\infty} 0 = a_m\pi$.

$\rightarrow \int_{-\pi}^{\pi} sin(nx)cos(mx)dx = 0$ (from **??**). Thus we obtain:

$$\int_{-\pi}^{\pi} f(x)cos(mx)dx = 0 + a_m\pi + 0$$

And thus:

$$a_m = \frac{\int_{-\pi}^{\pi} f(x)cos(mx)dx}{\pi}$$

Which, by changing the index from m to n gives:

$$a_n = \frac{\int_{-\pi}^{\pi} f(x)cos(nx)dx}{\pi} \tag{206}$$

57.0.5 b_n coefficients

Using similar arguments to **??** we obtain:

$$b_n = \frac{\int_{-\pi}^{\pi} f(x)sin(nx)dx}{\pi} \tag{207}$$

58 Definition

Define a number i such that

Definition 58.1.

$$\boxed{i^2 = -1}$$

59 Euler's equation

This formula is the underlying connection between imaginary numbers and the trigonometric functions.

Theorem 59.1. *Euler's equation*

$$\boxed{e^{ix} = cos(x) + i \cdot sin(x)} \tag{208}$$

Proof 1 through Taylor series expansion

Taking the taylor series of $cos(x), sin(x), e^{ix}$ *we obtain*

$$cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - ...$$

$$sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - ...$$

$$e^{ix} = \sum_{k=0}^{\infty} \frac{(ix)^k}{k!}$$

Note that we can split the taylor series of e^{ix} *into a real part and an imaginary part*

$$e^{ix} = \sum_{k=0}^{\infty} \frac{(ix)^k}{k!} = \Big(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - ...\Big) + i\Big(x - \frac{x^3}{3!} + \frac{x^5}{5!} - ...\Big)$$

$$e^{ix} = cos(x) + i \cdot sin(x)$$

Proof 2 through partial fraction decomposition

Evaluating the integral of $\int \frac{1}{x^2+1}$ *two different ways*

First way, change of variables

$$\textit{Let } tan(\theta) = x, \textit{ then } dx = \frac{1}{cos^2(\theta)} d\theta$$

$$\int \frac{1}{x^2+1} dx = \int \frac{1}{\frac{sin^2(\theta)}{cos^2(\theta)} + 1} \frac{1}{cos^2(\theta)} d\theta = \int \frac{1}{sin^2(\theta) + cos^2(\theta)} d\theta = \int d\theta = \theta + C_1$$

$$\int \frac{1}{x^2+1} dx = arctan(x) + C_1$$

Second way, partial fractions decomposition

$$\int \frac{1}{x^2+1}dx = \int \frac{1}{(x+i)(x-i)}dx = \frac{i}{2}\cdot\int \frac{1}{x+i}dx - \frac{i}{2}\cdot\int \frac{1}{(x-i)}dx$$
$$\int \frac{1}{x^2+1}dx = \frac{i}{2}\cdot ln(x+i) - \frac{i}{2}ln(x-i) + C_2$$
$$\int \frac{1}{x^2+1}dx = \frac{i}{2}ln\Big(\frac{x+i}{x-i}\Big) + C_2$$

Equating the two expressions

$$\frac{i}{2}ln\Big(\frac{x+i}{x-i}\Big) + C_2 = arctan(x) + C_1$$

simplifying

$$-ln\Big(\frac{x+i}{x-i}\Big) + C = 2i\cdot arctan(x)$$
$$ln\Big(\frac{x-i}{x+i}\Big) + C = 2i\cdot arctan(x)$$
$$K\frac{x-i}{x+i} = e^{2i\cdot arctan(x)}$$

Using a change of variables of $tan(\theta) = x$

$$K\frac{tan(\theta)-i}{tan(\theta)+i} = e^{2i\cdot\theta}$$

simplifying

$$e^{2i\cdot\theta} = K\frac{(tan(\theta)-i)^2}{tan^2(\theta)+1} = K\frac{tan^2(\theta)-2i\cdot tan(\theta)-1}{\frac{1}{cos^2(\theta)}} = K(sin^2(\theta) - i\cdot 2sin(\theta)cos(\theta) - cos^2(\theta)$$

Using the double angle sin and cos formulas

$$e^{2i\cdot\theta} = K(-cos(2\theta) - isin(2\theta)$$

Plugging in $\theta = 0$ *we get*

$$K = -1$$

Replacing 2θ *with* x

$$e^{i\cdot x} = cos(x) + isin(x)$$

Proof 3 through differential equations

Consider the soluions of the differential equation $y' = i\cdot y$, *with* $y(0) = 1$

Notice that both $y(x) = e^{ix}$ *and* $y(x) = cos(x) + isin(x)$ *both satisfy the equation*

Prove that the solution is unique

let $y_1(x), y_2(x)$ *be two solutions to the above equation, then define*

$$f(x) = y_1(x) - y_2(x)$$

Notice that $f(0) = 1 - 1 = 0$

$$f'(x) = y_1'(x) - y_2'(x) = i \cdot f(x)$$

To prove that f' is 0, hence f is constant, we define

$$g(x) = e^{-ix} f(x)$$

Notice that $g(0) = 1 \cdot f(0) = 0$

$$g'(x) = -i \cdot g(x) + e^{-ix} f'(x)$$

$$g'(x) = -i \cdot g(x) + i \cdot e^{-ix} f(x)$$

$$g'(x) = -i \cdot g(x) + i \cdot g(x) = 0$$

$$g(x) = 0 \quad \forall x$$

$$e^{-ix} f(x) = 0 \quad \forall x$$

Assuming e^{ix} *is not 0 for any x*

$$f(x) = 0 \quad \forall x$$

$$y_1(x) - y_2(x) = 0 \quad \forall x$$

$$e^{i \cdot x} = cos(x) + isin(x) \quad \forall x$$

60 Properties

Starting from Euler's equation

Definition 60.1.

$$e^{ix} = cos(x) + i \cdot sin(x)$$

Using this equation we can plug in the value of $x = \frac{\pi}{2}$ and obtain

$$e^{i\frac{\pi}{2}} = cos(\frac{\pi}{2}) + i \cdot sin(\frac{\pi}{2}) = 0 + i \cdot 1 = i \tag{209}$$

Therefore we can claim

Definition 60.2.

$$i = e^{i\frac{\pi}{2}}$$

60.1 Reciprocal

$$\frac{1}{i} = \frac{1 \cdot i}{i \cdot i} = \frac{i}{-1} = -i \tag{210}$$

60.2 Square Root

$$\sqrt{i} = (e^{i\frac{\pi}{2}})^{\frac{1}{2}} = e^{i\frac{\pi}{4}} = cos(\frac{\pi}{4}) + i \cdot sin(\frac{\pi}{4}) = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \tag{211}$$

60.3 i to the power of i

$$i^i = (e^{i\frac{\pi}{2}})^i = e^{-\frac{\pi}{2}} \approx 0.208 \tag{212}$$

61 Basic Inverse Trig Functions

If we assume we have this as a given.

Definition 61.1 (Euler's Formula)**.**

$$e^{ix} = \cos(x) + i \cdot \sin(x) \tag{213}$$

Theorem 61.2 (Sine Definition)**.**

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i} = -i\frac{e^{ix} - e^{-ix}}{2} \tag{214}$$

Proof.

$$e^{-ix} = \cos(-x) + i \cdot \sin(-x) = \cos(x) - i \cdot \sin(x)$$
$$e^{ix} - e^{-ix} = 2i \cdot \sin(x)$$
$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i} = -i\frac{e^{ix} - e^{-ix}}{2}$$

□

Theorem 61.3 (Cosine Definition)**.**

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2} \tag{215}$$

Proof. Same as above. □

Theorem 61.4 (Tangent Definition)**.**

$$\tan(x) = -i\frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}} \tag{216}$$

Proof. Same as above. □

Theorem 61.5 (Cotangent Definition)**.**

$$\cot(x) = i + \frac{2i}{e^{2ix} - 1} \tag{217}$$

Proof.

$$\tan(x) = -i\frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}}$$
$$\cot(x) = i\frac{e^{ix} + e^{-ix}}{e^{ix} - e^{-ix}}$$
$$\cot(x) = i\frac{(e^{ix} - e^{-ix}) + (2e^{-ix})}{e^{ix} - e^{-ix}}$$
$$\cot(x) = i + 2i\frac{e^{-ix}}{e^{ix} - e^{-ix}}$$
$$\cot(x) = i + \frac{2i}{e^{2ix} - 1}$$

□

If we for now ignore the non-uniqueness of the square root and logarithm of complex numbers, we can obtain

Theorem 61.6 (Arcsine Function)**.**

$$\arcsin(x) = -i \cdot \ln(ix + \sqrt{1 - x^2}) \tag{218}$$

Proof. Let $y = \arcsin(x)$

$$\sin(y) = \frac{e^{iy} - e^{-iy}}{2i}$$
$$\sin(\arcsin(x)) = x = \frac{e^{iy} - e^{-iy}}{2i}$$
$$2ix = e^{iy} - e^{-iy}$$
$$e^{2iy} - 2ixe^{iy} - 1 = 0$$
$$e^{iy} = \frac{2ix \pm \sqrt{-4x^2 + 4}}{2}$$
$$e^{iy} = ix \pm \sqrt{1 - x^2}$$
$$iy = \ln(ix + \sqrt{1 - x^2})$$
$$y = \frac{\ln(ix + \sqrt{1 - x^2})}{i} = -i \cdot \ln(ix + \sqrt{1 - x^2})$$
$$\arcsin(x) = -i \cdot \ln(ix + \sqrt{1 - x^2})$$

□

Theorem 61.7 (Arccosine Function)**.**

$$\arccos(x) = -i \cdot \ln(x + \sqrt{x^2 - 1}) \tag{219}$$

Proof. Same as above. □

Theorem 61.8 (Arctangent Function)**.**

$$\arctan(x) = \frac{i}{2} \cdot \ln\left(\frac{i + x}{i - x}\right) \tag{220}$$

Proof. Same as above. □

Theorem 61.9 (Arcsecant Function)**.**

$$\operatorname{arcsec}(x) = -i \ln\left(\frac{1 \pm \sqrt{1 - x^2}}{x}\right) \tag{221}$$

Proof. Same as above. □

Theorem 61.10 (Arccosecant Function)**.**

$$\operatorname{arccsc}(x) = -i \ln\left(\frac{i}{x} + \sqrt{1 - \frac{1}{x^2}}\right) \tag{222}$$

Proof. Same as above. □

Theorem 61.11 (Arccotangent Function)**.**

$$\operatorname{arccot}(x) = \frac{i}{2} \cdot \ln\left(\frac{x-i}{x+i}\right) \tag{223}$$

Proof. Same as above. □

The same can be applied to hyperbolic functions, using

Definition 61.12 (Hyperbolic Functions)**.**

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \tag{224}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2} \tag{225}$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} \tag{226}$$

For each normal trigonometry identities there is a similar hyperbolic trigonometry identity.

The relation of the two are

Theorem 61.13 (Relation to Circular Functions)**.**

$$\cosh(ix) = \cos(x) \qquad \sinh(ix) = i \cdot \sin(x)$$
$$\tanh(ix) = i \cdot \tan(x) \qquad \cos(ix) = \cosh(x)$$
$$\sin(ix) = i \cdot \sinh(x) \qquad \tan(ix) = i \cdot \tanh(x)$$

62 Definition

The Fibonacci sequence is defined as:

$$\begin{aligned} f_0 &= 0 \\ f_1 &= 1 \\ f_{n+2} &= f_{n+1} + f_n \quad \forall \quad n \in \mathbb{Z}^+ \end{aligned} \tag{227}$$

The first few Fibonacci numbers then would be:

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

63 Ratios

Taking the ratios of consecutive Fibonacci numbers we find this pattern:
(Insert continued fractions Reference)

$$\begin{aligned} \frac{f_1}{f_0} &= 1 \\ \frac{f_2}{f_1} &= 2 = 1 + 1 \\ \frac{f_3}{f_2} &= \frac{3}{2} = 1 + \frac{1}{1+1} \\ \frac{f_4}{f_3} &= \frac{5}{3} = 1 + \frac{2}{3} = 1 + \frac{1}{1+\frac{1}{1+1}} \\ \frac{f_5}{f_4} &= \frac{8}{5} = 1 + \frac{3}{5} = 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+1}}} \end{aligned}$$

Introducing a new notation for continued fractions: (See Continued Fraction Latex document)

$$[a_0; a_1, a_2, a_3, a_4, ...] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \frac{1}{\ldots}}}}}} \tag{228}$$

Note that this notation has the following property:

$$A + \frac{1}{[a_0; a_1, a_2, a_3, a_4, ...]} = A + \frac{1}{a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \frac{1}{\ldots}}}}}}} = [A; a_0, a_1, a_2, a_3, a_4, \ldots] \tag{229}$$

This notation can be used for all fractions, for example:

$$\frac{8}{5} = [1; 1, 1, 1, 1, 1] = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}}}$$

Theorem 63.1.

$$\frac{f_{n+1}}{f_n} = \underbrace{[1;1,1,1,..,1]}_{n+1}$$

Proof by induction

Assume true for k

$$\frac{f_{k+1}}{f_k} = \underbrace{[1;1,1,1,..,1]}_{k+1}$$

For k+1

$$\frac{f_{k+2}}{f_{k+1}} = \frac{f_{k+1}+f_k}{f_{k+1}} = 1 + \frac{1}{\frac{f_{k+1}}{f_k}}$$

Using equation **??**

$$\frac{f_{k+2}}{f_{k+1}} = 1 + \frac{1}{\frac{f_{k+1}}{f_k}} = \underbrace{[1;1,1,1,..,1]}_{k+2}$$

Theorem 63.2. *If* $\lim_{n\to\infty} \underbrace{[1;1,1,1,..,1]}_{n}$ *converges then* $\lim_{n\to\infty} \frac{f_{n+1}}{f_n} = \frac{1+\sqrt{5}}{2}$

Proof

Assume convergence

$$\lim_{n\to\infty} \underbrace{[1;1,1,1,..,1]}_{n} = x$$

Then

$$x = \lim_{n\to\infty} \frac{f_{n+1}}{f_n} = \lim_{n\to\infty} 1 + \frac{1}{\frac{f_n}{f_{n-1}}} = 1 + \frac{1}{\lim_{n\to\infty} \frac{f_n}{f_{n-1}}} = 1 + \frac{1}{x}$$

Therefore

$$x = 1 + \frac{1}{x}$$
$$x^2 - x - 1 = 0$$
$$x = \frac{1 \pm \sqrt{5}}{2}$$

Since we know the value is positive we get

$$x = \lim_{n\to\infty} \frac{f_{n+1}}{f_n} = \frac{1+\sqrt{5}}{2}$$

This is only a preview to the relation of the Fibonacci numbers to the golden ratio, and it's characteristic equation.

64 Negative

If we ignore the condition of $n > 0$ we can can use the functional form definition of Fibonacci numbers to define negative indices:

$$f_{n-2} = f_n - f_{n-1} \tag{230}$$

Therefore the first few will be:

$$\begin{aligned}
f_{-1} &= f_1 - f_0 = 1 - 0 = 1 \\
f_{-2} &= f_0 - f_{-1} = 0 - 1 = -1 \\
f_{-3} &= f_{-1} - f_{-2} = 1 - (-1) = 2 \\
f_{-4} &= f_{-2} - f_{-3} = -1 - 2 = -3 \\
f_{-5} &= f_{-3} - f_{-4} = 2 - (-3) = 5
\end{aligned}$$

Theorem 64.1.

$$f_{-n} = (-1)^{n+1} f_n$$

Proof by induction

Assume true for k, and k+1

$$\begin{aligned}
f_{-k} &= (-1)^{k+1} f_k \\
f_{-k-1} &= (-1)^{k+2} f_{k+1}
\end{aligned}$$

Proving for k+2

$$\begin{aligned}
f_{-k-2} &= f_{-k} - f_{-k-1} \\
&= (-1)^{k+1} f_k - (-1)^{k+2} f_{k+1} \\
&= (-1)^{k+3} f_k + (-1)^{k+3} f_{k+1} \\
&= (-1)^{k+3} f_{k+2}
\end{aligned}$$

65 Closed Form (Binet's Formula)

Normally this is proved by taking the characteristic equation of this sequence definition. However since we haven't proved the characteristic equation yet, see Characteristic Equation of recurrence relations, we will investigate a particular function behaviours, without any motivation.

Recall the equation from the ratio limits $x^2 - x - 1 = 0$, with solutions $x = \frac{1 \pm \sqrt{5}}{2}$.

Theorem 65.1. *If* $x^2 - x - 1 = 0$ *then* $x^n = f_n \cdot x + f_{n-1} \quad \forall n \in \mathbb{Z}^+$

Proof by induction

For n=1 and n=2

$$\begin{aligned}
x &= 1 \cdot x + 0 = f_1 \cdot x + f_0 \\
x &= 1 \cdot x + 1 = f_2 \cdot x + f_1
\end{aligned}$$

Assume true for k

$$x^k = f_k \cdot x + f_{k-1}$$

Proving for k+1

$$\begin{aligned} x^{k+1} &= x \cdot x^k \\ &= f_k \cdot x^2 + f_{k-1} \cdot x \\ &= f_k \cdot (x+1) + f_{k-1} \cdot x \\ &= (f_k + f_{k-1}) \cdot x + f_k \\ &= f_{k+1} \cdot x + f_k \end{aligned}$$

Theorem 65.2. *Let* $x_1 = \frac{1+\sqrt{5}}{2}$ *and* $x_2 = \frac{1-\sqrt{5}}{2}$ *then*

$$f_n = \frac{x_1^n - x_2^n}{\sqrt{5}} \quad \forall n \in \mathbb{Z}^+$$

Proof

Since both x_1 *and* x_2 *satisfy Theorem 65.1 we get*

$$\begin{aligned} x_1^n &= x_1 \cdot f_n + f_{n-1} \\ x_2^n &= x_2 \cdot f_n + f_{n-1} \end{aligned}$$

Subtracting both equations we get:

$$\begin{aligned} x_1^n - x_2^n &= (x_1 - x_2) \cdot f_n \\ &= (\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}) \cdot f_n \\ &= \frac{2\sqrt{5}}{2} \cdot f_n \\ &= \sqrt{5} \cdot f_n \end{aligned}$$

Therefore:

$$f_n = \frac{x_1^n - x_2^n}{\sqrt{5}}$$

66 General Fibonacci

We can slightly generalize the Fibonacci sequence by keeping the same recurrence relation, but not fixing the initial values.

$$p_1 = a \tag{231}$$
$$p_2 = b \tag{232}$$

$$p_{n+2} = p_{n+1} + p_n \quad \forall \quad n \in \mathbb{Z}^+ \tag{233}$$

The first few General Fibonacci numbers then would be:

$$a, b, a+b, a+2b, 2a+3b, 3a+5b, 5a+8b, \ldots$$

Theorem 66.1. *Let* $x_1 = \frac{1+\sqrt{5}}{2}$ *and* $x_2 = \frac{1-\sqrt{5}}{2}$ *then*

$$p_n = p_1 \cdot f_{n-2} + p_2 \cdot f_{n-1} \quad \forall\ n \geq 2$$

or in another form

$$p_n = p_1 \cdot \frac{x_1^{n-2} - x_2^{n-2}}{\sqrt{5}} + p_2 \cdot \frac{x_1^{n-1} - x_2^{n-1}}{\sqrt{5}} \quad \forall\ n \geq 2$$

Proof by Induction

For n=2 and n=3

$$p_2 = a \cdot f_0 + b \cdot f_1 = b$$
$$p_3 = a \cdot f_1 + b \cdot f_2 = a + b$$

Assume true for k-1 and k

$$p_{k-1} = a \cdot f_{k-3} + b \cdot f_{k-2}$$
$$p_k = a \cdot f_{k-2} + b \cdot f_{k-1}$$

Proving for k+1

$$\begin{aligned} p_{k+1} &= p_k + p_{k-1} \\ &= a \cdot f_{k-2} + b \cdot f_{k-1} + a \cdot f_{k-3} + b \cdot f_{k-2} \\ &= a \cdot (f_{k-2} + f_{k-3}) + b \cdot (f_{k-1} + f_{k-2}) \\ &= a \cdot f_{k-1} + b \cdot f_k \end{aligned}$$

66.1 Relation to Fibonacci

Note that if

$$p_1 = a = f_1 = 1$$
$$p_2 = b = f_2 = 1$$

Then we get back the same Fibonacci closed form as before.

$$\begin{aligned} p_n &= 1 \cdot f_{n-2} + 1 \cdot f_{n-1} \\ &= f_n \end{aligned} \tag{234}$$

Alternatively, if someone hates themselves or wants to practise their algebra skills this can be proves through just the closed form as shown below:

$$\begin{aligned} p_n &= 1 \cdot \frac{x_1^{n-2} - x_2^{n-2}}{\sqrt{5}} + 1 \cdot \frac{x_1^{n-1} - x_2^{n-1}}{\sqrt{5}} \\ &= \frac{x_1^{n-2} - x_2^{n-2} + x_1^{n-1} - x_2^{n-1}}{\sqrt{5}} \\ &= \frac{x_1^{n-2}(x_1 + 1) - x_2^{n-2}(x_2 + 1)}{\sqrt{5}} \\ &= \frac{x_1^{n-2}(x_1^2) - x_2^{n-2}(x_2^2)}{\sqrt{5}} \\ &= \frac{x_1^n - x_2^n}{\sqrt{5}} \\ &= f_n \end{aligned}$$

66.2 Relation to Lucas Numbers

Note that there exist other variations of Fibonacci numbers such as Lucas Numbers which are defined as

$$\begin{aligned} p_0 &= 2 \\ p_1 &= 1 \end{aligned} \tag{235}$$

Which implies that $p_2 = 3$.

The following lemma will become useful in simplifying the closed form of Lucas numbers.

Theorem 66.2. *Let x_1, and x_2 be solutions to $x^2 = x + 1$. $x_1 = \frac{1+\sqrt{5}}{2}$ and $x_2 = \frac{1-\sqrt{5}}{2}$ then we get*

$$\begin{aligned} 3x_1 + 1 &= \sqrt{5}x_1^2 \\ 3x_2 + 1 &= -\sqrt{5}x_2^2 \end{aligned}$$

Proof

$$\begin{aligned} 3x + 1 &= 3\frac{1 \pm \sqrt{5}}{2} + 1 \\ &= \frac{5 \pm 3\sqrt{5}}{2} \\ \sqrt{5}x^2 &= \sqrt{5}(x+1) \\ &= \sqrt{5}(\frac{1 \pm \sqrt{5}}{2} + 1) \\ &= \sqrt{5}\frac{3 \pm \sqrt{5}}{2} \\ &= \frac{3\sqrt{5} \pm 5}{2} \end{aligned}$$

Theorem 66.3. *Let $x_1 = \frac{1+\sqrt{5}}{2}$ and $x_2 = \frac{1-\sqrt{5}}{2}$ then the closed form solution for the Lucas Numbers are*

$$\begin{aligned} p_n &= 1 \cdot f_{n-2} + 3 \cdot f_{n-1} \\ &= x_1^n + x_2^n \end{aligned} \tag{236}$$

Proof

$$\begin{aligned} p_n &= 1 \cdot \frac{x_1^{n-2} - x_2^{n-2}}{\sqrt{5}} + 3 \cdot \frac{x_1^{n-1} - x_2^{n-1}}{\sqrt{5}} \\ &= \frac{x_1^{n-2} - x_2^{n-2} + 3x_1^{n-1} - 3x_2^{n-1}}{\sqrt{5}} \\ &= \frac{x_1^{n-2}(3x_1 + 1) - x_2^{n-2}(3x_2 + 1)}{\sqrt{5}} \end{aligned}$$

Using Theorem 66.2

$$\begin{aligned} &= \frac{\sqrt{5}x_1^{n-2}(x_1^2) + \sqrt{5}x_2^{n-2}(x_2^2)}{\sqrt{5}} \\ &= x_1^n + x_2^n \end{aligned}$$

67 Series

We can examine the sequential sum of this generalized Fibonacci sequence.

Theorem 67.1. *Let $P_n = p_1 + p_2 + \cdots + p_n$ and $F_n = f_1 + f_2 + \cdots + f_n$, then:*

$$P_n = f_n \cdot p_1 + F_{n-1} \cdot p_2 \qquad \forall n \geq 2$$

Proof by Induction. For base cases $n = 2$ and $n = 3$:

$$P_2 = f_2 \cdot p_1 + F_1 \cdot p_2 = 1 \cdot p_1 + 1 \cdot p_2 = p_1 + p_2$$
$$P_3 = f_3 \cdot p_1 + F_2 \cdot p_2 = 2 \cdot p_1 + (1+1) \cdot p_2 = p_1 + p_2 + p_3$$

Assume the statement holds for k:

$$P_k = f_k \cdot p_1 + F_{k-1} \cdot p_2$$

Proving for $k + 1$:

$$\begin{aligned} P_{k+1} &= P_k + p_{k+1} \\ &= (f_k \cdot p_1 + F_{k-1} \cdot p_2) + (p_k + p_{k-1}) \\ &= (f_k \cdot p_1 + F_{k-1} \cdot p_2) + \big((p_1 \cdot f_{k-2} + p_2 \cdot f_{k-1}) + (p_1 \cdot f_{k-3} + p_2 \cdot f_{k-2})\big) \\ &= (f_k + f_{k-2} + f_{k-3}) \cdot p_1 + (F_{k-1} + f_{k-1} + f_{k-2}) \cdot p_2 \\ &= (f_k + f_{k-1}) \cdot p_1 + (F_{k-1} + f_k) \cdot p_2 \\ &= f_{k+1} \cdot p_1 + F_k \cdot p_2 \end{aligned}$$

□

67.1 Fibonacci Sum

We have expressed the sum of the general Fibonacci sequence in terms of the sum to the Fibonacci sequence, so now we can investigate the sum of the Fibonacci sequence.

Theorem 67.2.

$$F_n = f_{n+2} - 1 \qquad \forall n \geq 1$$

Proof by Induction. For $n = 1$:

$$F_1 = f_3 - 1 = 2 - 1 = 1 = f_1$$

Assume the statement holds for k:

$$F_k = f_{k+2} - 1$$

Proving for $k + 1$:

$$\begin{aligned} F_{k+1} &= F_k + f_{k+1} \\ &= (f_{k+2} - 1) + f_{k+1} \\ &= (f_{k+2} + f_{k+1}) - 1 \\ &= f_{k+3} - 1 \end{aligned}$$

□

67.2 Generalized Fibonacci Sum

Combining these two theorems we obtain:

Theorem 67.3.

$$P_n = f_n \cdot p_1 + (f_{n+1} - 1) \cdot p_2 \qquad \forall n \geq 2$$

Proof. Directly substituting $F_{n-1} = f_{n+1} - 1$:

$$P_n = f_n \cdot p_1 + F_{n-1} \cdot p_2 = f_n \cdot p_1 + (f_{n+1} - 1) \cdot p_2$$

□

Note if we chose p_1 and p_2 such that $p_1 = p_2 = c$ we obtain:

$$P_n = c \cdot f_{n+2} - c \tag{237}$$

68 General Fibonacci Negative Indices

If we again ignore the restriction of only positive indices, using the functional form we can obtain the following relation.

Theorem 68.1.

$$p_{-n} = (-1)^{n+1} f_{n+2} \cdot p_1 + (-1)^n f_{n+1} \cdot p_2$$

Proof

$$\begin{aligned} p_{-n} &= f_{-n-2} \cdot p_1 + f_{-n-1} \cdot p_2 \\ &= (-1)^{n+3} f_{n+2} \cdot p_1 + (-1)^{n+4} f_{n+1} \cdot p_2 \\ &= (-1)^{n+1} f_{n+2} \cdot p_1 + (-1)^n f_{n+1} \cdot p_2 \end{aligned}$$

69 Ratio Limit

Expanding on section 63, let us consider $\lim_{n\to\infty} \frac{p_{n+\alpha}}{p_n}$

Theorem 69.1.

$$\lim_{n\to\infty} \frac{p_{n+\alpha}}{p_n} = \left(\frac{1+\sqrt{5}}{2}\right)^{\alpha}$$

Proof

$$
\begin{aligned}
K &= \lim_{n\to\infty} \frac{p_{n+\alpha}}{p_n} \\
&= \lim_{n\to\infty} \frac{p_1 \cdot \frac{x_1^{n+\alpha-2}-x_2^{n+\alpha-2}}{\sqrt{5}} + p_2 \cdot \frac{x_1^{n+\alpha-1}-x_2^{n+\alpha-1}}{\sqrt{5}}}{p_1 \cdot \frac{x_1^{n-2}-x_2^{n-2}}{\sqrt{5}} + p_2 \cdot \frac{x_1^{n-1}-x_2^{n-1}}{\sqrt{5}}} \\
&= \lim_{n\to\infty} \frac{p_1 \cdot (x_1^{n+\alpha-2} - x_2^{n+\alpha-2}) + p_2 \cdot (x_1^{n+\alpha-1} - x_2^{n+\alpha-1})}{p_1 \cdot (x_1^{n-2} - x_2^{n-2}) + p_2 \cdot (x_1^{n-1} - x_2^{n-1})}
\end{aligned}
$$

Note that since $|x_2| < 1$ *then* $\lim_{r\to\infty} x_2^r = 0$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{p_1 \cdot x_1^{n+\alpha-2} + p_2 \cdot x_1^{n+\alpha-1}}{p_1 \cdot x_1^{n-2} + p_2 \cdot x_1^{n-1}} \\
&= \lim_{n\to\infty} \frac{p_1 \cdot x_1^{\alpha} + p_2 \cdot x_1^{\alpha+1}}{p_1 \cdot 1 + p_2 \cdot x_1} \\
&= x_1^{\alpha} \cdot \lim_{n\to\infty} \frac{p_1 \cdot 1 + p_2 \cdot x_1}{p_1 \cdot 1 + p_2 \cdot x_1} \\
&= x_1^{\alpha} \\
&= \Big(\frac{1+\sqrt{5}}{2}\Big)^{\alpha}
\end{aligned}
$$

70 Sum of every other Fibonacci number

Some other sums.

70.1 Even indices

If we sum all of the even indices we obtain:

Theorem 70.1.

$$\Sigma_{i=1}^{n} p_{2i} = p_{2n+1} - p_1 \ \forall\ n \geq 1$$

Proof by Induction

For n=1

$$
\begin{aligned}
\sum_{i=1}^{1} p_{2i} &= p_3 - p_1 \\
&= p_2 + p_1 - p_1 \\
&= p_2
\end{aligned}
$$

Assume true for k

$$\sum_{i=1}^{k} p_{2i} = p_{2k+1} - p_1$$

Proving for k+1

$$\begin{aligned}\sum_{i=1}^{k+1} p_{2i} &= \sum_{i=1}^{k} p_{2i} + p_{2k+2} \\ &= (p_{2k+1} - p_1) + p_{2k+2} \\ &= (p_{2k+1} + p_{2k+2}) - p_1 \\ &= p_{2k+3} - p_1\end{aligned}$$

70.2 Odd indices

If we sum all of the odd indices we obtain:

Theorem 70.2.

$$\sum_{i=1}^{n} p_{2i-1} = p_{2n} - p_2 + p_1 \ \forall \ n \geq 1$$

Proof by Induction

For n=1

$$\begin{aligned}\sum_{i=1}^{1} p_{2i-1} &= p_2 - p_2 + p_1 \\ &= p_1\end{aligned}$$

Assume true for k

$$\sum_{i=1}^{k} p_{2i-1} = p_{2k} - p_2 + p_1$$

Proving for k+1

$$\begin{aligned}\sum_{i=1}^{k+1} p_{2i-1} &= \sum_{i=1}^{k} p_{2i-1} + p_{2k+1} \\ &= (p_{2k} - p_2 + p_1) + p_{2k+1} \\ &= (p_{2k} + p_{2k+1}) - p_2 + p_1 \\ &= p_{2k+2} - p_2 + p_1\end{aligned}$$

71 Relation of Squares

Theorem 71.1.

$$f_{2n+1}^2 - f_{2n+1}f_{2n} - f_{2n}^2 = 1$$

Proof by Induction

For n=0

$$\begin{aligned}f_1^2 - f_1 f_0 - f_0^2 &= f_1 - 0 - 0 \\ &= 1\end{aligned}$$

Assume true for k

$$f_{2k+1}^2 - f_{2k+1}f_{2k} - f_{2k}^2 = 1$$

Proving for k+1

$$\begin{aligned}
1 &= f_{2k+1}^2 - f_{2k+1}f_{2k} - f_{2k}^2 \\
&= f_{2k+1}^2 - f_{2k+1}(f_{2k+2} - f_{2k+1}) - (f_{2k+2} - f_{2k+1})^2 \\
&= f_{2k+1}^2 - f_{2k+1}f_{2k+2} + f_{2k+1}^2 - f_{2k+2}^2 - f_{2k+1}^2 + 2f_{2k+2}f_{2k+1} \\
&= f_{2k+1}^2 + f_{2k+1}f_{2k+2} - f_{2k+2}^2 \\
&= (f_{2k+3} - f_{2k+2})^2 + (f_{2k+3} - f_{2k+2})f_{2k+2} - f_{2k+2}^2 \\
&= f_{2k+3}^2 - 2f_{2k+3}f_{2k+2} + f_{2k+2}^2 + f_{2k+3}f_{2k+2} - f_{2k+2}^2 - f_{2k+2}^2 \\
&= f_{2k+3}^2 - f_{2k+3}f_{2k+2} - f_{2k+2}^2 \\
&= f_{2(k+1)+1}^2 - f_{2(k+1)+1}f_{2(k+1)} - f_{2(k+1)}^2
\end{aligned}$$

72 Sum of squared Fibonacci numbers

Theorem 72.1.

$$\sum_{i=1}^{n} p_i^2 = p_n p_{n+1} - p_1 p_2 + p_1^2 \ \forall \ n \geq 1$$

Proof by Induction

For n=1

$$\begin{aligned}
\sum_{i=1}^{1} p_i^2 &= p_1 p_2 - p_1 p_2 + p_1^2 \\
&= p_1^2
\end{aligned}$$

Assume true for k

$$\sum_{i=1}^{k} p_i^2 = p_k p_{k+1} - p_1 p_2 + p_1^2$$

Proving for k+1

$$\begin{aligned}
\sum_{i=1}^{k+1} p_i^2 &= \sum_{i=1}^{k} p_i^2 + p_{k+1}^2 \\
&= (p_k p_{k+1} - p_1 p_2 + p_1^2) + p_{k+1}^2 \\
&= (p_k p_{k+1} + p_{k+1}^2) - p_1 p_2 + p_1^2 \\
&= (p_{k+1}(p_k + p_{k+1})) - p_1 p_2 + p_1^2 \\
&= (p_{k+1} p_{k+2} - p_1 p_2 + p_1^2
\end{aligned}$$

73 Generating Functions

Let us define a function with power series coefficients defined as the Fibonacci numbers

$$S_n(x) = \sum_{k=0}^{n} f_k x^k = x + x^2 + 2x^3 + 3x^4 + 5x^5 + 8x^7 + \cdots + f_n x^n \qquad (238)$$

Note that using the sums we have showed before we get:

$$S_n(1) = f_{n+2} - 1 \tag{239}$$

73.1 Finite Series

$$\begin{aligned}
S_n(x) &= \sum_{k=0}^{n} f_k x^k \\
&= f_1 x + \sum_{k=2}^{n} f_k x^k \\
&= f_1 x + \sum_{k=2}^{n} (f_{k-1} + f_{k-2}) x^k \\
&= f_1 x + \sum_{k=2}^{n} f_{k-1} x^k + \sum_{k=2}^{n} f_{k-2} x^k \\
&= f_1 x + x \sum_{k=2}^{n} f_{k-1} x^{k-1} + x^2 \sum_{k=2}^{n} f_{k-2} x^{k-2} \\
&= f_1 x + x\Big(\sum_{k=0}^{n} f_k x^k - f_n x^n - x f_0 \Big) + x^2 \Big(\sum_{k=0}^{n} f_k x^k - f_n x^n - f_{n-1} x^{n-1} \Big) \\
&= f_1 x + x S_n(x) + x^2 S_n(x) - f_n x^{n+1} - x^2 f_0 - f_n x^{n+2} - f_{n-1} x^{n+1}
\end{aligned} \tag{240}$$

Therefore we can write

$$S_n(x)(1 - x - x^2) = f_1 x - x^2 f_0 - f_{n+1} x^{n+1} - f_n x^{n+2} \tag{241}$$

For the Fibonacci sequence since $f_0 = 0$, and $f_1 = 1$ we obtain:

$$S_n(x) = \frac{x - f_{n+1} x^{n+1} - f_n x^{n+2}}{1 - x - x^2} \tag{242}$$

73.2 Infinite Series

Therefore

$$\lim_{n \to \infty} S_n(1) = \infty \tag{243}$$

Let's try to investigate for what values of x does the series $\lim_{n\to\infty} S_n(x)$ converge.

$$
\begin{aligned}
S_\infty(x) &= \lim_{n\to\infty} S_n(x) \\
&= \sum_{k=0}^{\infty} f_k x^k \\
&= f_1 x + \sum_{k=2}^{\infty} f_k x^k \\
&= f_1 x + \sum_{k=2}^{\infty} (f_{k-1} + f_{k-2}) x^k \\
&= f_1 x + \sum_{k=2}^{\infty} f_{k-1} x^k + \sum_{k=2}^{\infty} f_{k-2} x^k \\
&= f_1 x + x \sum_{k=2}^{\infty} f_{k-1} x^{k-1} + x^2 \sum_{k=2}^{\infty} f_{k-2} x^{k-2} \\
&= f_1 x + x \sum_{k=0}^{\infty} f_k x^k - x f_0 + x^2 \sum_{k=0}^{\infty} f_k x^k \\
&= f_1 x + x S_\infty(x) + x^2 S_\infty(x)
\end{aligned}
\tag{244}
$$

Therefore we can write

$$
S_\infty(x)(1 - x - x^2) = f_1 x \tag{245}
$$

For the Fibonacci sequence since $f_1 = 1$ we obtain:

$$
S_\infty(x) = \frac{x}{1 - x - x^2} \tag{246}
$$

By the ratio convergence test [Insert reference here] we get that

$$
\lim_{k\to\infty} \frac{f_{k+1}|x|^{k+1}}{f_k |x|^k} = \phi|x| < 1 \tag{247}
$$

This implies that this series converges for

$$
|x| < \frac{1}{\phi} \tag{248}
$$

Diverges for

$$
|x| > \frac{1}{\phi} \tag{249}
$$

And since we have the generating function we can test for equality and see that the series diverges at equality as well, due to the denominator being 0.

73.3 Reciprocal power series

We can obtain a very similar relation if we define a function with inverse power series coefficients defined as the Fibonacci numbers

$$
r_n(x) = \sum_{k=0}^{n} \frac{f_k}{x^k} = \frac{1}{x} + \frac{1}{x^2} + \frac{2}{x^3} + \cdots + \frac{f_n}{x^n} \tag{250}
$$

Following the same argument we obtain the following infinite series representation.

$$r_{\infty}(x) = \frac{x}{x^2 - x - 1} for\ x > 0 \tag{251}$$

74 Base Fibonacci

TODO

75 Definition

The Tribonacci sequence is defined as:

$$\begin{aligned} t_0 &= 0 \\ t_1 &= 1 \\ t_2 &= 1 \\ t_n &= t_{n-1} + t_{n-2} + t_{n-3} \quad \forall \quad n \in \mathbb{Z}^+ \end{aligned} \tag{252}$$

The first few Tribonacci numbers then would be:

$$0, 1, 1, 2, 4, 7, 13, 24, 44, 81, \ldots$$

76 Ratios

Taking the ratios of consecutive Tribonacci numbers we find this pattern:

$$\begin{aligned} \frac{t_2}{t_1} &= 1 \\ \frac{t_3}{t_2} &= \frac{2}{1} \\ \frac{t_4}{t_3} &= \frac{2}{1} \\ \frac{t_5}{t_4} &= \frac{7}{4} \\ \frac{t_6}{t_5} &= \frac{13}{7} \\ \frac{t_7}{t_6} &= \frac{26}{13} \end{aligned}$$

Theorem 76.1. $\lim_{n\to\infty} \frac{t_{n+1}}{t_n} = \frac{1}{3} \cdot \left(1 + \sqrt[3]{19 + 3\sqrt{33}} + \sqrt[3]{19 - 3\sqrt{33}}\right)$

Proof TODO

77 Closed Form Expression

Using the characteristic equation of this sequence definition (see Characteristic Equation of recurrence relations), we have $x^3 - x^2 - x - 1 = 0$, with distinct roots of α, β, γ.

Theorem 77.1.

$$t_n = c_1 \cdot \alpha^n + c_2 \cdot \beta^n + c_3 \cdot \gamma^n \quad \forall n \in \mathbb{Z}^+$$

Where the constants c_1, c_2, c_3 are derived using the initial conditions.

Theorem 77.2.

$$t_n = \frac{\alpha^n}{-\alpha^2 + 4\alpha - 1} + \frac{\beta^n}{-\beta^2 + 4\beta - 1} + \frac{\gamma^n}{-\gamma^2 + 4\gamma - 1} \quad \forall n \in \mathbb{Z}^+$$

Proof. Using the initial conditions:

$$\begin{aligned} c_1 + c_2 + c_3 &= 0 && (n=0) \\ \alpha c_1 + \beta c_2 + \gamma c_3 &= 1 && (n=1) \\ \alpha^2 c_1 + \beta^2 c_2 + \gamma^2 c_3 &= 1 && (n=2) \end{aligned}$$

Rewriting this as a matrix equation gives a classic Vandermonde system:

$$\begin{bmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Using Gaussian elimination:

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ \alpha & \beta & \gamma & 1 \\ \alpha^2 & \beta^2 & \gamma^2 & 1 \end{array}\right) &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & \beta-\alpha & \gamma-\alpha & 1 \\ 0 & \beta^2-\alpha^2 & \gamma^2-\alpha^2 & 1 \end{array}\right) \\ &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{\gamma-\alpha}{\beta-\alpha} & \frac{1}{\beta-\alpha} \\ 0 & 1 & \frac{\gamma^2-\alpha^2}{\beta^2-\alpha^2} & \frac{1}{\beta^2-\alpha^2} \end{array}\right) \\ &\sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{\gamma-\alpha}{\beta-\alpha} & \frac{1}{\beta-\alpha} \\ 0 & 0 & \frac{\gamma^2-\alpha^2}{\beta^2-\alpha^2} - \frac{\gamma-\alpha}{\beta-\alpha} & \frac{1}{\beta^2-\alpha^2} - \frac{1}{\beta-\alpha} \end{array}\right) \end{aligned}$$

Therefore we can solve for c_3:

$$\begin{aligned} c_3 &= \frac{\frac{1}{\beta^2-\alpha^2} - \frac{1}{\beta-\alpha}}{\frac{\gamma^2-\alpha^2}{\beta^2-\alpha^2} - \frac{\gamma-\alpha}{\beta-\alpha}} \\ &= \frac{(\beta-\alpha) - (\beta^2-\alpha^2)}{(\gamma^2-\alpha^2)(\beta-\alpha) - (\gamma-\alpha)(\beta^2-\alpha^2)} \\ &= \frac{\beta-\alpha}{(\beta-\alpha)(\gamma-\alpha)} \cdot \frac{1-(\beta+\alpha)}{(\gamma+\alpha)-(\beta+\alpha)} \\ &= \frac{1-(\beta+\alpha)}{(\gamma-\beta)(\gamma-\alpha)} \\ &= \frac{1-(\beta+\alpha+\gamma)+\gamma}{(\gamma-\beta)(\gamma-\alpha)} \end{aligned}$$

Since α, β and γ satisfy $x^3 - x^2 - x - 1 = 0$, Vieta's formulas give:

$$\alpha + \beta + \gamma = 1, \qquad \alpha\beta\gamma = 1$$

Using these identities:

$$\begin{aligned} c_3 &= \frac{\gamma}{\gamma^2 - (\alpha+\beta)\gamma + \alpha\beta} \\ &= \frac{\gamma}{\gamma^2 - (\alpha+\beta+\gamma)\gamma + \gamma^2 + \frac{\alpha\beta\gamma}{\gamma}} \\ &= \frac{\gamma}{2\gamma^2 - \gamma + \frac{1}{\gamma}} \\ &= \frac{1}{2\gamma - 1 + \frac{1}{\gamma^2}} \end{aligned}$$

Rewriting $\frac{1}{\gamma^2}$ using $\gamma^3 - \gamma^2 - \gamma - 1 = 0 \implies \frac{1}{\gamma^2} = 2\gamma - \gamma^2$:

$$c_3 = \frac{1}{-\gamma^2 + 4\gamma - 1}$$

By symmetry:

$$c_1 = \frac{1}{-\alpha^2 + 4\alpha - 1}, \qquad c_2 = \frac{1}{-\beta^2 + 4\beta - 1}$$

Hence:

$$t_n = \frac{\alpha^n}{-\alpha^2 + 4\alpha - 1} + \frac{\beta^n}{-\beta^2 + 4\beta - 1} + \frac{\gamma^n}{-\gamma^2 + 4\gamma - 1} \quad \forall n \in \mathbb{Z}^+$$

□

78 Negative Indices

If we ignore the condition of $n > 0$ we can can use the functional form definition of Tribonacci numbers to define negative indices:

$$t_{n-3} = t_n - t_{n-1} - t_{n-2} \tag{253}$$

Therefore the first few will be:

$$\begin{aligned}
t_{-1} &= t_2 - t_1 - t_0 = 1 - 1 - 0 = 0 \\
t_{-2} &= t_1 - t_0 - t_{-1} = 1 - 0 - 0 = 1 \\
t_{-3} &= t_0 - t_{-1} - t_{-2} = 0 - 0 - 1 = -1 \\
t_{-4} &= t_{-1} - t_{-2} - t_{-3} = 0 - 1 - (-1) = 0 \\
t_{-5} &= t_{-2} - t_{-3} - t_{-4} = 1 - (-1) - 0 = 2 \\
t_{-6} &= t_{-3} - t_{-4} - t_{-5} = -1 - 0 - 2 = -3
\end{aligned}$$

Theorem 78.1.

$$t_{-n} = t_{n-1}^2 - t_{n-2} \cdot t_n$$

Proof by induction. Assume true for $k, k+1$, and $k+2$:

$$\begin{aligned}
t_{-k} &= t_{k-1}^2 - t_{k-2} \cdot t_k \\
t_{-k-1} &= t_{k-2}^2 - t_{k-3} \cdot t_{k-1} \\
t_{-k-2} &= t_{k-3}^2 - t_{k-4} \cdot t_{k-2}
\end{aligned}$$

Using the linear recurrence $t_{-k-3} = t_{-k} - t_{-k-1} - t_{-k-2}$, algebraic substitution confirms the relation for $k+3$. □

Abstract

We present a rigorous, self-contained mathematical analysis and derivation of Project Euler Problem 301. By employing Bouton's classic theorem from Combinatorial Game Theory, we express the winning/losing positions of three-heap Nim via the bitwise XOR sum (Nim-sum). We analyze the specific family of configurations $(n, 2n, 3n)$ and establish an exact algebraic bridge between bitwise addition and standard arithmetic, reducing the condition $X(n, 2n, 3n) = 0$ to carryless addition $n \,\&\, 2n = 0$. We prove that this condition is equivalent to the absence of consecutive ones in the binary representation of n, and we enumerate such integers on the interval $[1, 2^{30}]$ via a combinatorial bijection with the Fibonacci sequence, arriving at the exact count $\mathcal{N}(2^{30}) = F_{32} = 2\,178\,309$.

Contents

79 Introduction and Problem Formulation

79.1 Rules of Impartial Normal-Play Nim

Nim is a classic impartial game played by two players with a finite collection of heaps containing discrete tokens (or stones). The game operates under the **normal play convention**:

(i) Players alternate turns.

(ii) On each turn, the active player chooses a single non-empty heap and removes any positive integer number of stones from it (up to the entire heap).

(iii) The game ends when no stones remain in any heap.

(iv) The first player unable to make a valid move (because all heaps are empty) loses; equivalently, the player who makes the final move wins.

A game state with three heaps is denoted by the triple $(n_1, n_2, n_3) \in \mathbb{N}_0^3$, where n_i represents the number of stones in the i-th heap.

79.2 Combinatorial Game States: $\mathcal{P}$-Positions and $\mathcal{N}$-Positions

In Combinatorial Game Theory (CGT), every position in a finite impartial game without cycles belongs to one of two mutually exclusive classes:

- **$\mathcal{P}$-positions (Previous-player winning / Losing for the next player):** Any valid move from a state in $\mathcal{P}$ leads to a state in $\mathcal{N}$. The terminal state $(0, 0, 0)$ is in $\mathcal{P}$.
- **$\mathcal{N}$-positions (Next-player winning / Winning for the next player):** There exists at least one legal move from any state in $\mathcal{N}$ to some state in $\mathcal{P}$.

The characteristic evaluation function $X(n_1, n_2, n_3)$ is defined as:

$$X(n_1, n_2, n_3) = \begin{cases} 0, & \text{if } (n_1, n_2, n_3) \in \mathcal{P} \quad \text{(second player can force a win)}, \\ \neq 0, & \text{if } (n_1, n_2, n_3) \in \mathcal{N} \quad \text{(first player can force a win)}. \end{cases} \tag{254}$$

79.3 Example: The Winning Strategy from Starting Position $(1, 2, 3)$

To illustrate the mechanism of $\mathcal{P}$-positions and the winning strategy, consider the starting state $(1, 2, 3)$. We will show that $(1, 2, 3) \in \mathcal{P}$, meaning that the second player can always force a win regardless of the first player's initial move.

The first player has six possible opening moves from $(1, 2, 3)$:

1. **Reduce Heap 1 ($1 \to 0$):** The state becomes $(0, 2, 3)$.

   Winning Response by Player 2: Player 2 reduces Heap 3 from $3 \to 2$, leaving $(0, 2, 2)$. This is a two-heap symmetric position. Whatever Player 1 takes from one heap of size 2, Player 2 mirrors by taking the exact same amount from the other heap, guaranteeing the last move.

2. **Reduce Heap 2 ($2 \to 1$):** The state becomes $(1, 1, 3)$.

   Winning Response by Player 2: Player 2 removes all stones from Heap 3 ($3 \to 0$), leaving $(1, 1, 0)$. Again, Player 2 adopts the mirror strategy on the two equal heaps of size 1.

3. **Reduce Heap 2 ($2 \to 0$):** The state becomes $(1, 0, 3)$.

   Winning Response by Player 2: Player 2 reduces Heap 3 from $3 \to 1$, leaving $(1, 0, 1)$, securing the symmetric mirror state.

4. **Reduce Heap 3 ($3 \to 2$):** The state becomes $(1, 2, 2)$.

   Winning Response by Player 2: Player 2 takes the single stone from Heap 1 ($1 \to 0$), leaving $(0, 2, 2)$.

5. **Reduce Heap 3 ($3 \to 1$):** The state becomes $(1, 2, 1)$.

   Winning Response by Player 2: Player 2 removes all stones from Heap 2 ($2 \to 0$), leaving $(1, 0, 1)$.

6. **Reduce Heap 3 ($3 \to 0$):** The state becomes $(1, 2, 0)$.

   Winning Response by Player 2: Player 2 reduces Heap 2 from $2 \to 1$, leaving $(1, 1, 0)$.

In every single branch, Player 2 transitions the game to a state of two equal non-empty heaps $(k, k, 0)$, from which any move by Player 1 allows Player 2 to maintain equality or win immediately. An example sequence of optimal play is:

$$\begin{aligned} \text{Player 1 moves:} \quad & (1, 2, 3) \longrightarrow (1, 2, 1) \\ \text{Player 2 responds:} \quad & (1, 2, 1) \longrightarrow (1, 0, 1) \\ \text{Player 1 moves:} \quad & (1, 0, 1) \longrightarrow (0, 0, 1) \\ \text{Player 2 responds:} \quad & (0, 0, 1) \longrightarrow (0, 0, 0) \quad \text{(Player 2 wins!)} \end{aligned}$$

Hence, $X(1, 2, 3) = 0$.

79.4 The Target Question

Project Euler Problem 301 asks:

$$\text{For how many positive integers } n \leq 2^{30} \text{ does } X(n, 2n, 3n) = 0? \tag{255}$$

80 Bouton's Theorem and the Nim-Sum

80.1 Definition: The Nim-Sum

Definition 80.1 (Nim-Sum)**.** *Let $a, b \in \mathbb{N}_0$ have binary expansions:*

$$a = \sum_{i=0}^{\infty} a_i 2^i, \quad b = \sum_{i=0}^{\infty} b_i 2^i \qquad (a_i, b_i \in \{0, 1\}). \tag{256}$$

*The **Nim-sum** of a and b, denoted $a \oplus b$, is defined by bitwise addition modulo 2:*

$$a \oplus b = \sum_{i=0}^{\infty} (a_i \oplus b_i) 2^i, \quad \text{where } a_i \oplus b_i = (a_i + b_i) \bmod 2. \tag{257}$$

The algebraic structure $(\mathbb{N}_0, \oplus)$ is an infinite elementary abelian 2-group:

(i) **Associativity and Commutativity:** $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ and $a \oplus b = b \oplus a$.

(ii) **Identity Element:** $a \oplus 0 = a$ for all $a \in \mathbb{N}_0$.

(iii) **Self-Inversion:** $a \oplus a = 0$ for all $a \in \mathbb{N}_0$.

80.2 Bouton's Theorem

Theorem 80.1 (Bouton, 1901)**.** *In an m-heap game of Nim with heap sizes $(h_1, h_2, \ldots, h_m) \in \mathbb{N}_0^m$, a position is a $\mathcal{P}$-position (losing for the player whose turn it is) if and only if the Nim-sum of all heap sizes is zero:*

$$\bigoplus_{j=1}^{m} h_j = 0. \tag{258}$$

Equivalently, $X(h_1, h_2, \ldots, h_m) = \bigoplus_{j=1}^{m} h_j$.

Proof. Let $\mathcal{Z} = \left\{(h_1, \ldots, h_m) \in \mathbb{N}_0^m : \bigoplus_{j=1}^{m} h_j = 0\right\}$. We prove by complete induction on the total stone count $S = \sum_{j=1}^{m} h_j$ that $\mathcal{Z}$ satisfies the axiomatic properties of $\mathcal{P}$-positions:

1. **Base Case ($S = 0$):** The only state is $(0, \ldots, 0)$. Its Nim-sum is $0 \oplus \cdots \oplus 0 = 0 \in \mathcal{Z}$. The active player has no moves and loses, so $(0, \ldots, 0) \in \mathcal{P}$.

2. **Every move from $\mathcal{Z}$ leads outside $\mathcal{Z}$ ($\mathcal{P} \to \mathcal{N}$):** Let $(h_1, \ldots, h_m) \in \mathcal{Z}$, so $\bigoplus_{j=1}^{m} h_j = 0$. A legal move strictly decreases one heap, say $h_k \to h_k'$ with $0 \leq h_k' < h_k$. The new Nim-sum V' is:

$$V' = \left(\bigoplus_{j \neq k} h_j\right) \oplus h_k' = \left(\bigoplus_{j=1}^{m} h_j\right) \oplus h_k \oplus h_k' = 0 \oplus h_k \oplus h_k' = h_k \oplus h_k'. \tag{259}$$

Since $h'_k \neq h_k$, $h_k \oplus h'_k \neq 0$. Thus, $V' \neq 0$, meaning the new state is not in $\mathcal{Z}$.

3. **From any state not in $\mathcal{Z}$, there exists a move into $\mathcal{Z}$ ($\mathcal{N} \to \mathcal{P}$):** Suppose $(h_1, \ldots, h_m) \notin \mathcal{Z}$, so $V = \bigoplus_{j=1}^{m} h_j \neq 0$. Let $d = \lfloor \log_2 V \rfloor$ denote the most significant bit (MSB) of V, so bit d of V is 1. Since bit d of V is the parity of the d-th bits of all h_j, there exists at least one heap h_k whose d-th bit is 1.

   We choose the replacement size $h'_k = h_k \oplus V$.

   - For all bit indices $i > d$, the i-th bit of V is 0, so bit i of h'_k matches bit i of h_k.
   - At bit index d, bit d of h_k is 1 and bit d of V is 1, so bit d of h'_k is $1 \oplus 1 = 0$.

   Consequently, $h'_k < h_k$, so reducing heap k to h'_k is a valid Nim move. The new Nim-sum is:

$$V' = V \oplus h_k \oplus h'_k = V \oplus h_k \oplus (h_k \oplus V) = 0. \tag{260}$$

   Thus the resulting state is in $\mathcal{Z}$.

By mathematical induction, $\mathcal{Z} = \mathcal{P}$, completing the proof. □

81 Algebraic Reduction of $X(n, 2n, 3n) = 0$

81.1 Condition Equivalence

For the configuration $(n, 2n, 3n)$, Theorem 80.1 states:

$$X(n, 2n, 3n) = 0 \iff n \oplus 2n \oplus 3n = 0. \tag{261}$$

Using the self-inverting group property of $\oplus$:

$$n \oplus 2n \oplus 3n = 0 \iff n \oplus 2n = 3n. \tag{262}$$

Observing that $3n = n + 2n$ in ordinary arithmetic, this becomes:

$$n \oplus 2n = n + 2n. \tag{263}$$

81.2 Arithmetic Addition vs. Bitwise XOR

Lemma 81.1 (Arithmetic and Bitwise Addition Identity)**.** *For any $a, b \in \mathbb{N}_0$,*

$$a + b = (a \oplus b) + 2(a \,\&\, b), \tag{264}$$

where $\&$ denotes the bitwise AND operator.

Proof. Let $a = \sum_{i=0}^{\infty} a_i 2^i$ and $b = \sum_{i=0}^{\infty} b_i 2^i$ be the binary representations of a and b, with $a_i, b_i \in \{0, 1\}$. For each single bit position $i \geq 0$, we test all four possible binary assignments for (a_i, b_i):

a_i	b_i	$a_i + b_i$	$a_i \oplus b_i$	$a_i \,\&\, b_i$	$(a_i \oplus b_i) + 2(a_i \,\&\, b_i)$
0	0	0	0	0	$0 + 2(0) = 0$
0	1	1	1	0	$1 + 2(0) = 1$
1	0	1	1	0	$1 + 2(0) = 1$
1	1	2	0	1	$0 + 2(1) = 2$

In every case, $a_i + b_i = (a_i \oplus b_i) + 2(a_i \,\&\, b_i)$ holds identically. Multiplying by 2^i and summing over all $i \geq 0$:

$$\sum_{i=0}^{\infty}(a_i + b_i)2^i = \sum_{i=0}^{\infty}(a_i \oplus b_i)2^i + 2\sum_{i=0}^{\infty}(a_i \,\&\, b_i)2^i. \tag{265}$$

By linearity of summation, this yields $a + b = (a \oplus b) + 2(a \,\&\, b)$. □

Corollary 81.1.1 (Carryless Addition Criterion)**.** *For any non-negative integers $a, b \in \mathbb{N}_0$,*

$$a \oplus b = a + b \iff a \,\&\, b = 0. \tag{266}$$

Proof. From Lemma 81.1, $(a + b) - (a \oplus b) = 2(a \,\&\, b)$. Since $a \,\&\, b \geq 0$, the difference vanishes if and only if $a \,\&\, b = 0$. □

Applying Corollary 81.1.1 to $a = n$ and $b = 2n$:

$$X(n, 2n, 3n) = 0 \iff n \,\&\, (2n) = 0. \tag{267}$$

82 The "No Consecutive Ones" Bitwise Characterization

82.1 Bit-Shift Analysis

Let $n \in \mathbb{N}$ have binary representation $n = \sum_{i=0}^{\infty} n_i 2^i$ with $n_i \in \{0, 1\}$. Multiplying n by 2 corresponds to an arithmetic left-shift by 1 bit:

$$2n = \sum_{i=0}^{\infty} n_i 2^{i+1} = \sum_{j=1}^{\infty} n_{j-1} 2^j. \tag{268}$$

Thus, the i-th bit of $2n$ is:

$$(2n)_0 = 0, \qquad (2n)_i = n_{i-1} \quad \text{for all } i \geq 1. \tag{269}$$

82.2 Bitwise AND Evaluation

The i-th bit of the bitwise AND product $n \,\&\, (2n)$ is:

$$(n \,\&\, 2n)_0 = n_0 \cdot (2n)_0 = n_0 \cdot 0 = 0, \tag{270}$$

$$(n \,\&\, 2n)_i = n_i \cdot (2n)_i = n_i \cdot n_{i-1} \quad \text{for all } i \geq 1. \tag{271}$$

Therefore, $n \,\&\, (2n) = 0$ if and only if every bit of the product is 0:

$$n \,\&\, (2n) = 0 \iff n_i \cdot n_{i-1} = 0 \quad \text{for all } i \geq 1. \tag{272}$$

Theorem 82.1 (Bit-Pattern Characterization)**.** *A positive integer n satisfies $X(n, 2n, 3n) = 0$ if and only if the binary expansion of n contains no adjacent ones (that is, n contains no substring `11` in base 2).*

83 Combinatorial Enumeration via Fibonacci Numbers

83.1 Binary Strings Without Consecutive Ones

Let $\mathcal{B}_k$ denote the set of binary words of length $k \geq 0$ over $\{0, 1\}$ containing no adjacent ones:

$$\mathcal{B}_k = \{w \in \{0,1\}^k : w \text{ does not contain the substring } \texttt{11}\}. \tag{273}$$

Let $a_k = |\mathcal{B}_k|$.

- **Base Cases:**
  - $k = 0$: The empty word ε contains no $\texttt{11} \implies a_0 = 1$.
  - $k = 1$: The words $\texttt{"0"}$ and $\texttt{"1"}$ both contain no $\texttt{11} \implies a_1 = 2$.
- **Recurrence for $k \geq 2$:** We partition $\mathcal{B}_k$ according to the prefix of each word $w \in \mathcal{B}_k$:
  1. **Words starting with $\texttt{0}$:** Any such word is of the form $\texttt{0}u$, where $u \in \mathcal{B}_{k-1}$. There are a_{k-1} such words.
  2. **Words starting with $\texttt{1}$:** Since $\texttt{11}$ is forbidden, the second character must be $\texttt{0}$. Any such word is of the form $\texttt{10}v$, where $v \in \mathcal{B}_{k-2}$. There are a_{k-2} such words.

  Since these two subsets are disjoint and partition $\mathcal{B}_k$:

$$a_k = a_{k-1} + a_{k-2} \quad \text{for all } k \geq 2. \tag{274}$$

83.2 Connection to the Fibonacci Sequence

Recall the standard Fibonacci sequence $(F_m)_{m \geq 1}$:

$$F_1 = 1, \quad F_2 = 1, \quad F_m = F_{m-1} + F_{m-2} \quad (m \geq 3). \tag{275}$$

Lemma 83.1. *For every integer $k \geq 0$,*

$$a_k = F_{k+2}. \tag{276}$$

Proof. We proceed by induction on k:

- For $k = 0$: $a_0 = 1 = F_2 = F_{0+2}$.
- For $k = 1$: $a_1 = 2 = F_3 = F_{1+2}$.
- Inductive step: Assume $a_j = F_{j+2}$ holds for all $j < k$ ($k \geq 2$). Then:

$$a_k = a_{k-1} + a_{k-2} = F_{k+1} + F_k = F_{k+2}. \tag{277}$$

By mathematical induction, $a_k = F_{k+2}$ for all $k \geq 0$. □

84 Counting Solutions on $[1, 2^K]$

84.1 Integers in the Half-Open Interval $[0, 2^K)$

Every integer n in the range $0 \leq n < 2^K$ can be represented as a binary string of length exactly K (by prepending leading zeros if necessary). Since leading zeros do not introduce `11` substrings, there is a natural bijection between $\{n \in \mathbb{N}_0 : 0 \leq n < 2^K \text{ and } n \,\&\, 2n = 0\}$ and $\mathcal{B}_K$. By Lemma 83.1:

$$\#\{n \in \mathbb{N}_0 : 0 \leq n < 2^K \text{ and } n \,\&\, 2n = 0\} = a_K = F_{K+2}. \tag{278}$$

84.2 Exclusion of $n = 0$

The set above includes $n = 0$ ($0 \,\&\, 0 = 0$). Removing $n = 0$ leaves:

$$\#\{n \in \mathbb{N} : 1 \leq n < 2^K \text{ and } n \,\&\, 2n = 0\} = F_{K+2} - 1. \tag{279}$$

84.3 Inclusion of the Upper Endpoint $n = 2^K$

Consider the boundary integer $n = 2^K$:

- The binary expansion of 2^K is $1\underbrace{00\ldots00}_{K \text{ zeros}}$.
- This representation contains only a single 1-bit, and thus contains no adjacent ones.
- Directly checking: $2^K \,\&\, (2 \cdot 2^K) = 2^K \,\&\, 2^{K+1} = 0$.

Hence, $n = 2^K$ is always a valid solution.

84.4 Total Solution Count $\mathcal{N}(2^K)$

Summing the contributions:

$$\mathcal{N}(2^K) = (F_{K+2} - 1) + 1 = F_{K+2}. \tag{280}$$

Theorem 84.1 (Closed-Form Solution)**.** *For any integer $K \geq 1$, the number of positive integers $n \leq 2^K$ such that $X(n, 2n, 3n) = 0$ is given by the $(K+2)$-th Fibonacci number:*

$$\mathcal{N}(2^K) = F_{K+2}. \tag{281}$$

85 Exact Numerical Evaluation for $K = 30$

Setting $K = 30$ in Theorem 84.1, the total number of solutions is:

$$\mathcal{N}(2^{30}) = F_{30+2} = F_{32}. \tag{282}$$

85.1 Table of Fibonacci Numbers (F_1 to F_{32})

m	F_m	m	F_m	m	F_m	m	F_m
1	1	9	34	17	1 597	25	75 025
2	1	10	55	18	2 584	26	121 393
3	2	11	89	19	4 181	27	196 418
4	3	12	144	20	6 765	28	317 811
5	5	13	233	21	10 946	29	514 229
6	8	14	377	22	17 711	30	832 040
7	13	15	610	23	28 657	31	1 346 269
8	21	16	987	24	46 368	**32**	**2 178 309**

85.2 Final Conclusion

The number of positive integers $n \leq 2^{30}$ for which $X(n, 2n, 3n) = 0$ is:

$$\boxed{\mathbf{2\,178\,309}}$$

86 PascalRowSum

Prove that:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + ... + \binom{n}{n} = 2^n$$

Draw Pascal's Triangle.

$$\binom{0}{0}$$

$$\binom{1}{0} \quad \binom{1}{1}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

Notice, the sum of the nth row seems to be 2^n.

86.0.1 Sidenote 1

Prove that

$$\binom{p}{0} = \binom{p+1}{0}$$

Proof:

$$\binom{p}{0} = \frac{p!}{0! \cdot (p-0)!} = \frac{p!}{p!} = 1 = \frac{(p+1)!}{0! \cdot (p+1-0)!} = \binom{p+1}{0}$$

86.0.2 Sidenote 2

Prove that

$$\binom{p}{p} = \binom{p+1}{p+1}$$

Proof:

$$\binom{p}{p} = \frac{p!}{0! \cdot (p-0)!} = \frac{p!}{p!} = 1 = \frac{(p+1)!}{0! \cdot (p+1-0)!} = \binom{p+1}{p+1}$$

86.0.3 Sidenote 3

Prove that

$$\binom{p}{q} + \binom{p}{q+1} = \binom{p+1}{q+1}$$

Proof:

$$\begin{aligned}\binom{p}{q} + \binom{p}{q+1} &= \frac{p!}{q! \cdot (p-q)!} + \frac{p!}{(q+1)! \cdot (p-q-1)!} \\ &= \frac{p! \cdot (q+1)}{(q+1)! \cdot (p-q)!} + \frac{p! \cdot (p-q)}{(q+1)! \cdot (p-q)!}\end{aligned}$$

$$= \frac{p! \cdot (q+1+p-q)}{(q+1)! \cdot (p-q)!} = \frac{p! \cdot (p+1)}{(q+1)! \cdot (p-q)!}$$
$$= \frac{(p+1)!}{(q+1)! \cdot (p-q)!} = \binom{p+1}{q+1}$$

Using induction:
for n=0:

$$\binom{0}{0} = \frac{0!}{0! \cdot 0!} = 1 = 2^0$$

Assume n=k works:

$$\binom{k}{0} + \binom{k}{1} + \binom{k}{2} + ... + \binom{k}{k-2} + \binom{k}{k-1} + \binom{k}{k} = 2^k$$

Prove for n=k+1

$$\binom{k+1}{0} + \binom{k+1}{1} + \binom{k+1}{2} + ... + \binom{k+1}{k-1} + \binom{k+1}{k} + \binom{k+1}{k+1}$$

Using sidenote 1, replace $\binom{k+1}{0}$ with $\binom{k}{0}$

Using sidenote 2, replace $\binom{k+1}{k+1}$ with $\binom{k}{k}$

Using sidenote 2, replace $\binom{k+1}{q+1}$ with $\binom{k}{q} + \binom{k}{q+1}$

$$= \binom{k}{0}+\binom{k}{0}+\binom{k}{1}+\binom{k}{1}+\binom{k}{2}+...+\binom{k}{k-2}+\binom{k}{k-1}+\binom{k}{k-1}+\binom{k}{k}+\binom{k}{k}$$

$$= 2 \cdot (\binom{k}{0} + \binom{k}{1} + \binom{k}{2} + ... + \binom{k}{k-2} + \binom{k}{k-1} + \binom{k}{k}) = 2 \cdot (2^k) = 2^{k+1}$$

Therefore, proven by induction.

87 Geometric Series

Prove that:

$$\sum_{i=1}^{n} a \cdot r^{i-1} = \frac{a \cdot (1 - r^{n+1})}{1 - r}$$

Proof:
Define S_n such that:

$$S_n = a + ar + ar^2 + ... + ar^{n-1} + ar^n$$

Note that

$$r \cdot S_n = ar + ar^2 + ... + ar^{n-1} + ar^n + ar^{n+1}$$

Such that

$$S_n - r \cdot S_n = a - ar^{n+1}$$

$$=> S_n = \frac{a \cdot (1 - r^{n+1})}{1 - r}$$

If $|r| < 1$:
Then

$$\sum_{i=1}^{\infty} a \cdot r^{i-1} = \lim_{n \to +\infty} \frac{a \cdot (1 - r^{n+1})}{1 - r} = \frac{a}{1 - r}$$

Note that by taking derivatives of both sides and a bit of reindexing:

$$\sum_{i=0}^{\infty} \frac{d}{dr} a \cdot r^i = \sum_{i=0}^{\infty} i \cdot a \cdot r^{i-1} = 0 + \sum_{i=1}^{\infty} i \cdot a \cdot r^{i-1} = \frac{a}{(1-r)^2}$$

$$=> \sum_{i=1}^{\infty} i \cdot r^i = \frac{r}{(1-r)^2} \text{ when } |r| < 1$$

88 Arithmetic Sequence

Definition 88.1. *Given a starting value of a_0 and a constant difference value of d the arithmetic sequence is defined as follows.*

$$\boxed{a_n = a_0 + n \cdot d}$$

88.1 Sum of Arithmetic sequence

Theorem 88.1.

$$\sum_{k=0}^{n} a_k = \frac{(n+1)}{2}(2a_0 + n \cdot d) \tag{283}$$

Proof

Define S_n such that:

$$S_n = \sum_{k=0}^{n} a_k$$

Note that

$$2S_n = \sum_{k=0}^{n} a_k + \sum_{k=0}^{n} a_{n-k} = \sum_{k=0}^{n} \Big(a_k + a_{n-k}\Big)$$

Using the definition we see $a_k + a_{n-k} = \big(a_0 + k \cdot d\big) + \big(a_0 + (n-k) \cdot d\big) = 2a_0 + n \cdot d$ which does not depend on k

$$2S_n = \sum_{k=0}^{n} \Big(2a_0 + n \cdot d\Big) = (n+1)\Big(2a_0 + n \cdot d\Big)$$

$$S_n = \frac{(n+1)}{2}(2a_0 + n \cdot d)$$

88.2 Example: Sum of Natural Numbers

Prove that

$$\sum_{i=0}^{n} i = 0 + 1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$$

Proof: Set $a_0 = 0$ and $d = 1$

$$\sum_{i=0}^{n} i = \frac{(n+1)}{2}(n)$$

89 Technique Recursive sum of powers

To solve for $\sum_{i=1}^{n} i^p$ consider $\sum_{i=1}^{n} (1+i)^{p+1} - i^{p+1} = (1+n)^{p+1} - 1$

89.1 Sum of Squares

Let p=2

$$\sum_{i=1}^{n}(1+i)^3 - i^3 = \sum_{i=1}^{n} 1 + 3i + 3i^2 + i^3 - i^3 = \sum_{i=1}^{n} 1 + 3i + 3i^2$$

and

$$\sum_{i=1}^{n}(1+i)^3 - i^3 = (n+1)^3 - 1$$

Hence

$$\sum_{i=1}^{n} 1 + 3\sum_{i=1}^{n} i + 3\sum_{i=1}^{n} i^2 = (n+1)^3 - 1$$

$$n + 3\frac{n(n-1)}{2} + 3\sum_{i=1}^{n} i^2 = (n+1)^3 - 1$$

$$3\sum_{i=1}^{n} i^2 = n^3 + 3n^2 + 3n + 1 - n - 3\frac{n(n-1)}{2}$$

$$3\sum_{i=1}^{n} i^2 = n(n+1)(n+\frac{1}{2})$$

$$\boxed{\sum_{i=1}^{n} i^2 = \frac{n}{3} \cdot (n+1) \cdot (n+\frac{1}{2})}$$

89.2 Sum of Cubes

Let p=3

$$\sum_{i=1}^{n}(1+i)^4 - i^4 = \sum_{i=1}^{n} 1 + 4i + 6i^2 + 4i^3 + i^4 - i^4 = \sum_{i=1}^{n} 1 + 4i + 6i^2 + 4i^3$$

and

$$\sum_{i=1}^{n}(1+i)^4 - i^4 = (n+1)^4 - 1$$

Hence

$$\sum_{i=1}^{n} 1 + 4\sum_{i=1}^{n} i + 6\sum_{i=1}^{n} i^2 + 4\sum_{i=1}^{n} i^3 = (n+1)^4 - 1$$

$$n + 4\frac{n(n-1)}{2} + 6\frac{n}{3} \cdot (n+1) \cdot (n+\frac{1}{2}) + 4\sum_{i=1}^{n} i^3 = (n+1)^4 - 1$$

$$4\sum_{i=1}^{n} i^3 = n^4 + 4n^3 + 6n^2 + 4n - n - 2n(n-1) - n \cdot (n+1) \cdot (2n+1)$$

Simplifying

$$4\sum_{i=1}^{n} i^3 = n^4 + 2n^3 + n^2$$

$$4\sum_{i=1}^{n} i^3 = n^2(n^2 + 2n + 1) = n^2(n+1)^2$$

$$\boxed{\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2}$$

90 The Quadratic Equation

Given a quadratic polynomial $ax^2 + bx + c = 0$ The two solutions of x that satisfy this equation are given by the quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{284}$$

91 Algebraic Proof

The most straight-forward proof is given by a technique referred to as 'Completing the Square'.

$$ax^2 + bx + c = 0$$

Dividing both sides by a

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Adding and subtracting $\frac{b^2}{(2a)^2}$

$$x^2 + \frac{b}{a}x + \frac{b^2}{(2a)^2} + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Recognizing the 'square'

$$(x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Rearranging and solving

$$(x + \frac{b}{2a})^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x = -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

92 History

Scipione del Ferro is thought to have been the first to discover the general solution to the cubic equation, but did not publish it, and kept it a secret until his deathbed, telling his student Antonio Fior. Fior then challenged Niccolo Fontana, also known as Tartaglia, in 1535 to solve a list of 30 depressed cubics. On the night of February 13, 1535, Tartaglia discovered the solution.

Tartaglia revealed the secret to Gerolamo Cardano on March 25, 1539, with the oath of Cardano to never publish the findings. Cardano, along with his pupil Ludovico Ferrari, discovered the solution of the general cubic equation.

While inspecting del Ferro's papers, they came across a solution to the depressed cubic in del Ferro's own handwriting. Cardano and Ferrari were no longer prohibited from publishing their results.

In 1545, Cardano published his book Ars Magna, the "Great Art."In it, he published the solution to the depressed cubic, with a preface crediting del Ferro with the original solution. He also published his solution of the general cubic and also Ferrari's solution of the quartic.

Lagrange published a paper in 1771, Reflections on the Algebraic Theory of Equations, in which he described why the techniques used to solve the quadratic, cubic and quartic equations could not be applied to solve the quintic equation.

Paolo Ruffini attempted to prove that the quintic equation was insolvable. He published 6 proofs on the subject from 1799 to 1813. However, other mathematicians found his work too long and difficult to understand, except for Cauchy. These proofs did end up having a few gaps in them, but were on the right track.

Niels Henrik Abel published his proof that the quintic equation was insolvable in 1824. Abel is credited for finally putting this question to rest.

Evariste Galois, who died at the age of 20 in 1832 in a duel, studied attempted to generalize how to determine when the roots could be solved in terms of radicals. He became the first mathematician to use the word 'group' and defined the normal subgroup.

93 Quadratic

Definition 93.1. *A quadratic polynomial*

$$ax^2 + bx + c = 0$$

Given a quadratic polynomial, the two solutions of x that satisfy this equation are given by the quadratic equation

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}} \tag{285}$$

This one equation has three solutions due to the cube root of complex numbers.

93.1 Algebraic Proof

The most straight-forward proof is given by a technique referred to as 'Completing the Square'.

$$ax^2 + bx + c = 0$$

Dividing both sides by a

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

Adding and subtracting $\frac{b^2}{(2a)^2}$

$$x^2 + \frac{b}{a}x + \frac{b^2}{(2a)^2} + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Recognizing the 'square'

$$(x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{(2a)^2} = 0$$

Rearranging and solving

$$(x + \frac{b}{2a})^2 = \frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x = -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

93.2 Discriminant

The discriminant of a quadratic equation is defined to be

$$\Delta = b^2 - 4ac \tag{286}$$

94 Cubic

Definition 94.1. *A cubic polynomial*

$$ax^3 + bx^2 + cx + d = 0$$

Given a cubic polynomial the three solutions of x that satisfy this equation are given by

$$\boxed{x = -\frac{b}{3a} + \frac{1}{\sqrt[3]{2} \cdot 3a}\left(\sqrt[3]{9abc - 27da^2 - 2b^3 + 3a\sqrt[2]{3\left(27a^2d^2 + 4b^3d - 18abcd + 4ac^3 - b^2c^2\right)}} + \sqrt[3]{9abc - 27da^2 - 2b^3 - 3a\sqrt[2]{3\left(27a^2d^2 + 4b^3d - 18abcd + 4ac^3 - b^2c^2\right)}}\right)} \tag{287}$$

This was first shown by Gerolamo Cardano in 1545

94.1 Depressed Cubic

Definition 94.2. *A depressed cubic is a polynomial in the form of*

$$x^3 + cx + d = 0$$

Theorem 94.1. *We can transform any cubic to a depressed cubic form.*
Proof

Starting from definition 94.1

$$ax^3 + bx^2 + cx + d = 0$$

Dividing by a, we define

$$a_2 = \frac{b}{a}, a_1 = \frac{c}{a}, a_0 = \frac{d}{a}$$

Rewriting

$$x^3 + a_2x^2 + a_1x + a_0 = 0$$

Completing the cube

$$x^3 + 3\Big(\frac{a_2}{3}\Big)x^2 + 3\Big(\frac{a_2}{3}\Big)^2x + \Big(\frac{a_2}{3}\Big)^3 - 3\Big(\frac{a_2}{3}\Big)^2x - \Big(\frac{a_2}{3}\Big)^3 + a_1x + a_0 = 0$$

Using $(x+t)^3 = x^3 + 3x^2t + 3xt^2 + t^3$

$$\Big(x + \frac{a_2}{3}\Big)^3 + x\Big(a_1 - 3\Big(\frac{a_2}{3}\Big)^2\Big) + \Big(a_0 - \Big(\frac{a_2}{3}\Big)^3\Big) = 0$$

Let $z = x + \frac{a_2}{3}$

$$z^3 + \Big(z - \frac{a_2}{3}\Big)\Big(a_1 - 3\Big(\frac{a_2}{3}\Big)^2\Big) + \Big(a_0 - \Big(\frac{a_2}{3}\Big)^3\Big) = 0$$

Simplifying and collecting like terms

$$z^3 + z\Big(a_1 - 3\Big(\frac{a_2}{3}\Big)^2\Big) + \left(- \frac{a_2}{3}\Big(a_1 - 3\Big(\frac{a_2}{3}\Big)^2\Big) + \Big(a_0 - \Big(\frac{a_2}{3}\Big)^3\Big)\right) = 0$$

Therefore after shifting with $z = x + \frac{b}{3a}$

$$c' = \frac{c}{a} - 3\Big(\frac{b}{3a}\Big)^2$$
$$d' = -\frac{b}{3a}\Big(\frac{c}{a} - 3\Big(\frac{b}{3a}\Big)^2\Big) + \Big(\frac{d}{a} - \Big(\frac{b}{3a}\Big)^3\Big)$$

Simplifying

$$c' = \frac{c}{a} - 3\Big(\frac{b}{3a}\Big)^2$$
$$d' = \frac{d}{a} - \Big(\frac{b}{3a}\Big)^3 - \frac{b}{3a}c'$$

Theorem 94.2. *Starting from the depressed cubic form, we can solve for the roots.*

Proof

Starting from definition 94.2

$$z^3 + cz + d = 0$$

Define $p = -\frac{c}{3}$ *and* $q = -\frac{d}{2}$

$$z^3 = 3pz + 2q$$

Consider $z = u + v$

$z^3 = (u+v)^3 = u^3 + 3u^2v + 3uv^2 + v^3 = u^3 + v^3 + 3uv(u+v) = 3uv(z) + u^3 + v^3$

Comparing these forms

$$3uv = 3p$$
$$u^3 + v^3 = 2q$$

This implies

$$v = \frac{p}{u}$$
$$u^3 + \left(\frac{p}{u}\right)^3 = 2q$$

Therefore we can create a polynomial equation to solve for this relation

$$u^6 + -2qu^3 + p^3 = 0$$

We can use the quadratic solution to solve for this relation

$$u^3 = \frac{2q \pm \sqrt{4q^2 - 4p^3}}{2} = q \pm \sqrt{q^2 - p^3}$$

Define the solutions for u

$$u_+ = \sqrt[3]{q + \sqrt{q^2 - p^3}}$$
$$u_- = \sqrt[3]{q - \sqrt{q^2 - p^3}}$$

Since $u_+^3 u_-^3 = p^3$*, from quadratic equation*

$$u_+ u_- = p$$

Using definition $v = \frac{p}{u}$

$$v_+ = \frac{p}{u_+} = \frac{pu_-}{u_+u_-} = \frac{pu_-}{p} = u_-$$

Similarly

$$v_- = u_+$$

Therefore the solution to $z^3 = 3pz + 2q$

$$z = u + v = u_+ + u_- = \sqrt[3]{q + \sqrt{q^2 - p^3}} + \sqrt[3]{q - \sqrt{q^2 - p^3}}$$

Re-writing in the original form

$$z = \sqrt[3]{-\frac{d}{2} + \sqrt{\frac{d^2}{4} + \frac{c^3}{27}}} + \sqrt[3]{-\frac{d}{2} - \sqrt{\frac{d^2}{4} + \frac{c^3}{27}}}$$

94.2 General Cubic

Combining all of these results to back out the solution for a general cubic equation in terms of the original variables, $ax^3 + bx^2 + cx + d = 0$, we first summarize the steps taken.

- shift solution by $z = x + \frac{b}{3a}$
- write solution in depressed cubic form
- solve the depressed cubic equation.

Undoing these substitutions backwards

$$z = \sqrt[3]{-\frac{d'}{2} + \sqrt[2]{\frac{d'^2}{4} + \frac{c'^3}{27}}} + \sqrt[3]{-\frac{d'}{2} - \sqrt[2]{\frac{d'^2}{4} + \frac{c'^3}{27}}}$$

Using $c' = \frac{c}{a} - 3\left(\frac{b}{3a}\right)^2$ and $d' = \frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}c'$

$$z = \sqrt[3]{-\frac{\left(\frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)\right)}{2} + \sqrt[2]{\frac{\left(\frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)\right)^2}{4} + \frac{\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)^3}{27}}}$$
$$+\sqrt[3]{-\frac{\left(\frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)\right)}{2} - \sqrt[2]{\frac{\left(\frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)\right)^2}{4} + \frac{\left(\frac{c}{a} - 3\left(\frac{b}{3a}\right)^2\right)^3}{27}}}$$

Simplifying

$$c' = \frac{c}{a} - 3\left(\frac{b}{3a}\right)^2 = \frac{3ac - b^2}{3a^2}$$
$$d' = \frac{d}{a} - \left(\frac{b}{3a}\right)^3 - \frac{b}{3a}c' = \frac{27da^2 + 2b^3 - 9abc}{27a^3}$$
$$z = \sqrt[3]{-\frac{27da^2 + 2b^3 - 9abc}{54a^3} + \sqrt[2]{\frac{\left(27da^2 + 2b^3 - 9abc\right)^2}{54^2a^6} + \frac{\left(3ac - b^2\right)^3}{3^6a^6}}}$$
$$+\sqrt[3]{-\frac{27da^2 + 2b^3 - 9abc}{54a^3} - \sqrt[2]{\frac{\left(27da^2 + 2b^3 - 9abc\right)^2}{54^2a^6} + \frac{\left(3ac - b^2\right)^3}{3^6a^6}}}$$

Factoring out 3a

$$z = \sqrt[3]{-\frac{27da^2 + 2b^3 - 9abc}{54a^3} + \frac{1}{(3a)^3}\sqrt[2]{\frac{\left(27da^2 + 2b^3 - 9abc\right)^2}{2^2} + \left(3ac - b^2\right)^3}}$$
$$+\sqrt[3]{-\frac{27da^2 + 2b^3 - 9abc}{54a^3} - \frac{1}{(3a)^3}\sqrt[2]{\frac{\left(27da^2 + 2b^3 - 9abc\right)^2}{2^2} + \left(3ac - b^2\right)^3}}$$

Simplifying $\frac{\left(27da^2+2b^3-9abc\right)^2}{2^2}+\left(3ac-b^2\right)^3=A+B$

$$A=\frac{27^2}{4}a^4d^2+27a^2b^3d-\frac{27\cdot 9}{2}a^3bcd+b^6-9ab^4c+\frac{9^2}{4}a^2b^2c^2$$

$$B=3^3a^3c^3-27a^2b^2c^2+9ab^4c-b^6$$

$$A+B=\frac{27^2}{4}a^4d^2+27a^2b^3d-\frac{27\cdot 9}{2}a^3bcd+3^3a^3c^3+\left(\frac{9}{4}-3\right)9a^2b^2c^2$$

$$A+B=\frac{27a^2}{4}\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)$$

Note that this quantity is the negative of the discriminant of the cubic, denoted by Δ.

Plugging this back in we obtain

$$z=\sqrt[3]{-\frac{27da^2+2b^3-9abc}{54a^3}+\frac{1}{2\cdot 3^2a^2}\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}$$
$$+\sqrt[3]{-\frac{27da^2+2b^3-9abc}{54a^3}-\frac{1}{2\cdot 3^2a^2}\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}$$

Simplfying more

$$z=\frac{1}{\sqrt[3]{2}\cdot 3a}\left(\sqrt[3]{9abc-27da^2-2b^3+3a\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}\right.$$
$$\left.+\sqrt[3]{9abc-27da^2-2b^3-3a\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}\right)$$

Undoing the first shifting of $z=x+\frac{b}{3a}$

$$x=-\frac{b}{3a}+\frac{1}{\sqrt[3]{2}\cdot 3a}\left(\sqrt[3]{9abc-27da^2-2b^3+3a\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}\right.$$
$$\left.+\sqrt[3]{9abc-27da^2-2b^3-3a\sqrt[2]{3\left(27a^2d^2+4b^3d-18abcd+4ac^3-b^2c^2\right)}}\right)$$

Writing the full solution in terms of the discriminant we get

$$x=-\frac{b}{3a}+\frac{1}{\sqrt[3]{2}\cdot 3a}\left(\sqrt[3]{9abc-27da^2-2b^3+3^{\frac{3}{2}}a\sqrt[2]{\Delta}i}+\sqrt[3]{9abc-27da^2-2b^3-3^{\frac{3}{2}}a\sqrt[2]{\Delta}i}\right)$$

94.3 Discriminant

The discriminant of a cubic equation $ax^3+bx^2+cx+d=0$ is defined to be

$$\Delta=18abcd-4b^3d+b^2c^2-4ac^3-27a^2d^2 \tag{288}$$

1. If $\Delta>0$, the cubic has 3 distinct real roots.
2. If $\Delta=0$, the cubic has multiple roots (and all roots are real).
3. If $\Delta<0$, the cubic has 1 real root and 2 complex conjugate roots.

95 Quartic Equations

A general quartic polynomial has the form:

$$ax^4 + bx^3 + cx^2 + dx + e = 0, \qquad (a \neq 0) \tag{289}$$

95.1 Depressing the Quartic

Dividing by a and applying the Tschirnhaus transformation $x = y - \frac{b}{4a}$ eliminates the cubic term, yielding the *depressed quartic*:

$$y^4 + py^2 + qy + r = 0 \tag{290}$$

where

$$\begin{aligned} p &= \frac{8ac - 3b^2}{8a^2}, \\ q &= \frac{b^3 - 4abc + 8a^2 d}{8a^3}, \\ r &= \frac{-3b^4 + 256a^3 e - 64a^2 bd + 16ab^2 c}{256a^4} \end{aligned}$$

95.2 Ferrari's Method of Solution

Rewrite equation (290) by moving the linear and constant terms to the right-hand side:

$$y^4 + py^2 = -qy - r$$

Complete the square on the left-hand side by adding $\left(\frac{p}{2}\right)^2$:

$$\left(y^2 + \frac{p}{2}\right)^2 = -qy - r + \frac{p^2}{4}$$

Now introduce an auxiliary parameter m. Adding $2m\left(y^2 + \frac{p}{2}\right) + m^2$ to both sides yields:

$$\left(y^2 + \frac{p}{2} + m\right)^2 = 2my^2 - qy + \left(m^2 + pm + \frac{p^2}{4} - r\right) \tag{291}$$

The right-hand side is a quadratic in y: $Ay^2 + By + C$ where $A = 2m$, $B = -q$, and $C = m^2 + pm + \frac{p^2}{4} - r$.

For the right-hand side to be a perfect square $(ky + l)^2$, its discriminant with respect to y must vanish ($B^2 - 4AC = 0$):

$$(-q)^2 - 4(2m)\left(m^2 + pm + \frac{p^2}{4} - r\right) = 0$$

Expanding this condition gives **Ferrari's Resolvent Cubic**:

$$\boxed{8m^3 + 8pm^2 + (2p^2 - 8r)m - q^2 = 0} \tag{292}$$

This is a cubic equation in m, which can be solved explicitly in radicals using Cardano's formula.

Let $m_0 \neq 0$ be any real (or complex) root of the resolvent cubic. Then the right-hand side of equation (291) factors as:

$$2m_0y^2 - qy + \left(m_0^2 + pm_0 + \frac{p^2}{4} - r\right) = \left(\sqrt{2m_0}y - \frac{q}{2\sqrt{2m_0}}\right)^2$$

Taking square roots on both sides gives two quadratic equations:

$$y^2 + \frac{p}{2} + m_0 = \pm\left(\sqrt{2m_0}y - \frac{q}{2\sqrt{2m_0}}\right) \tag{293}$$

Rearranging into standard quadratic form:

$$y^2 - \sqrt{2m_0}y + \left(\frac{p}{2} + m_0 + \frac{q}{2\sqrt{2m_0}}\right) = 0 \tag{294}$$

$$y^2 + \sqrt{2m_0}y + \left(\frac{p}{2} + m_0 - \frac{q}{2\sqrt{2m_0}}\right) = 0 \tag{295}$$

Solving each quadratic with the standard quadratic formula yields the four roots y_1, y_2, y_3, y_4, and substituting back $x_k = y_k - \frac{b}{4a}$ provides the exact solutions to the original quartic equation.

Remark 95.1 (The Abel-Ruffini Theorem)**.** *By the Abel-Ruffini Theorem (1799/1824) and Galois Theory, the quartic is the highest-degree general polynomial equation solvable in radicals; general polynomials of degree $n \geq 5$ cannot be solved in terms of radicals.*

96 Sylvester Matrices, Resultants, and General Discriminants

96.1 The Sylvester Matrix

Let $P(x)$ and $Q(x)$ be polynomials with degrees $n \geq 1$ and $m \geq 1$:

$$P(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0, \qquad (a_n \neq 0) \tag{296}$$

$$Q(x) = b_mx^m + b_{m-1}x^{m-1} + \cdots + b_1x + b_0, \qquad (b_m \neq 0) \tag{297}$$

Definition 96.1 (Sylvester Matrix)**.** *The **Sylvester matrix** $S(P, Q)$ is the $(n + m) \times (n + m)$ matrix formed by filling m rows with the coefficients of $P(x)$ and n rows with the coefficients of $Q(x)$, shifted successively:*

$$S(P,Q) = \begin{pmatrix} a_n & a_{n-1} & \cdots & a_1 & a_0 & 0 & \cdots & 0 \\ 0 & a_n & a_{n-1} & \cdots & a_1 & a_0 & \cdots & 0 \\ \vdots & & \ddots & & & & \ddots & \vdots \\ 0 & \cdots & 0 & a_n & a_{n-1} & \cdots & a_1 & a_0 \\ b_m & b_{m-1} & \cdots & b_1 & b_0 & 0 & \cdots & 0 \\ 0 & b_m & b_{m-1} & \cdots & b_1 & b_0 & \cdots & 0 \\ \vdots & & \ddots & & & & \ddots & \vdots \\ 0 & \cdots & 0 & b_m & b_{m-1} & \cdots & b_1 & b_0 \end{pmatrix} \tag{298}$$

96.2 The Resultant

Definition 96.2 (Resultant)**.** *The* ***resultant*** *of polynomials P and Q, denoted* $\operatorname{Res}(P, Q)$*, is the determinant of their Sylvester matrix:*

$$\operatorname{Res}(P, Q) = \det S(P, Q) \tag{299}$$

Theorem 96.1 (Fundamental Property of Resultants)**.** *Let* $P(x) = a_n \prod_{i=1}^{n}(x - \alpha_i)$ *and* $Q(x) = b_m \prod_{j=1}^{m}(x - \beta_j)$*. Then:*

$$\operatorname{Res}(P, Q) = a_n^m b_m^n \prod_{i=1}^{n}\prod_{j=1}^{m}(\alpha_i - \beta_j) = a_n^m \prod_{i=1}^{n} Q(\alpha_i) = (-1)^{nm} b_m^n \prod_{j=1}^{m} P(\beta_j) \tag{300}$$

In particular, $\operatorname{Res}(P, Q) = 0$ *if and only if* $P(x)$ *and* $Q(x)$ *share a common root in* $\mathbb{C}$*.*

96.3 The General Polynomial Discriminant

A polynomial $P(x)$ has a multiple root if and only if $P(x)$ and its derivative $P'(x)$ share a root, which occurs precisely when $\operatorname{Res}(P, P') = 0$.

Definition 96.3 (Polynomial Discriminant)**.** *For a polynomial* $P(x) = a_n \prod_{i=1}^{n}(x - \alpha_i)$*, the* ***discriminant*** $\Delta(P)$ *is defined as:*

$$\Delta(P) = a_n^{2n-2} \prod_{1 \le i < j \le n} (\alpha_i - \alpha_j)^2 \tag{301}$$

In terms of the resultant with its derivative $P'(x)$*:*

$$\boxed{\Delta(P) = \frac{(-1)^{\frac{n(n-1)}{2}}}{a_n} \operatorname{Res}(P, P')} \tag{302}$$

96.3.1 Example 1: Quadratic Discriminant

For $P(x) = ax^2 + bx + c$, $P'(x) = 2ax + b$. The Sylvester matrix is 3×3:

$$S(P, P') = \begin{pmatrix} a & b & c \\ 2a & b & 0 \\ 0 & 2a & b \end{pmatrix}$$

Evaluating the determinant:

$$\det S(P, P') = a(b^2) - b(2ab) + c(4a^2) = ab^2 - 2ab^2 + 4a^2c = -a(b^2 - 4ac)$$

Since $n = 2$, $(-1)^{n(n-1)/2} = (-1)^1 = -1$, yielding:

$$\Delta(P) = \frac{-1}{a}\left[-a(b^2 - 4ac)\right] = b^2 - 4ac$$

96.3.2 Example 2: Cubic Discriminant

For $P(x) = ax^3 + bx^2 + cx + d$, $P'(x) = 3ax^2 + 2bx + c$. The Sylvester matrix is 5×5:

$$S(P, P') = \begin{pmatrix} a & b & c & d & 0 \\ 0 & a & b & c & d \\ 3a & 2b & c & 0 & 0 \\ 0 & 3a & 2b & c & 0 \\ 0 & 0 & 3a & 2b & c \end{pmatrix}$$

Since $n = 3$, $(-1)^{n(n-1)/2} = (-1)^3 = -1$. Evaluating $\det S(P, P')$ directly gives:

$$\Delta(P) = -\frac{1}{a} \det S(P, P') = 18abcd - 4b^3 d + b^2 c^2 - 4ac^3 - 27a^2 d^2$$

matching exactly the formula in equation (288).

97 Ramanujan Identity

Prove that:

$$\alpha\sum_{k=1}^{\infty}\frac{k}{e^{2\alpha k}-1}+\beta\sum_{k=1}^{\infty}\frac{k}{e^{2\beta k}-1}=\frac{\alpha+\beta}{24}-\frac{1}{4}$$

$$\alpha>0,\beta>0,\alpha\cdot\beta=\pi^2$$

$$f(t)=\begin{cases}\frac{|t|}{e^{2\alpha|t|}-1} & t\neq 0\\ \frac{1}{2\alpha} & t=0\end{cases}$$

$$\hat{f}(s)=\int_{-\infty}^{\infty}f(t)e^{-i2\pi st}dt=\int_{-\infty}^{\infty}f(t)\cos(2\pi st)dt+i\int_{-\infty}^{\infty}f(t)\sin(2\pi st)dt=2\int_{0}^{\infty}\frac{t\cos(2\pi st)}{e^{2\alpha t}-1}dt \text{ and } s\neq 0$$

$$\hat{f}(0)=2\int_{0}^{\infty}\frac{t}{e^{2\alpha t}-1}dt=\frac{2}{(2\alpha)^2}\zeta(2)=\frac{\pi^2}{12\alpha^2}\text{ (Insert Reference)}$$

Notice both the function and it's Fourier Transform are even.
By Poisson Summation formula (Include Reference):

$$\hat{f}(0)+2\sum_{m=1}^{\infty}\hat{f}(m)=f(0)+2\sum_{n=1}^{\infty}f(n)$$

Hence applying this:

$$\frac{1}{2\alpha}+2\sum_{n=1}^{\infty}\frac{n}{e^{2\alpha n}-1}=\frac{\pi^2}{12\alpha^2}+4\sum_{m=1}^{\infty}\int_{0}^{\infty}\frac{t\cos(2\pi mt)}{e^{2\alpha t}-1}dt$$

Examining

$$\int_{0}^{\infty}\frac{t\cos(2\pi mt)}{e^{2\alpha t}-1}dt$$

By considering

$$\int_{0}^{\infty}\frac{\sin(cx)}{e^{x}-1}dx=\int_{0}^{\infty}\frac{e^{-x}\sin(cx)}{1-e^{-x}}dx=\int_{0}^{\infty}\sin cx\sum_{k=1}^{\infty}e^{-kx}dx$$

By the Dominant Convergence Theorem: (Include Reference)

$$\int_{0}^{\infty}\sin(cx)\sum_{k=1}^{\infty}e^{-kx}dx=\sum_{k=1}^{\infty}\int_{0}^{\infty}\sin(cx)e^{-kx}dx$$

By the LaPlace Transform of $\sin at$:

$$\int_{0}^{\infty}\sin(ct)e^{-sx}dt=\frac{c}{c^2+s^2}\text{ (Insert Reference)}$$

$$\sum_{k=1}^{\infty}\int_{0}^{\infty}\sin(cx)e^{-kx}dx=\sum_{k=1}^{\infty}\frac{c}{c^2+k^2}$$

By the Tangent Identity (Insert Reference)

$$\sum_{n=-\infty}^{\infty} \frac{1}{\alpha^2+n^2} = \frac{\pi}{\alpha}\coth(\pi\alpha)$$

Where

$$\coth(x) = \frac{\cosh(x)}{\sinh(x)} = \frac{e^x+e^{-x}}{e^x-e^{-x}} = \frac{e^{2x}+1}{e^{2x}-1}$$

Hence:

$$\sum_{n=-\infty}^{\infty} \frac{1}{\alpha^2+n^2} = \frac{\pi}{\alpha}\cdot\frac{e^{2\pi\alpha}+1}{e^{2\pi\alpha}-1}$$

Hence:

$$\sum_{n=1}^{\infty} \frac{1}{\alpha^2+n^2} = \frac{1}{2}\cdot(\frac{\pi}{\alpha}\cdot\frac{e^{2\pi\alpha}+1}{e^{2\pi\alpha}-1} - \frac{1}{\alpha^2})$$

$$=> \sum_{k=1}^{\infty} \frac{c}{c^2+k^2} = \frac{\pi}{2}\cdot\frac{e^{2\pi c}+1}{e^{2\pi c}-1} - \frac{1}{2c}$$

$$=> \int_0^{\infty} \frac{\sin(cx)}{e^x-1}dx = \frac{\pi}{2}\cdot\frac{e^{2\pi c}+1}{e^{2\pi c}-1} - \frac{1}{2c}$$

Taking the derivative of both sides using Leibniz Rule

$$\frac{d}{dc}\int_0^{\infty} \frac{\sin(cx)}{e^x-1}dx = \int_0^{\infty} \frac{\partial}{\partial c}\frac{\sin(cx)}{e^x-1}dx = \int_0^{\infty} \frac{x\cos(cx)}{e^x-1}dx$$

$$= \frac{d}{dc}(\frac{\pi}{2}\cdot\frac{e^{2\pi c}+1}{e^{2\pi c}-1} - \frac{1}{2c}) = \frac{\pi}{2}\cdot\frac{2\pi e^{2\pi c}}{e^{2\pi c}-1} - \frac{\pi}{2}\cdot\frac{2\pi e^{4\pi c}+2\pi e^{2\pi c}}{(e^{2\pi c}-1)^2} + \frac{1}{2c^2} = \frac{-2\pi^2 e^{2\pi c}}{(e^{2\pi c}-1)^2} + \frac{1}{2c^2}$$

Therefore making a change of variables $x = 2\alpha t$

$$\int_0^{\infty} \frac{x\cos(cx)}{e^x-1}dx = \int_0^{\infty} \frac{2\alpha t\cos(2\alpha ct)}{e^{2\alpha t}-1}2\alpha dt = (2\alpha)^2\int_0^{\infty} \frac{t\cos(2\alpha ct)}{e^{2\alpha t}-1}dt = \frac{-2\pi^2 e^{2\pi c}}{(e^{2\pi c}-1)^2} + \frac{1}{2c^2}$$

Letting $c = \pi\cdot\frac{m}{\alpha}$

$$(2\alpha)^2\int_0^{\infty} \frac{t\cos(2\pi m\cdot t)}{e^{2\alpha t}-1}dt = \frac{-2\pi^2 e^{2\pi^2\cdot\frac{m}{\alpha}}}{(e^{2\pi^2\cdot\frac{m}{\alpha}}-1)^2} + \frac{1}{2\pi^2\cdot\frac{m^2}{\alpha^2}} = \frac{-2\pi^2 e^{2\pi^2\cdot\frac{m}{\alpha}}}{(e^{2\pi^2\cdot\frac{m}{\alpha}}-1)^2} + \frac{\alpha^2}{2\pi^2 m^2}$$

Plugging this result back into our original statement:

$$\frac{1}{2\alpha} + 2\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n}-1} = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty}(\frac{-2\pi^2 e^{2\pi^2\cdot\frac{m}{\alpha}}}{\alpha^2(e^{2\pi^2\cdot\frac{m}{\alpha}}-1)^2} + \frac{1}{2\pi^2 m^2}) = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty} \frac{-2\pi^2 e^{2\pi^2\cdot\frac{m}{\alpha}}}{\alpha^2(e^{2\pi^2\cdot\frac{m}{\alpha}}-1)^2} + \frac{1}{2\pi^2}\zeta(2)$$

$$\frac{1}{2\alpha} + 2\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n}-1} = \frac{\pi^2}{12\alpha^2} + \sum_{m=1}^{\infty} \frac{-2\pi^2 e^{2\pi^2\cdot\frac{m}{\alpha}}}{\alpha^2(e^{2\pi^2\cdot\frac{m}{\alpha}}-1)^2} + \frac{1}{12}$$

Let $\beta = \frac{\pi^2}{\alpha}$

$$=> \frac{1}{2\alpha} + 2\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + \frac{1}{12} - \frac{2\pi^2}{\alpha^2} \sum_{m=1}^{\infty} \frac{e^{2m\beta}}{(e^{2m\beta} - 1)^2}$$

Examining $\sum_{m=1}^{\infty} \frac{e^{2m\beta}}{(e^{2m\beta}-1)^2}$

$$\sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} = \sum_{n=1}^{\infty} \frac{ne^{-2\beta n}}{1 - e^{-2\beta n}} = \sum_{n=1}^{\infty} (n \sum_{m=1}^{\infty} e^{-2\beta \cdot n \cdot m})$$

$$\sum_{n=1}^{\infty} (n \sum_{m=1}^{\infty} e^{-2\beta \cdot n \cdot m}) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} ne^{-2\beta \cdot n \cdot m} = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} ne^{-2\beta \cdot n \cdot m}$$

Insert Reference from Geometric Series Section

$$\sum_{m=1}^{\infty} \frac{e^{-2\beta \cdot m}}{(1 - e^{-2\beta \cdot m})^2} = \sum_{m=1}^{\infty} \frac{e^{2\beta \cdot m}}{(e^{2\beta \cdot m} - 1)^2} = \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1}$$

Hence combining this fact we obtain:

$$=> \frac{1}{2\alpha} + 2\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\pi^2}{12\alpha^2} + \frac{1}{12} - \frac{2\pi^2}{\alpha^2} \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1}$$

$$\frac{1}{2\alpha} + 2\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\beta}{12\alpha} + \frac{1}{12} - \frac{2\beta}{\alpha} \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1}$$

$$\frac{1}{2} + 2\alpha\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} = \frac{\beta}{12} + \frac{\alpha}{12} - 2\beta \sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1}$$

$$\boxed{\alpha\sum_{n=1}^{\infty} \frac{n}{e^{2\alpha n} - 1} + \beta\sum_{n=1}^{\infty} \frac{n}{e^{2\beta n} - 1} = \frac{\beta + \alpha}{24} - \frac{1}{4}}$$

Where $\alpha \cdot \beta = \pi^2$
The special case of $\alpha = \beta = \pi$:

$$\boxed{\sum_{n=1}^{\infty} \frac{n}{e^{2\pi n} - 1} = \frac{1}{24} - \frac{1}{8\pi}}$$

98 Overview of matrix derivatives

$$\frac{\partial x^T A x}{\partial x} = x^T (A^T + A) \tag{303}$$

$$\frac{\partial x^T a}{\partial x} = a^T \tag{304}$$

99 Overview of LaGrange Multipliers

For our purposes we only need the simplified version of LaGrange Multipliers

If

$$\max F(\vec{x})$$

subject to

$$g(\vec{x}) = \vec{c}$$

Then

$$\mathcal{L}(\vec{x}, \vec{\lambda}) = F(\vec{x}) - \overrightarrow{\lambda}^T \big(g(\vec{x}) - \vec{c}\big)$$

$$\frac{\partial \mathcal{L}}{\partial \vec{\lambda}} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \vec{x}} = 0$$

100 Proof

Consider the problem of finding a vector on the unit sphere that maximizes the quadratic form of a **symmetric** square matrix A. Written as an optimization problem we get:

$$\begin{aligned} \max{}_x \; & x^T A x \\ & x^T x = 1 \end{aligned} \tag{305}$$

Since the unit sphere is a compact and bounded space in R^n we know that a maximum (and a minimum) exist. We can use LaGrange multipliers to uniquely solve this problem. Using the method we get the function:

$$g(x, \lambda) = x^T A x - \lambda \cdot (x^T x - 1)$$

Taking derivatives:

$$\frac{\partial g(x, \lambda)}{\partial x} = 2x^T A - 2\lambda \cdot x^T$$

$$\frac{\partial g(x, \lambda)}{\partial \lambda} = -(x^T x - 1)$$

Therefore the optimal values of x and λ will satisfy:

$$A^T x_{opt} = A x_{opt} = \lambda_{opt} \cdot x_{opt} \tag{306}$$

$$x_{opt}^T x_{opt} = 1 \tag{307}$$

Denote $x_1 = x_{opt}$ and $\lambda_1 = \lambda_{opt}$. We can choose to name these as our first eigenvector and first eigenvalue respectively.

Let's attempt to find a perpendicular vector to the one we found that maximizes quadratic form of A in a different direction.

$$\begin{aligned} \max_x \; & x^T A x \\ & x^T x = 1 \\ & x^T x_1 = 0 \end{aligned} \tag{308}$$

Solving we get:

$$g(x, \lambda, \mu) = x^T A x - \lambda \cdot (x^T x - 1) - \mu \cdot x^T x_1$$

Taking derivatives:

$$\frac{\partial g(x, \lambda, \mu)}{\partial x} = 2x^T A - 2\lambda \cdot x^T - \mu \cdot x_1^T = 0$$

Taking transposes and using $A^T = A$:

$$2Ax - 2\lambda \cdot x - \mu \cdot x_1 = 0$$

Taking the dot product with respect to x_1^T we get:

$$(2Ax)x_1^T - (2\lambda x)x_1^T - (\mu x_1)x_1^T = 0$$

Using

$$(2Ax)x_1^T = 2A^T(xx_1^T) = 0$$
$$x_1 x_1^T = 1$$

We get

$$(2Ax)x_1^T - (2\lambda x)x_1^T - (\mu x_1)x_1^T = 0 + 0 - \mu = 0$$

Therefore the optimal values of x and λ will satisfy:

$$Ax_{opt} = \lambda_{opt} \cdot x_{opt} \tag{309}$$

Denote $x_2 = x_{opt}$ and $\lambda_2 = \lambda_{opt}$. We can choose to name these as our second eigenvector and second eigenvalue respectively.

The same argument extends for consecutive eigenvalues and eigenvectors. Interpreting this proof leads to the concept of Principle Component Analysis.

101 Motivating Example

Suppose there are three assets, a, b, c , by knowing the volatility of each of the assets, and the correlation between assets a and b, as well as the correlation between assets b and c, what is the interval of valid correlations between and assets a and c?

This document presents one general method of addressing this problem, and solves the case of three assets analytically.

102 Definition

Definition 102.1. *A positive-definite matrix is defined as TODO.*

Definition 102.2. *A semi-positive-definite matrix is defined as TODO.*

Definition 102.3. *A principle minor is defined as TODO.*

103 Background Theory

We make use of some known properties of positive-definite (PD) and positive-semi-definite (PSD) matrices.

Theorem 103.1. *A PD matrix must have all positive eigenvalues.*
Proof:

From the definition of PD

$$TODO$$

Theorem 103.2. *A PD matrix must have all positive principle minors.*
Proof:

From the definition of PD

$$TODO$$

104 Main Idea

All principle majors of a PD matrix, must be positive. Therefore, compute them all, and figure out bounds on the unknown quantities.

104.1 Two Asset

The case $n = 2$ is trivial, due to the definition of correlation being bounded by -1 and 1.

$$|\rho_{a,b}| \leq 1$$
$$|\frac{\sigma_{a,b}}{\sigma_a \cdot \sigma_b}| \leq 1$$

Since σ_a and σ_b are positive by definition

$$|\sigma_{a,b}| \leq \sigma_a \cdot \sigma_b$$

We show that the positive principle minors method arrives at the same bound. Let matrix C be a 2x2 matrix,

$$C = \begin{pmatrix} \sigma_a^2 & \sigma_{ab} \\ \sigma_{ab} & \sigma_b^2 \end{pmatrix} \tag{310}$$

104.2 3x3

We can

$$C = \begin{pmatrix} 1 & \rho_{ab} & \rho_{\mathbf{ac}} \\ \rho_{ab} & 1 & \rho_{bc} \\ \rho_{\mathbf{ac}} & \rho_{bc} & 1 \end{pmatrix} \tag{311}$$

Theorem 104.1.

$$\boxed{|\rho_{\mathbf{ac}} - \rho_{ab} \cdot \rho_{bc}| \leq \sqrt{(1-\rho_{ab}^2) \cdot (1-\rho_{bc}^2)}} \tag{312}$$

Proof:

Det(C) ¿= 0

TODO

104.3 4x4

We can

$$C = \begin{pmatrix} 1 & \rho_{ab} & \rho_{ac} & x \\ \rho_{ab} & 1 & \rho_{bc} & \rho_{bd} \\ \rho_{ac} & \rho_{bc} & 1 & \rho_{cd} \\ x & \rho_{bd} & \rho_{cd} & 1 \end{pmatrix} \tag{313}$$

Theorem 104.2. *Bounds on $x = \rho_{ad}$*

$$\boxed{\begin{gathered} |\rho_{\mathbf{ad}} - \frac{(\rho_{a,b} \cdot \rho_{b,d} + \rho_{a,c} \cdot \rho_{c,d}) - \rho_{b,c} \cdot (\rho_{a,b} \cdot \rho_{c,d} + \rho_{a,c} \cdot \rho_{b,d})}{1-\rho_{b,c}^2}| \leq (1-\rho_{b,c}^2)\sqrt{D} \\ D = (1-\rho_{a,b}^2 - \rho_{a,c}^2 - \rho_{b,c}^2 + 2 \cdot \rho_{a,b} \cdot \rho_{a,c} \cdot \rho_{b,c}) \cdot (1-\rho_{b,c}^2 - \rho_{b,d}^2 - \rho_{c,d}^2 + 2 \cdot \rho_{b,c} \cdot \rho_{b,d} \cdot \rho_{c,d}) \end{gathered}}$$

Proof:

$$\begin{gathered} Det(C) \geq 0 \\ 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2) \\ +2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c}) \\ -2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d}) \\ +(\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2) \\ +2x(\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) \\ -2x(\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c}) \\ +x^2(\rho_{b,c}^2 - 1) \geq 0 \end{gathered}$$

This is just a quadradic equation

$$a = \rho_{b,c}^2 - 1$$
$$b = 2(\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) - 2(\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c})$$
$$c = 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2)$$
$$+2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c})$$
$$-2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d})$$
$$+(\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2)$$
$$ax^2 + bx + c \geq 0$$
$$|2ax + b| \leq \sqrt{b^2 - 4ac}$$

Dividing all sides by 2

$$a = \rho_{b,c}^2 - 1$$
$$b = (\rho_{a,b}\rho_{b,d} + \rho_{a,c}\rho_{c,d}) - (\rho_{a,b}\rho_{b,c}\rho_{c,d} + \rho_{a,c}\rho_{b,d}\rho_{b,c})$$
$$c = 1 - (\rho_{c,d}^2 + \rho_{b,c}^2 + \rho_{b,d}^2 + \rho_{a,b}^2 + \rho_{a,c}^2)$$
$$+2 \cdot (\rho_{b,c}\rho_{b,d}\rho_{c,d} + \rho_{a,b}\rho_{b,c}\rho_{a,c})$$
$$-2(\rho_{a,b}\rho_{b,d}\rho_{a,c}\rho_{c,d})$$
$$+(\rho_{a,b}^2\rho_{c,d}^2 + \rho_{a,c}^2\rho_{b,d}^2)$$
$$|ax + b| \leq \sqrt{b^2 - ac}$$

Simplifying the square root

$$b^2 - ac = (a,b)^2(b,d)^2 + (a,c)^2(c,d)^2 + (b,c)^2(c,d)^2 + (b,c)^2(b,d)^2$$
$$+(b,c)^2(a,b)^2 + (b,c)^2(a,c)^2 - (a,b)^2(c,d)^2 - (a,c)^2(b,d)^2$$
$$+(2 * (b,c)^2 + (c,d)^2 + (b,d)^2 + (a,b)^2 + (a,c)^2) - 1$$
$$+4(a,b)(b,c)^2(c,d)(a,c)(b,c) - 2((b,c)(b,d)(c,d) + (a,b)(b,c)(a,c)) - (b,c)^4$$
$$-2((a,b)^2(b,d)(b,c)(c,d) + (a,b)(b,d)^2(b,c)(a,c) + (a,c)(c,d)^2(a,b)(b,c)$$
$$+(a,c)^2(c,d)(b,c)(b,d)) - 2((b,c)^3(b,d)(c,d) + (b,c)^3(a,b)(a,c))$$

105 A Quadratic Function

Given a quadratic function $y = ax^2 + bx + c$, to find the slope of the tangent line for all x, we need to solve for m such that $y_t = mx + b$ intersects the quadratic function.

$$ax^2 + bx + c = mx + k \tag{314}$$

Since we know the quadratic function is strictly convex, we know that the tangent line will only intersect the quadratic curve at one and only one point.

$$ax^2 + bx + c = mx + k$$

Rearranging

$$ax^2 + (b - m)x + (c - k) = 0$$

Using the quadratic equation

$$x = \frac{m - b \pm \sqrt{(b - m)^2 - 4a(c - k)}}{2a}$$

Since we know that there is only one intersection between these two curve, we know that the term inside the square root needs to be 0, hence making both solution of x the same.

$$x = \frac{m - b}{2a}$$

Solving for m

$$m = 2ax + b$$

Notice that this is the same as the derivative of $ax^2 + bx + c$.

106 A reciprocal function

Given a reciprocal function $y = \frac{a}{x}$, to find the slope of the tangent line for all x, we need to solve for m such that $y_t = mx + b$ intersects the quadratic function.

$$\frac{a}{x} = mx + k \tag{315}$$

Since we know the reciprocal function is strictly concave, we know that the tangent line will only intersect the reciprocal curve at one and only one point.

$$\frac{a}{x} = mx + k$$

Rearranging

$$mx^2 + kx - a = 0$$

Using the quadratic equation

$$x = \frac{-k \pm \sqrt{k^2 + 4ma}}{2m}$$

Since we know that there is only one intersection between these two curve, we know that the term inside the square root needs to be 0, hence making both solution of x the same.

$$x = \frac{-k}{2m}$$

Substituting for k

$$x = \frac{-(\frac{a}{x} - mx)}{2m}$$

Simplifying

$$2mx = mx - \frac{a}{x}$$

Solving for m

$$m = -\frac{a}{x^2}$$

Notice that this is the same as the derivative of $\frac{a}{x}$.

107 Defining logs simple

107.1 Exponating rules

$$a^x \cdot a^y = a^{x+y} \tag{316}$$

$$a^x \cdot a^{-x} = a^x \cdot \frac{1}{a^x} = 1 = a^{x-x} = a^0 \tag{317}$$